

impedance from side 1 to side 2. Under open-circuit test conditions

$$|V_2|^2 a^2 G_c = (11.5 \times 10^3)^2 \times 36 \times G_c = 66.7 \times 10^3 \text{ W}$$

$$G_c = 14.0 \times 10^{-6} \text{ S}$$

$$|Y| = \frac{|I_2|}{|V_2|} \times \frac{1}{a^2} = \frac{30.4}{11,500} \times \frac{1}{36} = 73.4 \times 10^{-6} \text{ S}$$

$$B_m = \sqrt{|Y|^2 - G_c^2} = 10^{-6} \sqrt{73.4^2 - 14.0^2} = 72.05 \times 10^{-6} \text{ S}$$

Under rated conditions the total loss is approximately the sum of short-circuit and open-circuit test losses, and since efficiency is the ratio of the output to the input kilowatts, we have

$$\text{Efficiency} = \frac{12,000}{12,000 + (105.8 + 66.7)} \times 100 = 98.6\%$$

This example illustrates the fact that G_c is so much smaller than B_m that it may be omitted. B_m is also very small so that I_E is often neglected entirely.

2.4 PER-UNIT IMPEDANCES IN SINGLE-PHASE TRANSFORMER CIRCUITS

The ohmic values of resistance and leakage reactance of a transformer depend on whether they are measured on the high- or low-voltage side of the transformer. If they are expressed in per unit, the base kilovoltamperes is understood to be the kilovoltampere rating of the transformer. The base voltage is understood to be the voltage rating of the low-voltage winding if the ohmic values of resistance and leakage reactance are referred to the low-voltage side of the transformer. Likewise, the base voltage is taken to be the voltage rating of the high-voltage winding if the ohmic values are referred to the high-voltage side of the transformer. The per-unit impedance of a transformer is the same regardless of whether it is determined from ohmic values referred to the high-voltage or low-voltage sides of the transformers, as shown by the following example.

Example 2.5. A single-phase transformer is rated 110/440 V, 2.5 kVA. Leakage reactance measured from the low-voltage side is 0.06 Ω . Determine leakage reactance in per unit.

Solution. From Eq. (1.46) we have

$$\text{Low-voltage base impedance} = \frac{0.110^2 \times 1000}{2.5} = 4.84 \Omega$$

In per unit

$$X = \frac{0.06}{4.84} = 0.0124 \text{ per unit}$$

If leakage reactance had been measured on the high-voltage side, the value would be

$$X = 0.06 \left(\frac{440}{110} \right)^2 = 0.96 \, \Omega$$

$$\text{High-voltage base impedance} = \frac{0.440^2 \times 1000}{2.5} = 77.5 \, \Omega$$

In per unit

$$X = \frac{0.96}{77.5} = 0.0124 \text{ per unit}$$

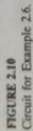
A great advantage in making per-unit computations is realized by the proper selection of different bases for circuits connected to each other through a transformer. To achieve the advantage in a single-phase system, *the voltage bases for the circuits connected through the transformer must have the same ratio as the turns ratio of the transformer windings*. With such a selection of voltage bases and the same kilovoltampere base, the per-unit value of an impedance will be the same when it is expressed on the base selected for its own side of the transformer as when it is referred to the other side of the transformer and expressed on the base of that side.

So, the transformer is represented completely by its impedance ($R + jX$) in per unit when magnetizing current is neglected. No per-unit voltage transformation occurs when this system is used, and the current will also have the same per-unit value on both sides of the transformer if magnetizing current is neglected.

Example 2.6. Three parts of a single-phase electric system are designated A , B , and C and are connected to each other through transformers, as shown in Fig. 2.10. The transformers are rated as follows:

A - B	10,000 kVA, 13.8/138 kV, leakage reactance 10%
B - C	10,000 kVA, 138/69 kV, leakage reactance 8%

If the base in circuit B is chosen as 10,000 kVA, 138 kV, find the per-unit impedance of the 300- Ω resistive load in circuit C referred to circuits C , B , and A . Draw the impedance diagram neglecting magnetizing current, transformer resistances, and line impedances.



Base voltage for circuit A: $0.1 \times 138 = 13.8 \text{ kV}$

Base voltage for circuit C: $0.5 \times 138 = 69 \text{ kV}$

$$\text{Base impedance of circuit C: } \frac{69^2 \times 1000}{10,000} = 476 \, \Omega$$

$$\frac{300}{476} = 0.63 \text{ per unit}$$

Because the selection of base in various parts of the system is determined by the turns ratio of the transformers, and because the base kilovoltamperes is the same in all parts of the system, the per-unit impedance of the load referred to any part of the system will be the same. This is verified as follows:

$$\text{Base impedance of circuit B; } \frac{138^2 \times 1000}{10,000} = 1900 \, \Omega$$

Impedance of load referred to circuit B: $300 \times 2^2 = 1200 \Omega$

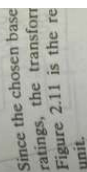
$$\text{Per-unit impedance of load referred to B: } \frac{1200}{1900} = 0.63 \text{ per unit}$$

Base impedance of circuit A;

$$\frac{13.8^2 \times 1000}{10,000} = 19 \Omega$$

Impedance of load referred to circuit A: $300 \times \frac{10,000}{10,000} = 19 \Omega$

Per-unit impedance of load referred to A; $\frac{12}{19} = 0.63$ per unit



Because of the advantages stated in the preceding paragraph, the single-phase system is almost always used. In this system, the kilovoltampere base is the kilovoltampere base of the base kilovolt to be assigned, and the other parts of the system diagram are the per-unit in-

2.5 THREE-PHASE

Three identical single-phase transformers with 1000 turns on the primary and 500 turns on the secondary are connected in a Δ - Δ configuration. The primary is connected to a 2400-V, 60-Hz source. The secondary is connected to a load consisting of three identical impedances of $10 \angle 30^\circ \Omega$ each. The load is connected in a Δ configuration. The transformer ratio is $1:2$. The primary current is $10 \angle 0^\circ$ A. The secondary current is $20 \angle 30^\circ$ A. The primary voltage is $2400 \angle 0^\circ$ V. The secondary voltage is $1200 \angle 30^\circ$ V. The primary power is $2400 \times 10 = 24,000$ W. The secondary power is $1200 \times 20 = 24,000$ W. The transformer efficiency is 100% .

For a single-phase of each winding, or a high-voltage winding then labeled H_2 and Figure 2.12 show form a Y-Y three-phase

simultaneously yields

$$\begin{aligned} Z_p &= \frac{1}{3}(Z_{pi} + Z_{pi} - Z_{pi}) \\ Z_n &= \frac{1}{3}(Z_{pi} + Z_{pi} - Z_{pi}) \\ Z_i &= \frac{1}{3}(Z_{pi} + Z_{pi} - Z_{pi}) \end{aligned} \quad (2.39)$$

The impedances of the three windings are connected to represent the equivalent circuit of the single-phase three-winding transformer with magnetizing current neglected, as shown in Fig. 2.19(b). The common point is fictitious and unrelated to the neutral of the system. The points p , s , and i are connected to the parts of the impedance diagrams representing the parts of the system connected to the primary, secondary, and tertiary windings, respectively, of the transformer. As in two-winding transformers, conversion to per-unit impedance requires the same kilovoltampere base for all three circuits and requires voltage bases in the three circuits that are in the same ratio as the rated line-to-line voltages of the three circuits of the transformer.

When three such transformers are connected for three-phase operation, the primary and secondary windings are usually Y -connected and the tertiary windings are connected in Δ to provide a path for the third harmonic of the exciting current.

Example 2.11. The three-phase ratings of a three-winding transformer are:

Primary	Y -connected, 66 kV, 15 MVA
Secondary	Y -connected, 13.2 kV, 10 MVA
Tertiary	Δ -connected, 2.3 kV, 5 MVA

Neglecting resistance, the leakage impedances are

$$\begin{aligned} Z_{pi} &= 7\% \text{ on } 15 \text{ MVA, } 66 \text{ kV base} \\ Z_{pi} &= 9\% \text{ on } 15 \text{ MVA, } 66 \text{ kV base} \\ Z_{ii} &= 8\% \text{ on } 10 \text{ MVA, } 13.2 \text{ kV base} \end{aligned}$$

Find the per-unit impedances of the per-phase equivalent circuit for a base of 15 MVA, 66 kV in the primary circuit.

Solution. With a base of 15 MVA, 66 kV in the primary circuit, the proper bases for the per-unit impedances of the equivalent circuit are 15 MVA, 66 kV for primary-circuit quantities, 15 MVA, 13.2 kV for secondary-circuit quantities, and 15 MVA, 2.3 kV for tertiary-circuit quantities.

Since Z_{pi} and Z_{ii} are measured in the i expressed on the proper base for the equivalent circuit, the required change in base is required for Z_{pi} . The required change in base is as follows:

$$Z_{pi} = 8\% \times \frac{15}{10} = 12\%$$

In per unit on the specified base

$$\begin{aligned} Z_p &= \frac{1}{3}(j0.07 + j0.09 - j0.1) \\ Z_i &= \frac{1}{3}(j0.07 + j0.12 - j0.1) \\ Z_n &= \frac{1}{3}(j0.09 + j0.12 - j0.1) \end{aligned}$$

Example 2.12. A constant-voltage source (an MW, 2.3 kV three-phase load and a 7.5 MW a subtransient reactance of $X'' = 20\%$. The three-winding transformer described in the load are connected to the secondary and impedance diagram of the system and and 66 kV, 15 MVA in the primary. Neglect c that of the resistive load.

Solution. The constant-voltage source can internal impedance.

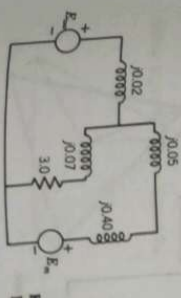
The resistance of the load is 1.0 pu tertiary. Expressed on a 15 MVA, 2.3 kV

$$R = 1.0 \times \frac{15}{5} = 3.0$$

The reactance of the motor on a

$$X'' = 0.20 \times \frac{15}{7} = 0.43$$

Figure 2.20 is the required diagram. V which occurs between the Y -connect



Since Z_{pi} and $Z_{pi'}$ are measured in the primary circuit, they are already expressed on the proper base for the equivalent circuit. No change of voltage base is required for $Z_{pi'}$. The required change in base megavoltamperes for Z_{pi} is made as follows:

$$Z_{pi} = 8\% \times \frac{15}{10} = 12\%$$

In per unit on the specified base

$$Z_p = \frac{1}{2}(j0.07 + j0.09 - j0.12) = j0.02 \text{ per unit}$$

$$Z_s = \frac{1}{2}(j0.07 + j0.12 - j0.09) = j0.05 \text{ per unit}$$

$$Z_t = \frac{1}{2}(j0.09 + j0.12 - j0.07) = j0.07 \text{ per unit}$$

Example 2.12. A constant-voltage source (infinite bus) supplies a purely resistive 5 MW, 2.3 kV three-phase load and a 7.5 MVA, 13.2 kV synchronous motor having a subtransient reactance of $X'' = 20\%$. The source is connected to the primary of the three-winding transformer described in Example 2.11. The motor and resistive load are connected to the secondary and tertiary of the transformer. Draw the impedance diagram of the system and mark the per-unit impedances for a base of 66 kV, 15 MVA in the primary. Neglect exciting current and all resistance except that of the resistive load.

Solution. The constant-voltage source can be represented by a generator having no internal impedance.

The resistance of the load is 1.0 per unit on a base of 5 MVA, 2.3 kV in the tertiary. Expressed on a 15 MVA, 2.3 kV base, the load resistance is

$$R = 1.0 \times \frac{15}{5} = 3.0 \text{ per unit}$$

The reactance of the motor on a base of 15 MVA, 13.2 kV is

$$X'' = 0.20 \frac{15}{7.5} = 0.40 \text{ per unit}$$

Figure 2.20 is the required diagram. We must remember, however, the phase shift which occurs between the Y-connected primary and the Δ-connected tertiary.

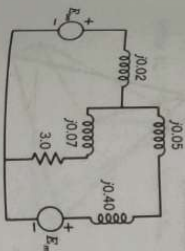


FIGURE 2.20
Impedance diagram for Example 2.11.

V_0 is zero and Eq. (7.9) reduces to the 1×1 matrix equation

$$[Y] V_m = I_m \quad (7.10)$$

corresponding to removal of row m and column m from the coefficient matrix. Despite its straightforward derivation, Eq. (7.9) and the procedure leading to it are important in more general situations. We note that the branch voltages V_b are transformed to the node voltages V_n and I_b and the branch currents I_b are likewise represented by the current injections I_n and I_p . The coefficient matrix relating the node voltages and currents of Eq. (7.9) follows from the fact that

$$\begin{bmatrix} m \\ n \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} m \\ n \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (7.11)$$

This 2×2 matrix, also seen in Eq. (1.64), is an important building block for representing more general networks, as we shall soon see. The row and column pointers identify each entry in the coefficient matrix by node numbers. For instance, in the first row and second column of Eq. (7.11) the entry -1 is identified with nodes m and n of Fig. 7.2 and the other entries are similarly identified.

Thus, the coefficient matrices of Eqs. (7.9) and (7.10) are simply storage matrices with row and column labels determined by the end nodes of the branches. Each branch of the network has a similar matrix labeled according to the nodes of the network to which that branch is connected. To obtain the overall nodal admittance matrix of the entire network, we simply combine the individual branch matrices by adding together elements with identical row and column labels. Such addition causes the sum of the branch currents flowing from each node of the network to equal the total current injected into that node, as required by Kirchhoff's current law. In the overall matrix, the off-diagonal element Y_{mn} is the negative sum of the admittances connected between nodes m and n and the diagonal entry Y_{nn} is the algebraic sum of the admittances connected to node n . Provided at least one of the network branches is connected to the reference node, the net result is V_{bus} of the system as shown in the following example.

Example 7.1. The single-line diagram of a small power system is shown in Fig. 7.3. The corresponding reactance diagram, with reactances specified in per unit, is shown in Fig. 7.4. A generator with emf equal to $1.25 \angle 0^\circ$ per unit is connected through a transformer to high-voltage node 3 , while a motor with internal voltage equal to $0.85 \angle -45^\circ$ is similarly connected to node 4 . Develop the nodal

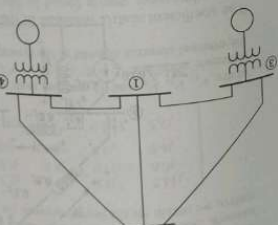


FIGURE 7.3 Single-line diagram of the four-bus system of Example 7.1. Reference node is not shown.

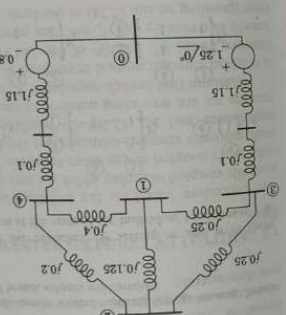


FIGURE 7.4 Reactance diagram for Fig. 7.3. Node 0 is reference; reactances and voltages are in per unit.

admittance matrix for each of the network branches and then write the nodal admittance equations of the system.

Solution. The reactances of the generator and the motor may be combined with their respective step-up transformer reactances. Then, by transformation of sources the combined reactances and shunt admittances shown in Fig. 7.5. We will treat the current sources as external injections at nodes 3 and 4 and name the seven passive branches according to the subscripts of their currents and voltages. For example, the branch between nodes 1 and 3 will be called branch c . The admittance of each branch is simply the reciprocal of all values in per unit. The two branches a and g connected to the reference node are characterized by Eq. (7.10), while Eq.

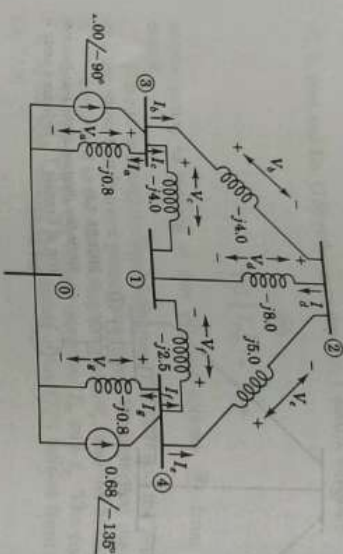


FIGURE 7.5
Per-unit admittance diagram for Fig. 7.4 with current sources replacing voltage sources. Branch names a to e correspond to the subscripts of branch voltages and currents.

(7.9) applies to each of the other five branches. By setting m and n in those equations equal to the node numbers at the ends of the individual branches of Fig. 7.5, we obtain

$$\begin{array}{llll} \textcircled{3} & \textcircled{3} & \textcircled{2} & \textcircled{3} \\ \textcircled{1} & \textcircled{3} & \textcircled{2} & \textcircled{1} \\ Y_a & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_b & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_c & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_e \end{array}$$

$$\begin{array}{llll} \textcircled{2} & \textcircled{1} & \textcircled{4} & \textcircled{2} \\ \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_d & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_e & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_f & \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \right] Y_f \end{array}$$

The order in which the labels are assigned is not important here, provided the columns and rows follow the *same* order. However, for consistency with later sections let us assign the node numbers in the directions of the branch currents of Fig. 7.5, which also shows the numerical values of the admittances. Combining together those elements of the above matrices having identical row and column labels gives

$$\begin{array}{cccc} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \left[\begin{array}{cccc} Y_a + Y_d + Y_f & -Y_d & -Y_c & -Y_f \\ -Y_d & (Y_b + Y_d + Y_e) & -Y_e & -Y_f \\ -Y_c & -Y_e & (Y_c + Y_e + Y_f) & 0 \\ -Y_f & -Y_f & 0 & (Y_e + Y_f + Y_f) \end{array} \right] \end{array}$$

which is the same as Y_{bus} of Eq. (1.62) since Figs. 1.23 and 7.5 are for the same network. Substituting the numerical values of the branch admittances into this matrix, we obtain for the overall network the nodal admittance equations

$$\begin{bmatrix} -j14.5 & j8.0 & j4.0 & j2.5 \\ j8.0 & -j17.0 & j4.0 & j5.0 \\ j4.0 & j4.0 & -j8.8 & 0.0 \\ j2.5 & j5.0 & 0.0 & -j8.3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \end{bmatrix}$$

where V_1 , V_2 , V_3 , and V_4 are the node voltages measured with respect to the reference node and $I_1 = 0$, $I_2 = 0$, $I_3 = 1.00 \angle -90^\circ$ and $I_4 = 0.68 \angle -135^\circ$ are the external currents injected at the system nodes.

The coefficient matrix obtained in the above example is exactly the same as the bus admittance matrix found in Sec. 1.12 using the usual rules for Y_{bus} formation. However, the approach based on the building-block matrix has advantages when extended to networks with mutually coupled branches, as we now demonstrate.

7.2 MUTUALLY COUPLED BRANCHES IN Y_{bus}

The procedure based on the building-block matrix is now extended to two mutually coupled branches which are part of a larger network but which are not inductively coupled to any other branches. In Sec. 2.2 the primitive equations of such mutually coupled branches are developed in the form of Eq. (2.24) for impedances and Eq. (2.26) for admittances. The notation is different here because we are now using numbers to identify nodes rather than branches.

Assume that branch impedance Z_a connected between nodes $\textcircled{m}$ and $\textcircled{n}$ is coupled through mutual impedance Z_M to branch impedance Z_b connected between nodes $\textcircled{p}$ and $\textcircled{q}$ of Fig. 7.6. The voltage drops V_a and V_b due to the branch currents I_a and I_b are then given by the primitive impedance equation corresponding to Eq. (2.24) in the form

$$\begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} Z_a & Z_M \\ Z_M & Z_b \end{bmatrix} \begin{bmatrix} I_a \\ I_b \end{bmatrix} \quad (7.12)$$

in which the coefficient matrix is symmetrical. As noted in Sec. 2.2, the mutual impedance Z_M is considered positive when currents I_a and I_b enter the terminals marked with dots in Fig. 7.6(a); the voltage drops V_a and V_b then have the polarities shown. Multiplying Eq. (7.12) by the inverse of the primitive impedance matrix

$$\begin{bmatrix} Z_a & Z_M \\ Z_M & Z_b \end{bmatrix}^{-1} = \frac{1}{Z_a Z_b - Z_M^2} \begin{bmatrix} Z_b & -Z_M \\ -Z_M & Z_a \end{bmatrix} = \begin{bmatrix} Y_a & Y_M \\ Y_M & Y_b \end{bmatrix} \quad (7.13)$$

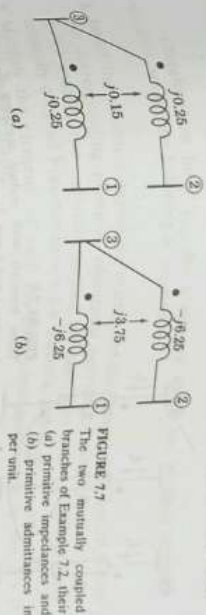


FIGURE 7.7 The two mutually coupled branches of Example 7.2, their (a) primitive impedances and (b) primitive admittances in per unit.

Solution. The primitive impedance matrix for the mutually coupled branches of Fig. 7.7(a) is inverted as a single entry to yield the primitive admittances of Fig. 7.7(b), that is,

$$\begin{bmatrix} j0.25 & j0.15 \\ j0.15 & j0.25 \end{bmatrix}^{-1} = \begin{bmatrix} -j6.25 & j3.75 \\ j3.75 & -j6.25 \end{bmatrix}$$

First, the rows and columns of the building-block matrix, which multiplies the primitive self-admittance between nodes ① and ③ are labeled ③ and ① in that order to correspond to the dot marking node ③. Next, the rows and columns of the 2×2 matrix multiplying the self-admittance between nodes ② and ③ are labeled ③ and ② in the order shown because node ③ is marked. Finally, the pointers of the matrices multiplying the mutual admittance are aligned with those of the self-admittances to form the 4×4 array similar to Eq. (7.20) as follows:

$$\begin{bmatrix} \text{③} & \text{①} & \text{③} & \text{②} \\ \text{③} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (-j6.25) & \text{③} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (j3.75) \\ \text{①} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (j3.75) & \text{①} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (-j6.25) \\ \text{②} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (j3.75) & \text{②} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} (-j6.25) \end{bmatrix}$$

Since there are only three nodes in Fig. 7.7, the required 3×3 matrix is found by adding the columns and rows of common node ③ to obtain

$$\begin{bmatrix} \text{①} & \text{②} & \text{③} \\ \text{①} \begin{bmatrix} -j6.25 & j3.75 & j6.25 - j3.75 \end{bmatrix} \\ \text{②} \begin{bmatrix} j3.75 & -j6.25 & -j3.75 + j6.25 \end{bmatrix} \\ \text{③} \begin{bmatrix} j6.25 - j3.75 & -j3.75 + j6.25 & -j6.25 - j6.25 + 2(j3.75) \end{bmatrix} \end{bmatrix}$$

nodal admittance matrix is $-Y_M$ and similar entries are made from the other elements of Eq. (7.19).

The nodal admittance matrix of the two coupled branches may be formed directly by inspection. This becomes clear when we write the coefficient matrix of Eq. (7.19) in the alternative form

$$\begin{bmatrix} \text{③} & \text{①} & \text{③} & \text{②} \\ \text{③} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M & \text{③} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M \\ \text{①} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M & \text{①} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M \\ \text{②} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M & \text{②} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} Y_M \end{bmatrix} \quad (7.20)$$

To obtain Eq. (7.20), we multiply each element of the primitive admittance matrix by the 2×2 building-block matrix. The labels assigned to the rows and columns of the multipliers in Eq. (7.20) are easily determined. First, we note that the self-admittance Y_n is measured between nodes ③ and ① with the dot at node ③. Hence, the 2×2 matrix multiplying Y_n in Eq. (7.20) has rows and columns labeled ③ and ① in that order. Then, the self-admittance Y_n between nodes ③ and ② is multiplied by the 2×2 matrix with labels ③ and ② in the order shown since node ③ is marked with a dot. Finally, the labels of the matrices multiplying the mutual admittance Y_M are assigned row by row and then column by column so as to align and agree with those already given to the self-admittances. In the nodal admittance matrix of Eqs. (7.19) and (7.20) the sum of the columns (and of the rows) adds up to zero. This is because none of the nodes ③, ①, ②, and ② has been considered as the reference node of the network. In the special case where one of the nodes, say, node ③, is in fact the reference, Y_n is zero and column n in Eq. (7.19) does not need to appear; furthermore, I_n does not have to be explicitly represented since the reference node current is not an independent quantity. Consequently, when node ③ is the reference, we may eliminate the row and column of that node from Eqs. (7.19) and (7.20).

It is important to note that nodes ③, ①, ②, and ② are often not distinct. For instance, suppose that nodes ③ and ② are one and the same node. In that case columns ③ and ② of Eq. (7.19) can be combined together since $Y_n = Y_n$, and the corresponding rows can be added because I_n and I_n are parts of the common injected current. The following example illustrates the situation.

Example 7.2. Two branches having impedances equal to $j0.25$ per unit are coupled through mutual impedance $Z_M = j0.15$ per unit, as shown in Fig. 7.7. Find

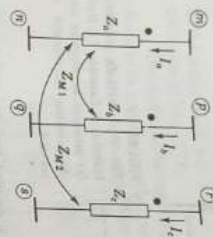


FIGURE 7.8 Three branches with mutual coupling Z_{M1} between branches a and b and Z_{M2} between branches b and c .

The new diagonal element, representing node ③, for instance, is the sum of the four elements $(-j6.25 - j6.25 + j3.75 + j3.75)$ in rows ③ and columns ③ of the previous matrix. The three nodal admittance equations in vector-matrix form are then written

$$\begin{bmatrix} -j6.25 & j3.75 & j2.50 \\ j3.75 & -j6.25 & j2.50 \\ j2.50 & j2.50 & -j5.00 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

where V_1 , V_2 , and V_3 are the voltages at nodes ①, ②, and ③ measured with respect to reference, while I_1 , I_2 , and I_3 are the external currents injected at the respective nodes.

As in Sec. 7.1, the coefficient matrix of the last equation can be combined with the nodal admittance matrices of the other branches of the network in order to obtain the nodal admittance matrix of the entire system.

For three or more coupled branches we follow the same procedure as above. For example, the three coupled branches of Fig. 7.8 have primitive impedance and admittance matrices given by

$$\begin{bmatrix} Z_a & Z_{M1} & Z_{M2} \\ Z_{M1} & Z_b & 0 \\ Z_{M2} & 0 & Z_c \end{bmatrix}^{-1} = \begin{bmatrix} Y_a & Y_{M1} & Y_{M2} \\ Y_{M1} & Y_b & Y_{M3} \\ Y_{M2} & Y_{M3} & Y_c \end{bmatrix} \quad (7.21)$$

The zeros in the Z -matrix arise because branches b and c are not directly coupled. For all nonzero values of the current I_a of Fig. 7.8, branches b and c are *indirectly* coupled through branch a , as shown by nonzero Y_{M3} of the primitive admittance matrix.

Therefore, to form Y_{bus} for a network which has mutually coupled branches, we do the following in sequence:

1. Invert the primitive impedance matrices of the network branches to obtain the corresponding primitive admittance matrices. A single branch has a 1×1 matrix, two mutually coupled branches have a 2×2 matrix, three mutually coupled branches have a 3×3 matrix, and so on.
2. Multiply the elements of each primitive admittance matrix by the 2×2 building-block matrix.
3. Label the two rows and the two columns of each *diagonal* building-block matrix with the end-node numbers of the corresponding self-admittance. For mutually coupled branches it is important to label in the order of the *marked* (dotted)—*then*—*unmarked* (undotted) node numbers.
4. Label the two rows of each *off-diagonal* building-block matrix with node numbers aligned and consistent with the row labels assigned in (3); then label the columns consistent with the column labels of (3).
5. Combine, by adding together, those elements with identical row and column labels to obtain the nodal admittance matrix of the overall network. If one of the nodes encountered is the reference node, omit its row and column to obtain the system Y_{bus} .

7.3 AN EQUIVALENT ADMITTANCE NETWORK

We have demonstrated how to write the nodal admittance equations for one branch or a number of mutually coupled branches which are part of a larger network. We now show that such equations can be interpreted as representing an equivalent admittance network with no mutually coupled elements. This may be useful when forming Y_{bus} for an original network having mutually coupled elements.

The currents injected into the nodes of Fig. 7.6 are described in terms of node voltages and admittances by Eq. (7.19). For example, the equation for the current I_m at node ③ is given by the first row of Eq. (7.19) as follows:

$$I_m = Y_a V_m - Y_a V_n + Y_M V_p - Y_M V_q \quad (7.22)$$

Adding and subtracting the term $Y_M V_m$ on the right-hand side of Eq. (7.22) and combining terms with common coefficients, we obtain the Kirchhoff's current equation at node ③

$$I_m = \underbrace{Y_a(V_m - V_n)}_{I_{m,n}} + \underbrace{(-Y_M)(V_m - V_p)}_{I_{m,p}} + \underbrace{Y_M(V_m - V_q)}_{I_{m,q}} \quad (7.23)$$

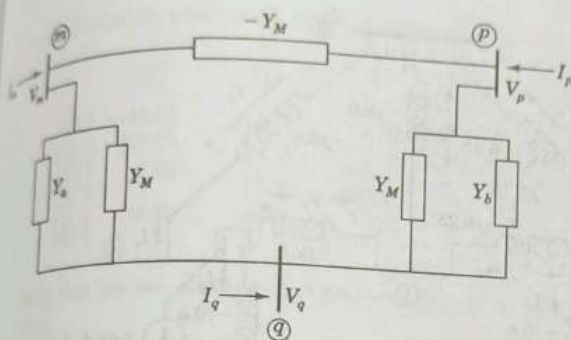


FIGURE 7.10
The nodal admittance network of two mutually coupled branches connected to nodes (m), (p), and (q).

Eq. (7.19). Accordingly, the standard rules of circuit analysis may be applied to the equivalent. For example, if the two coupled branches are physically connected among three independent nodes as in Example 7.2, we may regard nodes (m) and (q) of Fig. 7.9(d) as one and the same node, which we simply join together, as shown in Fig. 7.10. The three-bus equivalent circuit of Fig. 7.10 then yields the nodal equations for the original branches.

Thus, each physical branch or mutually coupled pair gives rise to an equivalent admittance network to which the usual rules of circuit analysis apply. The following example illustrates the role of the equivalent circuit in forming Y_{bus} .

Example 7.3. Replace branches *b* and *c* between node-pairs (1)–(3) and (2)–(3) of Fig. 7.5 by the mutually coupled branches of Fig. 7.7. Then, find Y_{bus} and the nodal equations of the new network.

Solution. The admittance diagram of the new network including the mutual coupling is shown in Fig. 7.11. From Example 7.2 we know that the mutually coupled branches have the nodal admittance matrix

$$\begin{array}{c} \textcircled{1} \quad \textcircled{2} \quad \textcircled{3} \\ \begin{bmatrix} -j6.25 & j3.75 & j2.50 \\ j3.75 & -j6.25 & j2.50 \\ j2.50 & j2.50 & -j5.00 \end{bmatrix} \end{array}$$

which corresponds to the equivalent circuit shown encircled in Fig. 7.12. The remaining portion of Fig. 7.12 is drawn from Fig. 7.5. Since mutual coupling is not evident in Fig. 7.12, we may apply the standard rules of Y_{bus} formation to the

$$(V_n - V_p) \quad (7.24)$$

$$I_{np}$$

$$(V_p - V_n) \quad (7.25)$$

$$I_{pn}$$

Figs. 7.9(b) and (c).
network because it
be partial networks
t in the form of the
(q) in Fig. 7.9(d).
it is equivalent in
6 since it satisfies

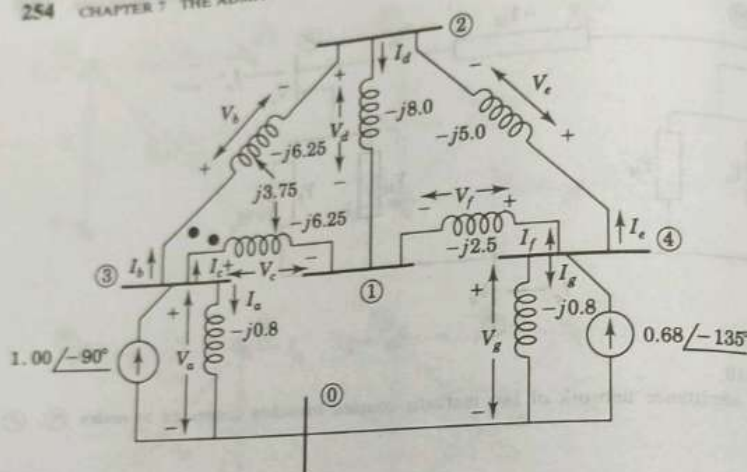


FIGURE 7.11 Per-unit admittance diagram for Example 7.3.

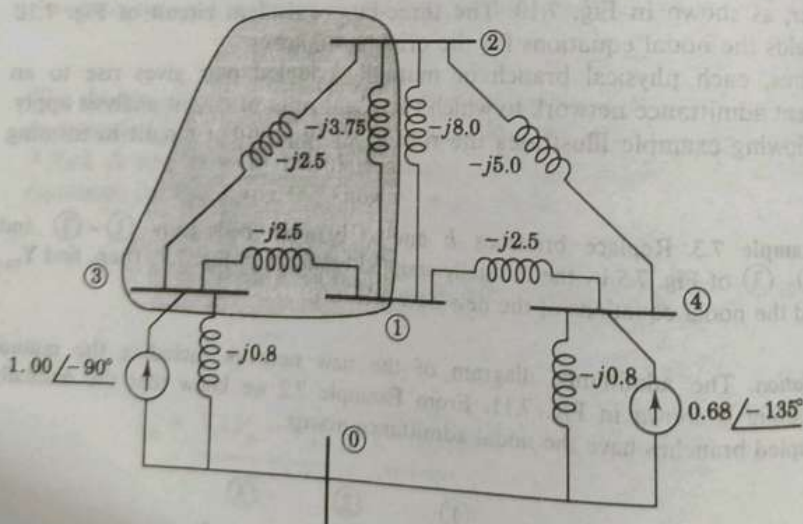


FIGURE 7.12 Nodal admittance network for Example 7.3. The shaded portion represents two mutually coupled branches connected between buses 1, 2, and 3.

overall network, which le

$$\begin{bmatrix} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{bmatrix} \begin{bmatrix} -j16.75 \\ j11.75 \\ j2.50 \\ j2.50 \end{bmatrix}$$

Note that the two adm

7.4 MODIFICATION

The building-block app
important insights into
admittance contributes
network of the overall
systematic means of c
network branches. We
according to the sequ
numbered, and within
the nodal admittance
easily see how to mo
other changes to the s
network to reflect the
and $\textcircled{n}$, we simply ad
from the symmetrical
new branch admittan
change matrix ΔY_{bus}

Again, we recognize
and $\textcircled{n}$. Using Eq
branch of the netwo
and $\textcircled{n}$ such that
the desired value.
connected between n
admittance $-Y_a$

256 elements of ΔY_{bus} from the existing Y_{bus} . Equation (7.20) shows that a pair of mutually coupled branches can be removed from the network by subtracting the entries in the change matrix

$$\Delta Y_{bus} = \begin{bmatrix} \textcircled{m} & \textcircled{n} & \textcircled{p} & \textcircled{q} \\ \textcircled{m} & Y_a & -Y_a & -Y_M \\ \textcircled{n} & -Y_a & Y_a & Y_M \\ \textcircled{p} & Y_M & -Y_M & Y_b \\ \textcircled{q} & -Y_M & Y_M & -Y_b \end{bmatrix} \quad (7.27)$$

from the rows and columns of Y_{bus} corresponding to the end nodes $\textcircled{m}$, $\textcircled{n}$, $\textcircled{p}$, and $\textcircled{q}$. Of course, if only one of the two mutually coupled branches is to be removed from the network, we could first remove all the entries for the mutually coupled pair from Y_{bus} using Eq. (7.27) and then add the entries for the branch to be retained using Eq. (7.26). Other strategies for modifying Y_{bus} to reflect network changes become clear from the insights developed in Secs. 7.1 through 7.3.

Example 7.4. Determine the bus admittance matrix of the network of Fig. 7.5 by removing the effects of mutual coupling from Y_{bus} of Fig. 7.11.

Solution. The Y_{bus} for the entire system of Fig. 7.11 including mutual coupling is found in Example 7.3 to be

$$Y_{bus} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & -j16.75 & j11.75 & j2.50 \\ \textcircled{2} & j11.75 & -j19.25 & j2.50 \\ \textcircled{3} & j2.50 & j2.50 & -j5.80 \\ \textcircled{4} & j2.50 & j5.00 & 0 \end{bmatrix} \begin{matrix} j2.50 \\ j5.00 \\ 0 \\ -j8.30 \end{matrix}$$

To remove completely the effect of mutual coupling from the network Y_{bus} , we proceed in two steps by (a) first removing the two mutually coupled branches altogether and (b) then restoring each of the two branches without mutual coupling between them. (a) To remove the two mutually coupled branches from the network, we subtract from the system Y_{bus} the entries in

$$\Delta Y_{bus,1} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & -j6.25 & j3.75 & j2.50 \\ \textcircled{2} & j3.75 & -j6.25 & j2.50 \\ \textcircled{3} & j2.50 & j2.50 & -j5.00 \\ \textcircled{4} & & & \end{bmatrix}$$

corresponding to the encircled portion of Fig. 7.12.

(b) Now we must reconnect to the network the uncoupled branches, each of which has an admittance $(j0.25)^{-1} = -j4.0$ per unit. Accordingly, to reconnect the branch between nodes $\textcircled{1}$ and $\textcircled{3}$, we add to Y_{bus} the change matrix

$$\Delta Y_{bus,2} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & 1 & & \\ \textcircled{2} & & & \\ \textcircled{3} & & & \\ \textcircled{4} & & & \end{bmatrix} \begin{matrix} (-j4.0) \\ \\ \\ \end{matrix}$$

and similarly for the branch between nodes $\textcircled{2}$ and $\textcircled{3}$ we add

$$\Delta Y_{bus,3} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & & & \\ \textcircled{2} & 1 & & \\ \textcircled{3} & & & \\ \textcircled{4} & & & \end{bmatrix} \begin{matrix} (-j4.0) \\ \\ \\ \end{matrix}$$

Appropriately subtracting and adding the three change matrices and the original Y_{bus} give the new bus admittance matrix for the uncoupled branches

$$Y_{bus(\text{new})} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & -j14.5 & j8.0 & j4.0 \\ \textcircled{2} & j8.0 & -j17.0 & j4.0 \\ \textcircled{3} & j4.0 & j4.0 & -j8.8 \\ \textcircled{4} & j2.5 & j5.0 & 0 \end{bmatrix} \begin{matrix} j2.5 \\ j5.0 \\ 0 \\ -j8.3 \end{matrix}$$

which agrees with Example 7.1.

7.5 THE NETWORK INCIDENCE MATRIX AND Y_{bus}

In Secs. 7.1 and 7.2 nodal admittance equations for each branch and mutually coupled pair of branches are derived independently from those of other branches in the network. The nodal admittance matrices of the individual branches are then combined together in order to build Y_{bus} of the overall system. Since we now understand the process, we may proceed to the more formal approach which treats all the equations of the system simultaneously rather than separately. We will use the example system of Fig. 7.11 to establish the general procedure.

Two of the seven branches in Fig. 7.11 are mutually coupled as shown. The mutually coupled pair is characterized by Eq. (7.14) and the other five

Example 7.5. Determine the per-unit bus admittance matrix of the example system of Fig. 7.11 using the tree shown in Fig. 7.13 with reference node ①.

Solution. The primitive admittance matrix Y_{pr} describing the branch admittances is given by Eq. (7.28) and the branch-to-node incidence matrix A for the spanning tree is given by Eq. (7.31). Therefore, performing the row-by-column multiplications for $A^T Y_{pr}$ indicated by

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -j0.8 & & & \\ & -j6.25 & j3.75 & \\ & j3.75 & -j6.25 & \\ & & & -j8.0 \\ & & & & -j5.0 \\ & & & & & -j2.5 \\ & & & & & & -j0.8 \end{bmatrix}$$

we obtain the intermediate result

$$A^T Y_{pr} = \begin{bmatrix} 0 & -j3.75 & j6.25 & j8.0 & 0 & j2.5 & 0 \\ 0 & j6.25 & -j3.75 & -j8.0 & j5.0 & 0 & 0 \\ -j0.8 & -j2.5 & -j2.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -j5.0 & -j2.5 & -j0.8 \end{bmatrix}$$

which we may now postmultiply by A to calculate

$$Y_{bus} = A^T Y_{pr} A = \begin{matrix} & \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \textcircled{1} & -j16.75 & j11.75 & j2.50 & j2.50 \\ \textcircled{2} & j11.75 & -j19.25 & j2.50 & j5.00 \\ \textcircled{3} & j2.50 & j2.50 & -j5.80 & 0 \\ \textcircled{4} & j2.50 & j5.00 & 0 & -j8.30 \end{matrix} \quad (4)$$

Since currents are injected at only buses ③ and ④, the nodal equations in matrix form are written

$$\begin{bmatrix} -j16.75 & j11.75 & j2.50 & j2.50 \\ j11.75 & -j19.25 & j2.50 & j5.00 \\ j2.50 & j2.50 & -j5.80 & 0 \\ j2.50 & j5.00 & 0 & -j8.30 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \end{bmatrix}$$

Example 7.6. Solve the node equations of Example 7.5 to find the bus voltages by inverting the bus admittance matrix.

Solution. Premultiplying both sides of the matrix nodal equation by the inverse of the bus admittance matrix (determined by using a standard program on a computer

of calculator) yields

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} j0.73128 & j0.69140 & j0.61323 & j0.63677 \\ j0.69140 & j0.71966 & j0.60822 & j0.64178 \\ j0.61323 & j0.60822 & j0.69890 & j0.55110 \\ j0.63677 & j0.64178 & j0.55110 & j0.69890 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \end{bmatrix}$$

performing the indicated multiplications, we obtain the per-unit results

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0.96903 \angle -18.4189^\circ \\ 0.96734 \angle -18.6028^\circ \\ 0.99964 \angle -15.3718^\circ \\ 0.94866 \angle -20.7466^\circ \end{bmatrix} = \begin{bmatrix} 0.91939 - j0.30618 \\ 0.91680 - j0.30859 \\ 0.96388 - j0.26499 \\ 0.88715 - j0.33605 \end{bmatrix}$$

7.6 THE METHOD OF SUCCESSIVE ELIMINATION

In industry-based studies of power systems the networks being solved are geographically extensive and often encompass many hundreds of substations, generating plants and load centers. The Y_{bus} matrices for these large networks of thousands of nodes have associated systems of nodal equations to be solved for a correspondingly large number of unknown bus voltages. For such solutions computer-based numerical techniques are required to avoid direct matrix inversion, thereby minimizing computational effort and computer storage. The method of successive elimination, called *Gaussian elimination*, underlies many of the numerical methods solving the equations of such large-scale power systems. We now describe this method using the nodal equations of the four-bus system

$$Y_{11}V_1 + Y_{12}V_2 + Y_{13}V_3 + Y_{14}V_4 = I_1 \quad (7.39)$$

$$Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3 + Y_{24}V_4 = I_2 \quad (7.40)$$

$$Y_{31}V_1 + Y_{32}V_2 + Y_{33}V_3 + Y_{34}V_4 = I_3 \quad (7.41)$$

$$Y_{41}V_1 + Y_{42}V_2 + Y_{43}V_3 + Y_{44}V_4 = I_4 \quad (7.42)$$

Gaussian elimination consists of reducing this system of four equations in the four unknowns V_1, V_2, V_3 , and V_4 to a system of three equations in three unknowns and then to a reduced system of two equations in two unknowns, and so on until there remains only one equation in one unknown. The final equation yields a value for the corresponding unknown, which is then substituted back in the reduced sets of equations to calculate the remaining unknowns. The successive elimination of unknowns in the forward direction is called *forward elimination* and the substitution process using the latest calculated values is

where the second set of calculated coefficients is given by

$$Y_{jk}^{(2)} = Y_{jk}^{(1)} - \frac{Y_{j2}^{(1)} Y_{k2}^{(1)}}{Y_{22}^{(1)}} \quad \text{for } j \text{ and } k = 3, 4 \quad (7.59)$$

and the net currents injected at buses (3) and (4) are

$$I_j^{(2)} = I_j^{(1)} - \frac{Y_{j2}^{(1)}}{Y_{22}^{(1)}} I_2^{(1)} \quad \text{for } j = 3, 4 \quad (7.60)$$

Equations (7.57) and (7.58) describe a further reduced equivalent network having only buses (3) and (4). Voltages V_3 and V_4 are exactly the same as those of the original four-bus network because the current injections $I_3^{(2)}$ and $I_4^{(2)}$ represent the effects of all the original current sources.

We now consider elimination of the variable V_3 .

Step 3

1. Divide Eq. (7.57) by the pivot $Y_{33}^{(2)}$ to obtain

$$V_3 + \frac{Y_{34}^{(2)}}{Y_{33}^{(2)}} V_4 = \frac{1}{Y_{33}^{(2)}} I_3^{(2)} \quad (7.61)$$

2. Multiply Eq. (7.61) by $Y_{43}^{(2)}$ and subtract the result from Eq. (7.58) to obtain

$$Y_{44}^{(3)} V_4 = I_4^{(3)} \quad (7.62)$$

in which we have defined

$$Y_{44}^{(3)} = Y_{44}^{(2)} - \frac{Y_{43}^{(2)} Y_{34}^{(2)}}{Y_{33}^{(2)}} \quad \text{and} \quad I_4^{(3)} = I_4^{(2)} - \frac{Y_{43}^{(2)} I_3^{(2)}}{Y_{33}^{(2)}} \quad (7.63)$$

Equation (7.62) describes the single equivalent branch admittance $Y_{44}^{(3)}$ with voltage V_4 from bus (4) to reference caused by the equivalent injected current $I_4^{(3)}$.

The final step in the forward elimination process yields V_4 .

Step 4

1. Divide Eq. (7.62) by $Y_{44}^{(3)}$ to obtain

$$V_4 = \frac{1}{Y_{44}^{(3)}} I_4^{(3)} \quad (7.64)$$

At this point we have found a value for bus voltage V_4 which can be substituted

back in Eq. (7.61) to obtain a value for V_3 . Continuing this process of back substitution using the values of V_3 and V_4 in Eq. (7.56), we obtain V_2 and then solve for V_1 from Eq. (7.47).

Thus, the gaussian-elimination procedure demonstrated here for a four-bus system provides a systematic means of solving large systems of equations without having to invert the coefficient matrix. This is most desirable when a large-scale power system is being analyzed. The following example numerically illustrates the procedure.

Example 7.7. Using gaussian elimination, solve the nodal equations of Example 7.5 to find the bus voltages. At each step of the solution find the equivalent circuit of the reduced coefficient matrix.

Solution. In Example 7.5 the nodal admittance equations in matrix form are found to be

$$\begin{array}{cccc} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \begin{bmatrix} -j16.75 & j11.75 & j2.50 & 0 \\ j11.75 & -j19.25 & j2.50 & 0 \\ j2.50 & j2.50 & -j5.80 & 0.00 \\ j2.50 & j5.00 & 0.00 & -j8.30 \end{bmatrix} & \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} & = & \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \end{bmatrix} \end{array}$$

Step 1

To eliminate the variable V_1 from rows 2, 3, and 4, we first divide the first row by the pivot $-j16.75$ to obtain

$$V_1 - 0.70149V_2 - 0.14925V_3 - 0.14925V_4 = 0$$

We now use this equation to eliminate the $j11.75$ entry in (row 2, column 1) of Y_{bus} , and in the process all the other elements of row 2 are modified. Equation (7.51) shows the procedure. For example, to modify the element $j2.50$ underlined in (row 2, column 3), subtract from it the product of the elements enclosed by rectangles divided by the pivot $-j16.75$; that is,

$$Y_{23}^{(1)} = Y_{23} - \frac{Y_{21} Y_{13}}{Y_{11}} = j2.50 - \frac{j11.75 \times j2.50}{-j16.75} = j4.25373 \text{ per unit}$$

Similarly, the other elements of new row 2 are

$$Y_{22}^{(1)} = -j19.25 - \frac{j11.75 \times j11.75}{-j16.75} = -j11.00746 \text{ per unit}$$

$$Y_{24}^{(1)} = j5.00 - \frac{j11.75 \times j2.50}{-j16.75} = j6.75373 \text{ per unit}$$

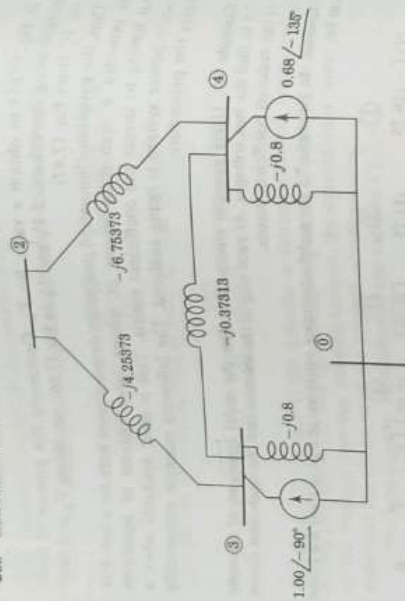


FIGURE 7.14 The equivalent three-bus network following Step 1 of Example 7.7

Modified elements of rows 3 and 4 are likewise found to yield

$$\begin{bmatrix} 1 & -0.70149 & -0.14925 & -0.14925 \\ 0 & -j11.00746 & j4.25373 & j6.75373 \\ 0 & j4.25373 & -j5.42686 & j0.37313 \\ 0 & j6.75373 & j0.37313 & -j7.92686 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \\ 0 \end{bmatrix}$$

Because $I_1 = 0$, no current is distributed from bus ① to the remaining buses ②, ③, and ④, and so the currents I_2^1 , I_3^1 , and I_4^1 in the right-hand-side vector have the same values before and after Step 1. The partitioned system of equations involving the unknown voltages V_2 , V_3 , and V_4 corresponds to the three-bus equivalent network constructed in Fig. 7.14 from the reduced coefficient matrix.

Step 2

Forward elimination applied to the partitioned 3×3 system of the last equation proceeds as in Step 1 to yield

$$\begin{bmatrix} 1 & -0.70149 & -0.14925 & -0.14925 \\ 0 & 1 & -0.38644 & -0.61356 \\ 0 & 0 & -j3.78305 & j2.98305 \\ 0 & 0 & j2.98305 & -j3.78305 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.00 \angle -90^\circ \\ 0.68 \angle -135^\circ \\ 0 \end{bmatrix}$$

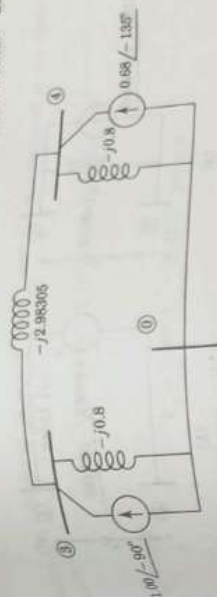


FIGURE 7.15 The equivalent two-bus network following Step 2 of Example 7.7

in which the injected currents $I_3^{(2)}$ and $I_4^{(2)}$ of Step 2 are also unchanged because $I_3^{(1)} = I_3 = 0$. At this stage we have eliminated V_1 and V_2 from the original 4×4 system of equations and there remains the 2×2 system in the variables V_3 and V_4 corresponding to Y_{bus} of Fig. 7.15. Note that nodes ① and ② are eliminated.

Step 3

Continuing the forward elimination, we find

$$\begin{bmatrix} 1 & -0.70149 & -0.14925 & -0.14925 \\ 0 & 1 & -0.38644 & -0.61356 \\ 0 & 0 & 1 & -0.78853 \\ 0 & 0 & 0 & -j1.43082 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.00 \angle -90^\circ \\ 0.26434 \angle 0^\circ \\ 1.35738 \angle -110.7466^\circ \end{bmatrix}$$

in which the entry for bus ③ in the right-hand-side vector is calculated to be

$$I_3^{(3)} = I_3^{(2)} - \frac{1.00 \angle -90^\circ}{3.78305 \angle -90^\circ} = 0.26434 \angle 0^\circ \text{ per unit}$$

and the modified current at bus ④ is given by

$$\begin{aligned} I_4^{(3)} &= I_4^{(2)} - \frac{Y_{43}^{(2)}}{Y_{33}^{(2)}} I_3^{(3)} \\ &= 0.68 \angle -135^\circ - \frac{j2.98305}{-j3.78305} 1.00 \angle -90^\circ \\ &= 1.35738 \angle -110.7466^\circ \text{ per unit} \end{aligned}$$

Figure 7.16(a) shows the single admittance resulting from Step 3.

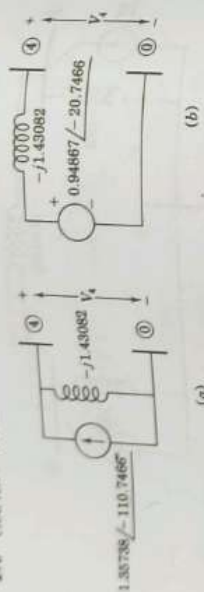


FIGURE 7.16

The equivalent circuits following (a) Step 3 and (b) Step 4 of Example 7.7.

Step 4

The forward-elimination process terminates with the calculation

$$V_4 = \frac{I_4^{(3)}}{Y_{44}^{(3)}} = \frac{1.35738 / -110.7466^\circ}{1.43082 / -90^\circ} = 0.94867 / -20.7466^\circ \text{ per unit}$$

corresponding to the source transformation of Fig. 7.16(b). Therefore, forward elimination leads to the triangular coefficient matrix given by

$$\begin{bmatrix} 1 & -0.70149 & -0.14925 & -0.14925 \\ 0 & 1 & -0.38644 & -0.61356 \\ 0 & 0 & 1 & -0.78853 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.26434 / 0^\circ \\ 0.94867 / -20.7466^\circ \end{bmatrix}$$

Since $V_4 = 0.94867 / -20.7466^\circ$, we now begin the back-substitution process to determine V_3 using the third row entries as follows:

$$V_3 - 0.78853V_4 = V_3 - 0.74805 / -20.7466^\circ = 0.26434 / 0^\circ$$

which yields

$$V_3 = 0.99965 / -15.3716^\circ \text{ per unit}$$

Back substitution for V_3 and V_4 in the second-row equation

$$V_2 - 0.38644V_3 - 0.61356V_4 = 0$$

yields

$$V_2 = 0.96734 / -18.6030^\circ \text{ per unit}$$

when the calculated values of V_2 , V_3 , and V_4 are substituted in the first-row equation

$$V_1 - 0.70149V_2 - 0.14925V_3 - 0.14925V_4 = 0$$

we obtain

$$V_1 = 0.96903 / -18.4189^\circ \text{ per unit}$$

and so the per-unit bus voltages are

$$V_1 = 0.96903 / -18.4189^\circ = 0.91939 - j0.30618$$

$$V_2 = 0.96734 / -18.6030^\circ = 0.91680 - j0.30859$$

$$V_3 = 0.99964 / -15.3716^\circ = 0.96388 - j0.26499$$

$$V_4 = 0.94867 / -20.7466^\circ = 0.88715 - j0.33605$$

which agree almost exactly with the results found in Example 7.6.

7.7 NODE ELIMINATION (KRON REDUCTION)

Section 7.6 shows that gaussian elimination removes the need for matrix inversion when solving the nodal equations of a large-scale power system. At the same time it is also shown that elimination of variables is identical to network reduction since it leads to a sequence of reduced-order network equivalents by node elimination at each step. This may be important in analyzing a large interconnected power system if there is special interest in the voltages at only some of the buses of the overall system. For instance, one electric utility company with interconnections to other companies may wish to confine its study of voltage levels to those substations within its own service territory. By selective numbering of the system buses, we may apply gaussian elimination so as to reduce the Y_{bus} equations of the overall system to a set which contains only those bus voltages of special interest. The coefficient matrix in the reduced-order set of equations then represents the Y_{bus} for an equivalent network containing only those buses which are to be retained. All other buses are eliminated in the mathematical sense that their bus voltages and current injections do not appear explicitly. Such reduction in size of the equation set leads to efficiency of computation and helps to focus more directly on that portion of the overall network which is of primary interest.

In gaussian elimination one bus-voltage variable at a time is sequentially removed from the original system of N equations in N unknowns. Following Step 1 of the procedure, the variable V_1 does not explicitly appear in the

We may use the triangular factors to solve the original system of equations by substituting the product LU for Y_{bus} in Eq. (7.38) to obtain

$$\begin{matrix} \underline{L} & \underline{U} & \underline{V} & = & \underline{I} \\ N \times N & N \times N & N \times 1 & & N \times 1 \end{matrix} \quad (7.71)$$

As an intermediate step in the solution of Eq. (7.71), we may replace the product UV by a new voltage vector V' such that

$$\begin{matrix} \underline{L} & \underline{V}' & = & \underline{I} \\ N \times N & N \times 1 & & N \times 1 \end{matrix} \quad \text{and} \quad \begin{matrix} \underline{U} & \underline{V} & = & \underline{V}' \\ N \times N & N \times 1 & & N \times 1 \end{matrix} \quad (7.72)$$

Expressing Eq. (7.72) in full format shows that the original system of Eq. (7.38) is now replaced by two triangular systems given by

$$\begin{bmatrix} Y_{11} & & & \\ Y_{21} & Y_{22}^{(1)} & & \\ Y_{31} & Y_{32}^{(1)} & Y_{33}^{(2)} & \\ Y_{41} & Y_{42}^{(1)} & Y_{43}^{(2)} & Y_{44}^{(3)} \end{bmatrix} \begin{bmatrix} V'_1 \\ V'_2 \\ V'_3 \\ V'_4 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} \quad (7.73)$$

$$\text{and} \quad \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} Y_{12} & Y_{13} & Y_{14} \\ Y_{22}^{(1)} & Y_{23}^{(1)} & Y_{24}^{(1)} \\ Y_{32}^{(1)} & Y_{33}^{(2)} & Y_{34}^{(2)} \\ Y_{42}^{(1)} & Y_{43}^{(2)} & Y_{44}^{(3)} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} V'_1 \\ V'_2 \\ V'_3 \\ V'_4 \end{bmatrix} \quad (7.74)$$

The lower triangular system of Eq. (7.73) is readily solved by forward substitution beginning with V'_1 . We then use the calculated values of V'_1 , V'_2 , V'_3 , and V'_4 to solve Eq. (7.74) by back substitution for the actual unknowns V_1 , V_2 , V_3 , and V_4 .

Therefore, when changes are made in the current vector I , the solution vector V is found in two sequential steps; the first involves forward substitution using L and the second employs back substitution using U .

Example 7.9. Using the triangular factors of Y_{bus} determine the voltage at bus ③ of Fig. 7.11 when the current source at bus ④ is changed to $I_4 = 0.60 \angle -120^\circ$ per unit. All other conditions of Fig. 7.11 are unchanged.

Solution. The Y_{bus} for the network of Fig. 7.11 is given in Example 7.3. The corresponding matrix L may be assembled column by column from Example 7.7 simply by recording the column which is eliminated from the coefficient matrix at each step of the forward-elimination procedure. Then, substituting for L and the new current vector I in the equation $LV' = I$, we obtain

$$\begin{bmatrix} -j16.75 & & & \\ j11.75 & -j11.00746 & & \\ j2.50 & j4.25373 & -j3.78305 & \\ j2.50 & j6.75373 & j2.98305 & -j1.43082 \end{bmatrix} \begin{bmatrix} V'_1 \\ V'_2 \\ V'_3 \\ V'_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.60 \angle -120^\circ \end{bmatrix}$$

Solution by forward substitution beginning with V'_1 yields

$$V'_1 = V'_2 = 0; \quad V'_3 = \frac{1.00 \angle -90^\circ}{3.78305 \angle -90^\circ} = 0.26434$$

$$V'_4 = \frac{0.60 \angle -120^\circ - (j2.98305)V'_3}{-j1.43082} = 0.93800 \angle -12.9163^\circ$$

The matrix U is shown directly following Step 4 of the forward elimination in Example 7.7. Substituting U and the calculated entries of V' in the equation $UV = V'$ gives

$$\begin{bmatrix} 1 & -0.70149 & -0.14925 & -0.14925 \\ & 1 & -0.38644 & -0.61356 \\ & & 1 & -0.78853 \\ & & & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.26434 \\ 0.93800 \angle -12.9163^\circ \end{bmatrix}$$

which we may solve by back substitution to obtain

$$V_4 = V'_4 = 0.93800 \angle -12.9163^\circ \text{ per unit}$$

$$V_3 = 0.26434 - (-0.78853)V_4 = 0.99904 \angle -9.5257^\circ \text{ per unit}$$

If desired, we may continue the back substitution using the values for V_3 and V_4 to evaluate $V_2 = 0.96118 \angle -11.5551^\circ$ per unit and $V_1 = 0.96324 \angle -11.4388^\circ$ per unit.

When the coefficient matrix Y_{bus} is symmetrical, which is almost always the case, an important simplification results. As can be seen by inspection of Eq. (7.68), when the first column of L is divided by Y_{11} , we obtain the first row of U ; when the second column of L is divided by $Y_{22}^{(1)}$, we obtain the second row of U , and so on for the other columns and rows of Eq. (7.68), provided $Y_{ii} = Y_{ii}^{(i)}$. Therefore, dividing the entries in each column of L by the principal diagonal element in that column yields U^T whenever Y_{bus} is symmetrical. We can then write

$$L = U^T D = \begin{bmatrix} \frac{1}{Y_{11}} & \frac{Y_{21}}{Y_{11}} & \frac{Y_{31}}{Y_{11}} & \frac{Y_{41}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{Y_{22}^{(1)}}{Y_{22}^{(1)}} & \frac{Y_{32}^{(1)}}{Y_{22}^{(1)}} & \frac{Y_{42}^{(1)}}{Y_{22}^{(1)}} \\ \frac{Y_{31}}{Y_{11}} & \frac{Y_{32}^{(1)}}{Y_{22}^{(1)}} & \frac{Y_{33}^{(2)}}{Y_{33}^{(2)}} & \frac{Y_{43}^{(2)}}{Y_{33}^{(2)}} \\ \frac{Y_{41}}{Y_{11}} & \frac{Y_{42}^{(1)}}{Y_{22}^{(1)}} & \frac{Y_{43}^{(2)}}{Y_{33}^{(2)}} & \frac{Y_{44}^{(3)}}{Y_{44}^{(3)}} \end{bmatrix} \begin{bmatrix} Y_{11} & 0 & 0 & 0 \\ 0 & Y_{22}^{(1)} & 0 & 0 \\ 0 & 0 & Y_{33}^{(2)} & 0 \\ 0 & 0 & 0 & Y_{44}^{(3)} \end{bmatrix} \quad (7.75)$$

where diagonal matrix D contains the diagonal elements of L . Substituting in Eq. (7.71) for L from Eq. (7.75), we obtain the nodal admittance equations in the form

$$Y_{bus} V = U^T D U V = I \quad (7.76)$$

Equation (7.76) may be solved for the unknown voltages V in three consecutive steps as follows:

$$U^T V'' = I \quad (7.77)$$

$$D V' = V'' \quad (7.78)$$

$$U V = V' \quad (7.79)$$

These equations will be recognized as an extension of Eqs. (7.72). The intermediate result V'' is first found from Eq. (7.77) by forward substitution. Next, each entry in V' is calculated from Eq. (7.78) by dividing the corresponding element of V'' by the appropriate diagonal element of D . Finally, the solution V is obtained from Eq. (7.79) by back substitution as demonstrated in Example 7.9.

Example 7.10. Using Eqs. (7.77) through (7.79), determine the solution vector V of unknown voltages for the system and operating conditions of Example 7.9.

Solution. Substituting in Eq. (7.77) from the current vector I and matrix U of Example 7.9, we obtain

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -0.70149 & 1 & 0 & 0 \\ -0.14925 & -0.38644 & 1 & 0 \\ -0.14925 & -0.61356 & -0.78853 & 1 \end{bmatrix} \begin{bmatrix} V_1'' \\ V_2'' \\ V_3'' \\ V_4'' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 0.60 \angle -120^\circ \end{bmatrix}$$

Straightforward solution of this system of equations yields

$$V_1'' = V_2'' = 0 \quad V_3'' = 1.00 \angle -90^\circ \text{ per unit}$$

$$V_4'' = 0.60 \angle -120^\circ + 0.78853 V_3'' = 1.34210 \angle -102.9164^\circ \text{ per unit}$$

Substituting for V'' in Eq. (7.78) leads to the diagonal system

$$\begin{bmatrix} -j16.75 & 0 & 0 & 0 \\ 0 & -j11.00746 & 0 & 0 \\ 0 & 0 & -j3.78305 & 0 \\ 0 & 0 & 0 & -j1.43082 \end{bmatrix} \begin{bmatrix} V_1' \\ V_2' \\ V_3' \\ V_4' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.00 \angle -90^\circ \\ 1.34210 \angle -102.9164^\circ \end{bmatrix}$$

the solution of which is exactly equal to V' of Example 7.9, and so the remaining steps of this example match those of Example 7.9.

7.9 SPARSITY AND NEAR-OPTIMAL ORDERING

Large-scale power systems have only a small number of transmission lines connected to each bulk-power substation. In the network graph for such systems the ratio of the number of branches to the number of nodes is about 1.5 and the corresponding Y_{bus} has mainly zero elements. In fact, if there are 750 branches in a 500-node network (excluding the reference node), then since each node has an associated diagonal element and each branch gives rise to two symmetrically placed off-diagonal elements, the total number of nonzero elements is $(500 + 2 \times 750) = 2000$. This compares to a total of 250,000 elements in Y_{bus} ; that is, only 0.8% of the elements in Y_{bus} are nonzero. Because of the small number of nonzero elements, such matrices are said to be *sparse*. From the viewpoint of computational speed, accuracy, and storage it is desirable to process only the nonzero entries in Y_{bus} and to avoid *filling-in* new nonzero elements in the course of gaussian elimination and triangular factorization. Ordering refers to the sequence in which the equations of a system are processed. When a sparse

voltage V_k^0 , and between buses (k) and (l) the Thévenin impedance of bus (l) Z_{lk}^0 and the open-circuit voltage $V_l^0 - V_k^0$ is evident. Finally, when the branch impedance Z_b is connected between buses (l) and (k) of Fig. 8.4(a), the resulting current I_b is given by

$$I_b = \frac{V_k^0 - V_l^0}{Z_{lk}^0 + Z_b} = \frac{V_k - V_l}{Z_b} \quad (8.27)$$

We use this equation in Sec. 8.3 to show how to modify Z_{bus} when a branch impedance is added between two buses of the network.

Example 8.1. A capacitor having a reactance of 5.0 per unit is connected between the reference node and bus (4) of the circuit of Examples 7.5 and 7.6. The original emfs and the corresponding external current injections at buses (3) and (4) are the same as in those examples. Find the current drawn by the capacitor.

Solution. The Thévenin equivalent circuit at bus (4) has an emf with respect to reference given by $V_4^0 = 0.94866 \angle -20.7466^\circ$ per unit, which is the voltage at bus (4) found in Example 7.6 before the capacitor is connected. The Thévenin impedance Z_{44} at bus (4) is calculated in Example 7.6 to be $Z_{44} = j0.69890$ per unit, and so Fig. 8.5(a) follows. Therefore, the current I_{cap} drawn by the capacitor is

$$I_{cap} = \frac{0.94866 \angle -20.7466^\circ}{-j5.0 + j0.69890} = 0.22056 \angle 69.2534^\circ \text{ per unit}$$

Example 8.2. If an additional current equal to $-0.22056 \angle 69.2534^\circ$ per unit is injected into the network at bus (4) of Example 7.6, find the resulting voltages at buses (1), (2), (3), and (4).

Solution. The voltage changes at the buses due to the additional injected current can be calculated by making use of the bus impedance matrix found in Example 7.6. The required impedances are in column 4 of Z_{bus} . The voltage changes due to the added current injection at bus (4) in per unit are

$$\Delta V_1 = -I_{cap} Z_{14} = -0.22056 \angle 69.2534^\circ \times j0.63677 = 0.14045 \angle -20.7466^\circ$$

$$\Delta V_2 = -I_{cap} Z_{24} = -0.22056 \angle 69.2534^\circ \times j0.64178 = 0.14155 \angle -20.7466^\circ$$

$$\Delta V_3 = -I_{cap} Z_{34} = -0.22056 \angle 69.2534^\circ \times j0.55110 = 0.12155 \angle -20.7466^\circ$$

$$\Delta V_4 = -I_{cap} Z_{44} = -0.22056 \angle 69.2534^\circ \times j0.69890 = 0.15415 \angle -20.7466^\circ$$

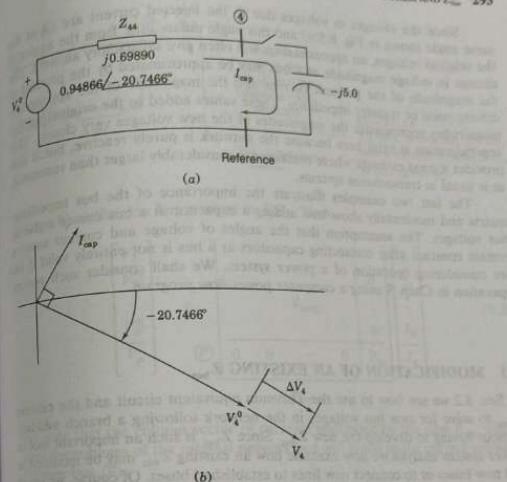


FIGURE 8.5 Circuit for Examples 8.1 and 8.2 showing: (a) Thévenin equivalent circuit; (b) phasor diagram at bus (4).

By superposition the resulting voltages are determined from Eq. (8.16) by adding these changes to the original bus voltages found in Example 7.6. The new bus voltages in per unit are

$$V_1 = 0.96903 \angle -18.4189^\circ + 0.14045 \angle -20.7466^\circ = 1.10938 \angle -18.7135^\circ$$

$$V_2 = 0.96734 \angle -18.6028^\circ + 0.14155 \angle -20.7466^\circ = 1.10880 \angle -18.8764^\circ$$

$$V_3 = 0.99964 \angle -15.3718^\circ + 0.12155 \angle -20.7466^\circ = 1.12071 \angle -15.9539^\circ$$

$$V_4 = 0.94866 \angle -20.7466^\circ + 0.15415 \angle -20.7466^\circ = 1.10281 \angle -20.7466^\circ$$

Example 8.3. Modify the bus impedance matrix of Example 7.6 to account for the connection of a capacitor having a reactance of 5.0 per unit between bus ④ and the reference node of the circuit of Fig. 7.9. Then, find V_4 using the impedances of the new matrix and the current sources of Example 7.6. Compare this value of V_4 with that found in Example 8.2.

Solution. We use Eq. (8.27) and recognize that Z_{orig} is the 4×4 matrix of Example 7.6. Thus, $Z_{44} = -j5.0$ per unit to find

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ 0 \end{bmatrix} = \begin{bmatrix} j0.63677 & j0.64178 & j0.55110 & j0.69890 & -j4.30110 \\ j0.64178 & j0.64178 & j0.55110 & j0.69890 & -j4.30110 \\ j0.55110 & j0.64178 & j0.55110 & j0.69890 & -j4.30110 \\ j0.69890 & j0.64178 & j0.55110 & j0.69890 & -j4.30110 \\ -j4.30110 & -j4.30110 & -j4.30110 & -j4.30110 & -j5.0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix}$$

The terms in the fifth row and column were obtained by repeating the fourth row and column of Z_{orig} and noting that

$$Z_{44} + Z_b = j0.69890 - j5.0 = -j4.30110$$

Then, eliminating the fifth row and column, we obtain for $Z_{bus(new)}$ from Eq. (8.33)

$$Z_{11(new)} = j0.73128 - \frac{j0.63677 \times j0.63677}{-j4.30110} = j0.82555$$

$$Z_{24(new)} = j0.64178 - \frac{j0.69890 \times j0.64178}{-j4.30110} = j0.74606$$

and other elements in a similar manner to give

$$Z_{bus(new)} = \begin{bmatrix} j0.82555 & j0.78641 & j0.69482 & j0.74024 \\ j0.78641 & j0.81542 & j0.69045 & j0.74606 \\ j0.69482 & j0.69045 & j0.76951 & j0.64065 \\ j0.74024 & j0.74606 & j0.64065 & j0.81247 \end{bmatrix}$$

The column matrix of currents by which the new Z_{bus} is multiplied to obtain the new bus voltages is the same as in Example 7.6. Since both I_1 and I_2 are zero while I_3 and I_4 are nonzero, we obtain

$$V_4 = j0.64065(1.00 \angle -90^\circ) + j0.81247(0.68 \angle -135^\circ)$$

$$= 1.03131 - j0.39066$$

$$= 1.10281 \angle -20.7466^\circ \text{ per unit}$$

as found in Example 8.2.

It is of interest to note that V_4 may be calculated directly from Eq. (8.27) by setting node ① equal to the reference node. We then obtain for $k = 4$ and $Z_{44} = Z_{44}$

$$V_4 = Z_{44} \frac{V_4^0}{Z_{44} + Z_b} = -j5.0 I_{cap}$$

$$= 1.10281 \angle -20.7466^\circ \text{ per unit}$$

since I_{cap} is already calculated in Example 8.1.

8.4 DIRECT DETERMINATION OF Z_{bus}

We could determine Z_{bus} by first finding V_{bus} and then inverting it, but this is not convenient for large-scale systems as we have seen. Fortunately, formulation of Z_{bus} using a direct building algorithm is a straightforward process on the computer.

At the outset we have a list of the branch impedances showing the buses to which they are connected. We start by writing the equation for one bus connected through a branch impedance Z_b to the reference as

$$[V_1] = \textcircled{1} [Z_b] [I_1] \quad (8.44)$$

and this can be considered as an equation involving three matrices, each of which has one row and one column. Now we might add a new bus connected to the first bus or to the reference node. For instance, if the second bus is connected to the reference node through Z_b , we have the matrix equation

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \textcircled{1} & \textcircled{2} \\ \textcircled{2} & \textcircled{1} \end{bmatrix} \begin{bmatrix} Z_a & 0 \\ 0 & Z_b \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \quad (8.45)$$

and we proceed to modify the evolving Z_{bus} matrix by adding other buses and branches following the procedures described in Sec. 8.3. The combination of these procedures constitutes the Z_{bus} building algorithm. Usually, the buses of a network must be renumbered internally by the computer algorithm to agree with the order in which they are to be added to Z_{bus} as it is built up.

Example 8.4. Determine Z_{bus} for the network shown in Fig. 8.8, where the impedances labeled 1 through 6 are shown in per unit. Preserve all buses.

Solution. The branches are added in the order of their labels and numbered subscripts on Z_{bus} will indicate intermediate steps of the solution. We start by

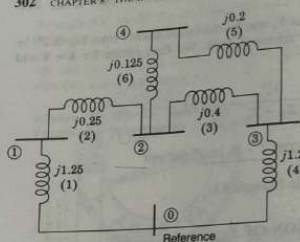


FIGURE 8.8 Network for Examples 8.4 and 8.5. Branch impedances are in per unit and branch numbers are in parentheses.

establishing bus ① with its impedance to the reference node and write

$$[V_1] = \textcircled{1} [j1.25] [I_1]$$

We then have the 1×1 bus impedance matrix

$$Z_{\text{bus},1} = \textcircled{1} [j1.25]$$

To establish bus ② with its impedance to bus ①, we follow Eq. (8.32) to write

$$Z_{\text{bus},2} = \begin{array}{c|c} \textcircled{1} & \textcircled{2} \\ \hline \textcircled{1} & j1.25 \quad j1.25 \\ \textcircled{2} & j1.25 \quad j1.50 \end{array}$$

The term $j1.50$ above is the sum of $j1.25$ and $j0.25$. The elements $j1.25$ in the new row and column are the repetition of the elements of row 1 and column 1 of the matrix being modified.

Bus ③ with the impedance connecting it to bus ② is established by writing

$$Z_{\text{bus},3} = \begin{array}{c|c|c} \textcircled{1} & \textcircled{2} & \textcircled{3} \\ \hline \textcircled{1} & j1.25 & j1.25 & j1.25 \\ \textcircled{2} & j1.25 & j1.50 & j1.50 \\ \textcircled{3} & j1.25 & j1.50 & j1.90 \end{array}$$

Since the new bus ③ is being connected to bus ②, the term $j1.90$ above is the sum of Z_{22} of the matrix being modified and the impedance Z_6 of the branch being connected to bus ② from bus ③. The other elements of the new row and column are the repetition of row 2 and column 2 of the matrix being modified since the new bus is being connected to bus ②.

If we now decide to add the impedance $Z_5 = j1.25$ from bus ③ to the reference node, we follow Eq. (8.32) to connect a new bus ④ through Z_5 and obtain the impedance matrix

$$Z_{\text{bus},4} = \begin{array}{c|c|c|c} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \hline \textcircled{1} & j1.25 & j1.25 & j1.25 & j1.25 \\ \textcircled{2} & j1.25 & j1.50 & j1.50 & j1.50 \\ \textcircled{3} & j1.25 & j1.50 & j1.90 & j1.90 \\ \textcircled{4} & j1.25 & j1.50 & j1.90 & j3.15 \end{array}$$

where $j3.15$ above is the sum of $Z_{33} + Z_5$. The other elements in the new row and column are the repetition of row 3 and column 3 of the matrix being modified since bus ④ is being connected to the reference node through Z_5 .

We now eliminate row p and column p by Kron reduction. Some of the elements of the new matrix from Eq. (8.33) are

$$Z_{11(\text{new})} = j1.25 - \frac{(j1.25)(j1.25)}{j3.15} = j0.75397$$

$$Z_{22(\text{new})} = j1.50 - \frac{(j1.50)(j1.50)}{j3.15} = j0.78571$$

$$Z_{23(\text{new})} = Z_{32(\text{new})} = j1.50 - \frac{(j1.50)(j1.90)}{j3.15} = j0.59524$$

When all the elements are determined, we have

$$Z_{\text{bus},5} = \begin{array}{c|c|c} \textcircled{1} & \textcircled{2} & \textcircled{3} \\ \hline \textcircled{1} & j0.75397 & j0.65476 & j0.49603 \\ \textcircled{2} & j0.65476 & j0.78571 & j0.59524 \\ \textcircled{3} & j0.49603 & j0.59524 & j0.75397 \end{array}$$

We now decide to add the impedance $Z_4 = j0.20$ from bus ③ to establish bus ④ using Eq. (8.32), and we obtain

$$Z_{\text{bus},6} = \begin{array}{c|c|c|c} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \hline \textcircled{1} & j0.75397 & j0.65476 & j0.49603 & j0.49603 \\ \textcircled{2} & j0.65476 & j0.78571 & j0.59524 & j0.59524 \\ \textcircled{3} & j0.49603 & j0.59524 & j0.75397 & j0.75397 \\ \textcircled{4} & j0.49603 & j0.59524 & j0.75397 & j0.95397 \end{array}$$

The off-diagonal elements of the new row and column are the repetition of row 1 and column 3 of the matrix being modified because the new bus (4) is being connected to bus (3). The new diagonal element is the sum of Z_{33} of the previous matrix and $Z_b = j0.20$.

Finally, we add the impedance $Z_b = j0.125$ between buses (2) and (4). If we let j and k in Eq. (8.41) equal 2 and 4, respectively, we obtain the elements for row 5 and column 5

$$Z_{15} = Z_{12} - Z_{14} = j0.65476 - j0.49603 = j0.15873$$

$$Z_{25} = Z_{22} - Z_{24} = j0.78571 - j0.59524 = j0.19047$$

$$Z_{35} = Z_{32} - Z_{34} = j0.59524 - j0.75397 = -j0.15873$$

$$Z_{45} = Z_{42} - Z_{44} = j0.59524 - j0.95397 = -j0.35873$$

and from Eq. (8.42)

$$\begin{aligned} Z_{55} &= Z_{22} + Z_{44} - 2Z_{24} + Z_b \\ &= j(0.78571 + 0.95397 - 2(0.59524)) + j0.125 = j0.67421 \end{aligned}$$

So, employing $Z_{bus,6}$ previously found, we write the 5×5 matrix

$$\textcircled{4} \begin{bmatrix} & & & & \textcircled{4} \\ & & & & j0.15873 \\ & & & & j0.19047 \\ & & & & -j0.15873 \\ & & & & -j0.35873 \\ j0.15873 & j0.19047 & -j0.15873 & -j0.35873 & j0.67421 \end{bmatrix} Z_{bus,6}$$

and from Eq. (8.43) we find by Kron reduction

$$Z_{bus} = \begin{matrix} & \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \\ \begin{matrix} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{matrix} & \begin{bmatrix} j0.71660 & j0.60992 & j0.53340 & j0.58049 \\ j0.60992 & j0.73190 & j0.64008 & j0.69659 \\ j0.53340 & j0.64008 & j0.71660 & j0.66951 \\ j0.58049 & j0.69659 & j0.66951 & j0.76310 \end{bmatrix} \end{matrix}$$

which is the bus impedance matrix to be determined. All calculations have been rounded off to five decimal places.

Since we shall again refer to these results, we note here that the reactance diagram of Fig. 8.8 is derived from Fig. 7.10 by omitting the sources and one of the mutually coupled branches. Also, the buses of Fig. 7.10 have been renumbered in Fig. 8.8 because the Z_{bus} building algorithm must begin with a bus connected to the reference node, as previously remarked.

The Z_{bus} building procedures are simple for a computer which first must determine the types of modification involved as each branch impedance is added. However, the operations must follow a sequence such that we avoid connecting an impedance between two new buses.

As a matter of interest, we can check the impedance values of Z_{bus} by the network calculations of Sec. 8.1.

Example 8.5. Find Z_{11} of the circuit of Example 8.4 by determining the impedance measured between bus (1) and the reference node when currents injected at buses (2), (3), and (4) are zero.

Solution. The equation corresponding to Eq. (8.12) is

$$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=I_3=I_4=0}$$

We recognize two parallel paths between buses (2) and (3) of the circuit of Fig. 8.8 with the resulting impedance of

$$\frac{(j0.125 + j0.20)(j0.40)}{j(0.125 + 0.20 + 0.40)} = j0.17931$$

This impedance in series with $(j0.25 + j1.25)$ combines in parallel with $j1.25$ to yield

$$Z_{11} = \frac{j1.25(j0.25 + j1.25 + j0.17931)}{j(1.25 + 0.25 + 1.25 + 0.17931)} = j0.71660$$

which is identical with the value found in Example 8.4.

Although the network reduction method of Example 8.5 may appear to be simpler by comparison with other methods of forming Z_{bus} , such is not the case because a different network reduction is required to evaluate each element of the matrix. In Example 8.5 the network reduction to find Z_{44} , for instance, is more difficult than that for finding Z_{11} . The computer could make a network reduction by node elimination but would have to repeat the process for each node.

required elements of column 3 of Z_{bus} :

$$Z_{43} = x_4$$

$$Z_{33} = x_3 - u_{34}Z_{43}$$

If all elements of $Z_{bus}^{(3)}$ are required, we can continue the calculations,

$$Z_{23} = x_2 - u_{23}Z_{33} - u_{24}Z_{43}$$

$$Z_{13} = x_1 - u_{12}Z_{23} - u_{13}Z_{33} - u_{14}Z_{43}$$

The computational effort in generating the required elements can be reduced by judiciously choosing the bus numbers.

In later chapters we shall find it necessary to evaluate terms like $(Z_{mn} - Z_{nn})$ involving differences between columns (m) and (n) of Z_{bus} . If the elements of Z_{bus} are not available explicitly, we can calculate the required differences by solving a system of equations such as

$$LUZ_{bus}^{(m-n)} = \begin{bmatrix} 0 \\ \vdots \\ 1_m \\ \vdots \\ -1_n \\ \vdots \\ 0 \end{bmatrix} \quad (8.52)$$

where $Z_{bus}^{(m-n)} = Z_{bus}^{(m)} - Z_{bus}^{(n)}$ is the vector formed by subtracting column n from column m of Z_{bus} , and $1_m = 1$ in row m and $-1_n = -1$ in row n of the vector shown.

In large-scale system calculations considerable computational efficiency can be realized by solving equations in the triangularized form of Eq. (8.52) while the full Z_{bus} need not be developed. Such computational considerations underlie many of the formal developments based on Z_{bus} in this text.

Example 8.6. The five-bus system shown in Fig. 8.9 has per-unit impedances as marked. The symmetrical bus admittance matrix for the system is given by

$$Y_{bus} = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \\ \textcircled{1} & -j30.0 & j10.0 & 0 & j20.0 & 0 \\ \textcircled{2} & j10.0 & -j26.2 & j16.0 & 0 & 0 \\ \textcircled{3} & 0 & j16.0 & -j36.0 & 0 & j20.0 \\ \textcircled{4} & j20.0 & 0 & 0 & -j20.0 & 0 \\ \textcircled{5} & 0 & 0 & j20.0 & 0 & -j20.0 \end{bmatrix}$$

and it is found that the triangular factors of Y_{bus} are

$$L = \begin{bmatrix} -j30.0 & & & & \\ j10.0 & -j22.866667 & & & \\ 0 & j16.000000 & -j24.804666 & & \\ j20.0 & j6.666667 & j4.664723 & -j3.845793 & \\ 0 & 0 & j20.000000 & j3.761164 & -j0.195604 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & -0.333333 & 0 & -0.666667 & 0 \\ & 1 & -0.699708 & -0.291545 & 0 \\ & & 1 & -0.188058 & -0.806300 \\ & & & 1 & -0.977995 \\ & & & & 1 \end{bmatrix}$$

Use the triangular factors to calculate $Z_{45} = (Z_{44} - Z_{45}) - (Z_{54} - Z_{55})$, the Thévenin impedance looking into the system between buses (4) and (5) of Fig. 8.9.

Solution. Since Y_{bus} is symmetrical, the reader should check that the row elements of U equal the column elements of L divided by their corresponding diagonal elements. With l 's representing the numerical values of L , forward solution of the system of equations

$$\begin{bmatrix} l_{11} & l_{12} & l_{13} & l_{14} & l_{15} \\ l_{21} & l_{22} & l_{23} & l_{24} & l_{25} \\ l_{31} & l_{32} & l_{33} & l_{34} & l_{35} \\ l_{41} & l_{42} & l_{43} & l_{44} & l_{45} \\ l_{51} & l_{52} & l_{53} & l_{54} & l_{55} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

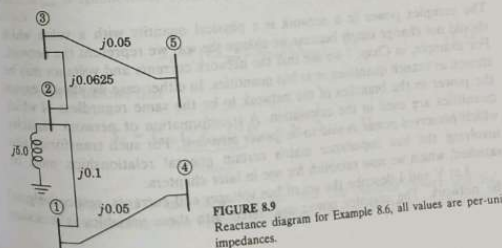


FIGURE 8.9 Reactance diagram for Example 8.6, all values are per-unit impedances.

CASE 8. Removing mutually coupled Z_b from existing buses (p) and (k) .

A single uncoupled branch of impedance Z_b can be removed from the network model by adding the negative of impedance Z_b between the same terminating buses. When the impedance Z_b to be removed is also mutually coupled to a second branch of impedance Z_a , the rule for modifying Z_{orig} is to add between the end buses of Z_b a branch, which has negative impedance $-Z_b$ and the same mutual coupling to Z_a as the original Z_b . The procedure is illustrated in Fig. 8.12. As shown, the mutually coupled branches Z_a and Z_b with mutual impedance Z_M are already included in Z_{orig} . In accordance with the above rule, we add the impedance $-Z_b$ having mutual coupling Z_M with the branch Z_a , and the voltage-drop equations for the three mutually coupled branches then are

$$V_a = Z_a I_a + Z_M I_b + Z_M I'_b \quad (8.90)$$

$$V_b = Z_M I_a + Z_b I_b + 0 \quad (8.91)$$

$$V_b = Z_M I_a + 0 + (-Z_b) I'_b \quad (8.92)$$

where the branch currents I_a , I_b , and I'_b are as shown in Fig. 8.12. Subtracting Eq. (8.92) from Eq. (8.91) yields

$$0 = Z_b(I_b + I'_b) \quad (8.93)$$

which shows that $(I_b + I'_b)$ is zero, and substituting this result in Eq. (8.90) gives

$$V_a = Z_a I_a \quad (8.94)$$

Because $(I_b + I'_b)$ equals zero, the net coupling effect of the two parallel

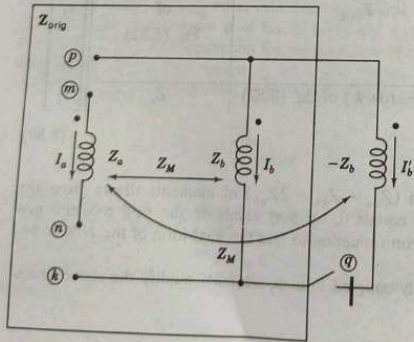


FIGURE 8.12 Removing mutually coupled branch Z_b from Z_{orig} . Initially, the switch is considered open and the branch $-Z_b$ is added from existing bus (p) to temporary bus (q) . The switch is then closed to connect bus (q) to bus (k) .

branches between buses (p) and (k) is zero. Consequently, the impedance Z_a between buses (m) and (n) can be made to stand alone as evidenced by Eq. (8.94). To do so, we follow exactly the same procedure as in Case 7, except that the elements of the new row and column of the temporary bus (q) are calculated sequentially using the modified formulas

$$Z_{jq} = Z_{qj} = Z_{pj} - Z_M[Y_a|Y_M] \left[\frac{Z_{mj} - Z_{nj}}{Z_{pj} - Z_{kj}} \right] \quad (8.95)$$

with index j ranging from 1 to N and

$$Z_{qq} = Z_{pq} - Z_M[Y_a|Y_M] \left[\frac{Z_{mq} - Z_{nq}}{Z_{pq} - Z_{kq}} \right] - (Y_a Z_M^2 + Z_b) \quad (8.96)$$

Using these elements in Eqs. (8.82) and (8.89), and Kron reducing the latter, gives the desired new bus impedance matrix from which the mutually coupled branch between buses (p) and (k) is omitted. Equations (8.95) and (8.96) (developed in Prob. 8.19) have admittances Y_a and Y_b , which are calculated from the impedance parameters Z_a , Z_b , and Z_M according to Eq. (7.10).

Table 8.2 summarizes the procedures of Cases 5 through 8.

Example 8.7. Between buses ① and ④ of Fig. 8.8 impedance Z_b equal to $j0.25$ per unit is connected so that it couples through mutual impedance $j0.15$ per unit to the branch impedance already connected between buses ① and ②. Modify the bus impedance matrix of Example 8.4 to include the addition of Z_b to Fig. 8.8.

Solution. Connecting Z_b between buses ① and ④ corresponds to the conditions of Case 7 above. Our calculations begin with Z_{orig} , the 4×4 matrix solution of Example 8.4 which includes the branch between buses ① and ②. To establish row q and column q for temporary bus (q) , we follow Eq. (8.87), recognizing that subscript $m = 1$, subscript $n = 2$, subscript $p = 1$, and we find that

$$Z_{qj} = Z_{1j} - \frac{Z_M}{Z_a}(Z_{1j} - Z_{2j}) = \left(1 - \frac{Z_M}{Z_a}\right)Z_{1j} + \frac{Z_M}{Z_a}Z_{2j}$$

The ratio (Z_M/Z_a) is given by $(j0.15/j0.25) = 0.6$, and substituting this value along with the elements of row 1 and row 2 of Z_{orig} in the above equation, we obtain

$$Z_{q1} = 0.4Z_{11} + 0.6Z_{21} = 0.4(j0.71660) + 0.6(j0.60992) = j0.65259$$

$$Z_{q2} = 0.4Z_{12} + 0.6Z_{22} = 0.4(j0.60992) + 0.6(j0.73190) = j0.68311$$

$$Z_{q3} = 0.4Z_{13} + 0.6Z_{23} = 0.4(j0.53340) + 0.6(j0.64008) = j0.59741$$

$$Z_{q4} = 0.4Z_{14} + 0.6Z_{24} = 0.4(j0.58049) + 0.6(j0.69659) = j0.65015$$

TABLE 8.2
Z_{bus} modifications; mutual coupling

Case	Add mutually coupled Z _b from	Z _{bus} (new)
5	Existing bus (p) to new bus (q) 	- Form the matrix $\begin{bmatrix} Z_{orig} & \text{col. } q \\ \text{row } q & Z_{qq} \end{bmatrix}$ $Z_{jq} = Z_{qj} = Z_{pj} - \frac{Z_M}{Z_a} (Z_{mj} - Z_{nq})$ $Z_{qq} = Z_{pq} - \frac{Z_M}{Z_a} (Z_{mq} - Z_{nq}) - \left(\frac{Z_b}{Z_a} - Z_b \right)$
6	Existing bus (p) to reference node 	- Repeat Case 5 and - Eliminate row (q) and column (q) by Kron reduction
7	Existing bus (p) to existing bus (k) 	- Repeat Case 5 - Then, form the matrix $\begin{bmatrix} Z_{orig} & \text{col. } q - \text{col. } k \\ \text{row } q - \text{row } k & Z_{qq} + Z_{kk} - 2Z_{qk} \end{bmatrix}$ - and - Eliminate last row and column by Kron reduction
8	Removal of mutually coupled line 	- Form the matrix of Case 5 using Eq. $Z_{jq} = Z_{qj} = Z_{pj} - Z_M [Y_a \ Y_M] \begin{bmatrix} Z_{mj} - Z_{nq} \\ Z_{pj} - Z_{nq} \end{bmatrix}$ $Z_{qq} = Z_{pq} - Z_M [Y_a \ Y_M] \begin{bmatrix} Z_{mq} - Z_{nq} \\ Z_{pq} - Z_{nq} \end{bmatrix} - (Y_a Z_M^2 + Z_b)$ - where $\begin{bmatrix} Y_a & Y_M \\ Y_M & Y_b \end{bmatrix} = \begin{bmatrix} Z_a & Z_M \\ Z_M & Z_b \end{bmatrix}^{-1}$ - Continue as in Case 7.

322

These elements constitute new row q and new column q except for the diagonal element Z_{qq}, which is found from Eq. (8.88) by setting the subscripts m, n, and p as above to yield

$$Z_{qq} = Z_{1q} - \frac{Z_M}{Z_a} (Z_{1q} - Z_{2q}) - \left(\frac{Z_b}{Z_a} - Z_b \right)$$

Substituting in this equation for Z_M = j0.15, Z_a = j0.25 and Z_b = j0.25 gives

$$Z_{qq} = 0.4Z_{1q} + 0.6Z_{2q} + j0.16$$

$$= 0.4(j0.65259) + 0.6(j0.68311) + j0.16 = j0.83090$$

Associating the newly calculated elements of row q and column q with Z_{orig} from Example 8.4 results in the 5 × 5 matrix

$$Z_{orig} = \begin{bmatrix} j0.71660 & j0.60992 & j0.53340 & j0.58049 & j0.65259 \\ j0.60992 & j0.73190 & j0.64008 & j0.69659 & j0.68311 \\ j0.53340 & j0.64008 & j0.71660 & j0.66951 & j0.59741 \\ j0.58049 & j0.69659 & j0.66951 & j0.76310 & j0.65015 \\ j0.65259 & j0.68311 & j0.59741 & j0.65015 & j0.83090 \end{bmatrix}$$

At this point in our solution the mutually coupled branch impedance Z_b has been incorporated into the network between bus (1) and bus (2). To complete the connection of Z_b to bus (2), we must find (V₂ - V₀) and then set it equal to zero. The first of these steps is accomplished by setting subscript k equal to 4 in Eq. (8.89), subtracting row 4 from row q of the above 5 × 5 matrix, and using the result to replace the existing row q and column q to obtain

$$Z_{orig} = \begin{bmatrix} j0.71660 & j0.60992 & j0.53340 & j0.58049 & j0.07210 \\ j0.60992 & j0.73190 & j0.64008 & j0.69659 & -j0.01348 \\ j0.53340 & j0.64008 & j0.71660 & j0.66951 & -j0.07210 \\ j0.58049 & j0.69659 & j0.66951 & j0.76310 & -j0.11295 \\ j0.07210 & -j0.01348 & -j0.07210 & -j0.11295 & j0.29370 \end{bmatrix}$$

The new diagonal element is calculated from

$$(Z_{qq} + Z_{44} - 2Z_{q4}) = j0.29370$$

The only remaining calculations involve setting (V_q - V₄) equal to zero in Eq.

(8.89) and eliminating the new row and column by Kron reduction to give

$$\mathbf{Z}_{\text{bus}} = \begin{array}{c} \begin{array}{c} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{array} \begin{array}{c} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{array} \begin{bmatrix} j0.69890 & j0.61323 & j0.55110 & j0.60822 \\ j0.61323 & j0.73128 & j0.63677 & j0.69140 \\ j0.55110 & j0.63677 & j0.69890 & j0.64178 \\ j0.60822 & j0.69140 & j0.64178 & j0.71966 \end{bmatrix} \end{array}$$

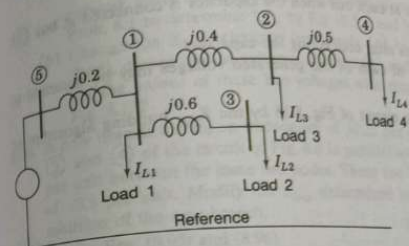
which is the desired bus impedance matrix. This result is the same as that shown in Example 7.6 except for the change in the bus numbers of Fig. 7.9 to those of Fig. 8.8.

8.8 SUMMARY

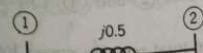
This chapter introduces the important $\mathbf{Z}_{\text{bus}}$ building algorithm, which starts by choosing a branch tied to the reference from a node and adding to this node a second branch connected from a new node. The starting node and the new node then have one row and one column each in the 2×2 bus impedance matrix which represents them. Next, a third branch connected to one or both of the first two chosen nodes is added to expand the evolving network and its $\mathbf{Z}_{\text{bus}}$ representation. In this manner the bus impedance matrix is built up one row and one column at a time until all branches of the physical network have been incorporated into $\mathbf{Z}_{\text{bus}}$. Whenever possible at any stage, it is computationally more efficient to select the next branch to be added between two nodes with rows and columns already included in the evolving $\mathbf{Z}_{\text{bus}}$. The elements of $\mathbf{Z}_{\text{bus}}$ can also be generated as needed using the triangular factors of $\mathbf{Y}_{\text{bus}}$, which is often the most attractive method computationally.

The variables used to analyze a power network can take on many different forms. However, regardless of the particular choice of representation, power in the physical network must not be arbitrarily altered when representing currents and voltages in the chosen format. Power invariance imposes requirements when transforming from one set of currents and voltages to another.

Mutually coupled branches, which do not generally arise except under unbalanced short-circuit (fault) conditions, can be added to and removed from $\mathbf{Z}_{\text{bus}}$ by algorithmic methods.



- 8.2. From the solution of Prob. 8.1, draw the Thévenin eq. Fig. 8.13 and use it to determine the current drawn by per unit connected between bus ④ and reference. Example 8.2, calculate the voltage changes at each capacitor.
- 8.3. Modify $\mathbf{Z}_{\text{bus}}$ of Prob. 8.1 to include a capacitor of reactance from bus ④ to reference and then calculate the modified $\mathbf{Z}_{\text{bus}}$. Check your answers using the results.
- 8.4. Modify the $\mathbf{Z}_{\text{bus}}$ determined in Example 8.4 for the new node connected to bus ③ through an impedance.
- 8.5. Modify the $\mathbf{Z}_{\text{bus}}$ determined in Example 8.4 by adding per unit between buses ① and ④ of the circuit.
- 8.6. Modify the $\mathbf{Z}_{\text{bus}}$ determined in Example 8.4 by removing between buses ② and ③ of the circuit of Fig. 8.
- 8.7. Find $\mathbf{Z}_{\text{bus}}$ for the circuit of Fig. 7.18 by the $\mathbf{Z}_{\text{bus}}$ Sec. 8.4. Assume there is no mutual coupling between.
- 8.8. For the reactance network of Fig. 8.14, find
 - (a) $\mathbf{Z}_{\text{bus}}$ by direct formulation,
 - (b) The voltage at each bus,
 - (c) The current drawn by a capacitor having a reactance from bus ③ to neutral,



The superscript (k) denotes the number of the iteration in which the voltage is

$$V_i^{(k)} = \frac{1}{P_{i, \text{sch}} - jQ_{i, \text{sch}}} \left[P_{i, \text{sch}} V_i^{(k-1)*} - \sum_{j=1}^N Y_{ij}^* V_j^{(k-1)} \right] \quad (9.19)$$

any bus i where P and Q are scheduled is

For a system of N buses the general equation for the calculated voltage at

of the uniform voltage profile assumed.

buses equal to $1.0/\theta^\circ$ per unit. Such initialization is called a *flat start* because

common practice to set the initial estimates of the unknown voltages at all load

values are of reasonable magnitude and do not differ too widely in phase. It is

Convergence upon an erroneous solution is usually avoided if the initial

power-flow equations is known as the *Gauss-Seidel iterative method*.

bus is less than some predetermined precision index. This process of solving the

carried out again and again until the amount of correction in voltage at every

calculated values are found for each state variable. Then, the entire process is

network (except at the slack bus) to complete the *first iteration* in which

The process is repeated at bus (i) and at each bus consecutively throughout the

$$V_i^{(1)} = \frac{1}{P_{i, \text{sch}} - jQ_{i, \text{sch}}} \left[P_{i, \text{sch}} V_i^{(0)*} - \sum_{j=1}^N Y_{ij}^* V_j^{(0)} \right] \quad (9.18)$$

obtain for the first calculated value at bus (i)

corrected voltage at the next bus. Therefore, substituting $V_i^{(1)}$ in Eq. (9.16), we

As the corrected voltage is found at each bus, it is used to calculate the

$V_i^{(0)}$ and $V_j^{(0)}$ at the other buses, and the actual voltages are not yet known.

the voltages on which this calculation for V_j depends are the estimated values

be the solution for V_j for the specific power-flow conditions, however, because

voltages but without regard to power at the other buses. This value would not

after several iterations and would be the correct value for V_j with the estimated

$V_i^{(0)}$ will not agree. Agreement would be reached to a good degree of accuracy

calculated value $V_i^{(1)}$ and the estimated value

in which all quantities in the right-hand-side expression are either fixed spec-

$$V_i^{(1)} = \frac{1}{P_{i, \text{sch}} - jQ_{i, \text{sch}}} \left[P_{i, \text{sch}} V_i^{(0)*} - \sum_{j=1}^N Y_{ij}^* V_j^{(0)} \right] \quad (9.17)$$

Solution of Eq. (9.15) gives the corrected voltage $V_i^{(1)}$ calculated from

$V_i^{(0)}$ and $V_j^{(0)}$ at the other buses.

scheduled slack bus voltage $V_1 = |V_1|/\delta_1$, and initial voltage estimates $V_i^{(0)}$

based on the scheduled real and reactive power at buses (2) , (3) , and (4) , the

directly from the equations as they appear. The solution proceeds by iteration

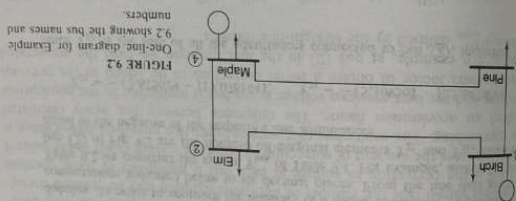
variables δ_2 to δ_4 and $|V_2|$ to $|V_4|$. However, we solve for the complex voltage

similar equation of bus (4) , we would obtain six equations in the six vari-

Line data for Example 9.2

Line, bus to	R per unit	X per unit	G per unit	B per unit	Total charging Mvar	Shunt Y
1-2	0.01008	0.05040	3.815629	-19.078144	10.25	0.05125
1-3	0.00744	0.03720	5.169561	-25.847809	7.75	0.03875
2-4	0.00744	0.03720	5.169561	-25.847809	7.75	0.03875
3-4	0.01272	0.06360	3.023705	-15.118528	12.75	0.06375

Base 100 MVA, 230 kV.



Example 9.2. Figure 9.2 shows the one-line diagram of a simple power system. Generators are connected at buses (1) and (4) while loads are indicated at all four buses. Base values for the transmission system are 100 MVA, 230 kV. The line data of Table 9.2 give per-unit series impedances and line-charging susceptances for the nominal- π equivalents of the four lines identified by the buses at which they terminate. The bus data in Table 9.3 list values for P_i , Q_i , and V_i at each bus. The Q values of load are calculated from the corresponding P values assuming a power factor of 0.85. The net scheduled values, $P_{i, \text{sch}}$ and $Q_{i, \text{sch}}$, are negative at the load buses (2) and (3) . Generated $Q_{i, \text{sch}}$ is not specified where voltage numbers.

Figure 9.2 showing the bus names and numbers.

Since Eq. (9.19) applies only at load buses where real and reactive power are specified, an additional step is necessary at voltage-controlled buses where voltage magnitude is to remain constant. Before investigating this additional step, let us look at an example of the calculations at a load bus.

Iteration. Thus, we see that the values for the voltages on the right-hand side of this equation are the most recently calculated values for the corresponding

currently being calculated and $(k-1)$ indicates the number of the preceding

TABLE 9.3
Bus data for Example 9.2

Bus	Generation		Load		Remarks
	P, MW	Q, Mvar	P, MW	Q, Mvar	
1	—	—	50	30.99	Slack bus
2	0	0	170	105.35	Load bus (inductive)
3	0	0	200	123.94	Load bus (inductive)
4	318	—	80	49.58	Voltage controlled (inductive)

The Q values of load are calculated from the corresponding P values assuming a power factor of 0.85.

magnitude is constant. In the voltage column the values for the load buses are bus-start estimates. The slack bus voltage magnitude $|V_1|$ and angle δ_1 and the power-flow study is to be made by the Gauss-Seidel method. Assuming that the iterative calculations start at bus (2), find the value of V_2 for the first iteration.

Solution. In order to approach the accuracy of a digital computer, we perform the computations indicated below to six decimal places. From the line data given in Table 9.2 we construct the system Y_{bus} of Table 9.4. For example, associated with bus (2) of Fig. 9.2 are the nonzero off-diagonal elements Y_{21} and Y_{24} , which are

$$Y_{21} = -(3.815629 - j19.078144); \quad Y_{24} = -(5.169561 - j25.847809)$$

Since Y_{22} is the sum of all the admittances connected to bus (2), including the

TABLE 9.4
Bus admittance matrix for Example 9.2†

Bus no.	①	②	③	④
①	8.985190 -j44.835953	+j19.078144 -3.815629	-5.169561 +j25.847809	0
②	+j19.078144 -3.815629	8.985190 -j44.835953	0	-5.169561 +j25.847809
③	-5.169561 +j25.847809	0	8.193267 -3.023705	+j15.118528 -40.863838
④	0	-5.169561 +j25.847809	+j15.118528 -40.863838	8.193267 -3.023705

†Per-unit values rounded to six decimal places

shunt susceptances for line charging of lines (2)-(1) and (2)-(4), we have

$$Y_{22} = (-Y_{21}) + j0.05125 + (-Y_{24}) + j0.03875 = 8.985190 - j44.835953$$

Substitution in Eq. (9.17) yields the per-unit voltage

$$V_2^{(1)} = \frac{1}{Y_{22}} \left[-1.7 + j1.0535 - 1.00(-3.815629 + j19.078144) - 1.02(-5.169561 + j25.847809) \right]$$

$$= \frac{1}{8.985190 - j44.835953} [-1.7 + j1.0535 + 9.088581 - j45.442909]$$

$$= 7.388581 - j44.389409 = 0.985364 - j0.032316$$

Experience with the Gauss-Seidel method of solution of power-flow problems has shown that the number of iterations required may be reduced considerably if the correction in voltage at each bus is multiplied by some constant that increases the amount of correction to bring the voltage closer to the value it is approaching. The multiplier that accomplishes this improved convergence is called an *acceleration factor*. The difference between the newly calculated voltage and the best previous voltage at the bus is multiplied by the appropriate acceleration factor to obtain a better correction to be added to the previous value. For example, at bus (2) in the first iteration we have the accelerated value $V_2^{(1),acc}$ defined by the straight-line formula

$$V_2^{(1),acc} = (1 - \alpha)V_2^{(0)} + \alpha V_2^{(1)} = V_2^{(0)} + \alpha(V_2^{(1)} - V_2^{(0)}) \quad (9.20)$$

in which α is the acceleration factor. More generally, for bus (i) during iteration k the accelerated value is given by

$$V_i^{(k),acc} = (1 - \alpha)V_i^{(k-1)} + \alpha V_i^{(k)} = V_i^{(k-1)} + \alpha(V_i^{(k)} - V_i^{(k-1)}) \quad (9.21)$$

If $\alpha = 1$, then the Gauss-Seidel computed value of V_i is stored as the current Gauss-Seidel value and the value to be stored is a weighted average of the value. If $0 < \alpha < 1$, then the value to be stored is an extrapolated value. In power-flow studies α is generally set at about 1.6 and cannot exceed 2 if convergence is to occur.

Substituting the results of Example 9.1 and an acceleration factor of 1.6 in Eq. (9.20), we find that

$$\begin{aligned} V_{2,\text{acc}}^{(1)} &= 1 + 1.6[(0.983564 - j0.032316) - 1] \\ &= 0.973703 - j0.051706 \text{ per unit} \end{aligned}$$

Using $V_{2,\text{acc}}^{(1)}$ in similar calculations for bus ③ gives first-iteration value

$$V_{3,\text{acc}}^{(1)} = 0.953949 - j0.066708 \text{ per unit}$$

Since bus ④ is voltage controlled, we must treat it differently as explained below. The acceleration factor for the real component of the correction may differ from that for the imaginary component. For any system optimum values for acceleration factors exist, and poor choice of factors may result in less rapid convergence or make convergence impossible. An acceleration factor of 1.6 for both the real and imaginary components is usually a good choice but studies may be made to determine the best choice for a particular system.

Voltage-controlled buses. When voltage magnitude rather than reactive power is specified at bus ①, the real and imaginary components of the voltage for each iteration are found by first computing a value for the reactive power. From Eq. (9.4) we have

$$Q_i = -\text{Im} \left\{ V_i^* \sum_{j=1}^N Y_{ij} V_j \right\} \quad (9.22)$$

which has the equivalent algorithmic expression

$$Q_i^{(k)} = -\text{Im} \left\{ V_i^{(k-1)*} \left[\sum_{j=1}^{i-1} Y_{ij} V_j^{(k)} + \sum_{j=i}^N Y_{ij} V_j^{(k-1)} \right] \right\} \quad (9.23)$$

where Im means "imaginary part of" and the superscripts indicate the relevant iteration. Reactive power $Q_i^{(k)}$ is evaluated by Eq. (9.23) for the best previous voltage values at the buses, and this value of $Q_i^{(k)}$ is substituted in Eq. (9.19) to find a new value of $V_i^{(k)}$. The components of the new $V_i^{(k)}$ are then multiplied by the ratio of the specified constant magnitude $|V_i|$ to the magnitude of $V_i^{(k)}$ found by Eq. (9.19). The result is the corrected complex voltage of the specified magnitude. In the four-bus example if bus ④ is voltage controlled, Eq. (9.23) yields the calculated value

$$Q_4^{(1)} = -\text{Im} \{ V_4^{(0)*} (Y_{41} V_1 + Y_{42} V_{2,\text{acc}}^{(1)} + Y_{43} V_{3,\text{acc}}^{(1)} + Y_{44} V_4^{(0)}) \} \quad (9.24)$$

wherein the calculated voltages of buses ② and ③ are accelerated values of the first iteration. Substituting $Q_4^{(1)}$ for $Q_{4,\text{sch}}$ in the equivalent of Eq. (9.19) for

bus ④ yields

$$V_4^{(1)} = \frac{1}{Y_{44}} \left[\frac{P_{4,\text{sch}} - jQ_4^{(1)}}{V_4^{(0)*}} - (Y_{41} V_1 + Y_{42} V_{2,\text{acc}}^{(1)} + Y_{43} V_{3,\text{acc}}^{(1)}) \right] \quad (9.25)$$

and all quantities on the right-hand side are now known. Since $|V_4|$ is specified, we correct the magnitude of $V_4^{(1)}$ as follows:

$$V_{4,\text{corr}}^{(1)} = |V_4| \frac{V_4^{(1)}}{|V_4^{(1)}|} \quad (9.26)$$

and proceed to the next step with stored value $V_{4,\text{corr}}^{(1)}$ of bus ④ voltage having the specified magnitude in the remaining calculations of the iteration.

As discussed in Sec. 9.1, either voltage magnitude or reactive power must be specified at every bus except the slack bus, where voltage is specified by both voltage magnitude and angle. At buses with generation the voltage magnitude is specified as well as the real power P_g supplied by the generator. The reactive power Q_g entering the network from the generator is then determined by the computer in solving the power-flow problem. From a practical viewpoint the Q_g output of the generator must be within definite limits given by the inequality

$$Q_{\min} \leq Q_g \leq Q_{\max}$$

where $Q_{\min}$ is the minimum and $Q_{\max}$ is the maximum limit imposed on the reactive power output of the generator at the bus. In the course of power-flow solution if the calculated value of Q_g is outside either limit, then Q_g is set equal to the limit violated, the originally specified voltage magnitude at the bus is relaxed, and the bus is then treated as a P - Q bus for which a new voltage is calculated by the computer program. In subsequent iterations the program endeavors to sustain the originally specified voltage at the bus while ensuring that Q_g is within the permitted range of values. This could well be possible since other changes may occur elsewhere in the system to support the local action of the generator excitation as it adjusts to satisfy the specified terminal voltage.

Example 9.3. To complete the first iteration of the Gauss-Seidel procedure, find the voltage at bus ④ of Example 9.2 computed with the originally estimated voltages at buses ② and ③ replaced by the accelerated values indicated above.

Solution. Table 9.4 shows Y_{41} equal to zero, and so Eq. (9.24) gives

$$Q_4^{(1)} = -\text{Im} \{ V_4^{(0)*} [Y_{42} V_{2,\text{acc}}^{(1)} + Y_{43} V_{3,\text{acc}}^{(1)} + Y_{44} V_4^{(0)}] \}$$

Substituting values for the indicated quantities in this equation, we obtain

$$Q_4^{(1)} = -\text{Im} \left\{ \frac{1.02[-5.169561 + j25.847809](0.973703 - j0.051706)}{(-3.023705 + j15.118528)(0.953949 - j0.066708)} + (8.193267 - j40.863838)(1.02) \right\}$$

$$= -\text{Im}[1.02[-5.573064 + j40.059396 + (8.193267 - j40.863838)1.02]]$$

$$= 1.654151 \text{ per unit}$$

This value for $Q_4^{(1)}$ is now substituted into Eq. (9.25) to yield

$$V_4^{(1)} = \frac{1}{Y_{44}} \left[\frac{P_{4,\text{sch}} - jQ_4^{(1)}}{V_4^{(0)}} - (Y_{42}V_2^{(1)} + Y_{43}V_3^{(1)}) \right]$$

$$= \frac{1}{Y_{44}} \left[\frac{2.38 - j1.654151}{1.02 - j0.0} - (-5.573066 + j40.059398) \right]$$

$$= \frac{7.906399 - j41.681115}{8.193267 - j40.863838} = 1.017874 - j0.010604 \text{ per unit}$$

Hence, $|V_4^{(1)}|$ equals 1.017929, and so we must correct the magnitude to 1.02

$$V_{4,\text{corr}}^{(1)} = \frac{1.02}{1.017929} (1.017874 - j0.010604)$$

$$= 1.019945 - j0.010625 \text{ per unit}$$

In this example $Q_4^{(1)}$ is found to be 1.654151 per unit in the first iteration. If the reactive power generation at bus (4) were limited below 1.654151 per unit, then the specified limit value would be used for $Q_4^{(1)}$ and bus (4) in that case would be treated as a load bus within the iteration. The same strategy is used within any other iteration in which the generator Q -limits are violated.

The Gauss-Seidel procedure is one method of solving the power-flow problem. However, today's industry-based studies generally employ the alternative Newton-Raphson iterative method, which is reliable in convergence, computationally faster, and more economical in storage requirements. The converged solution of Examples 9.2 and 9.3 agrees with the results given in Fig. 9.4 found later by the Newton-Raphson method.

9.3 THE NEWTON-RAPHSON METHOD

Taylor's series expansion for a function of two or more variables is the basis for the Newton-Raphson method of solving the power-flow problem. Our study of

the method begins by a discussion of the solution of a problem involving only two equations and two variables. Then, we see how to extend the analysis to the solution of power-flow equations.

Let us consider the equation of a function h_1 of two variables x_1 and x_2 equal to a constant b_1 expressed as

$$g_1(x_1, x_2, u) = h_1(x_1, x_2, u) - b_1 = 0 \quad (9.27)$$

and a second equation involving another function h_2 such that

$$g_2(x_1, x_2, u) = h_2(x_1, x_2, u) - b_2 = 0 \quad (9.28)$$

where b_2 is also a constant. The symbol u represents an independent control, which is considered constant in this chapter. As in Eqs. (9.9) and (9.10), the functions g_1 and g_2 are introduced for convenience to allow us to discuss the differences between calculated values of h_1 and h_2 and their respective specified values b_1 and b_2 .

For a specified value of u let us estimate the solutions of these equations to be $x_1^{(0)}$ and $x_2^{(0)}$. The zero superscripts indicate that these values are initial estimates and not the actual solutions x_1^* and x_2^* . We designate the correction $\Delta x_1^{(0)}$ and $\Delta x_2^{(0)}$ as the values to be added to $x_1^{(0)}$ and $x_2^{(0)}$ to yield the correct solutions x_1^* and x_2^* . So, we can write

$$g_1(x_1^*, x_2^*, u) = g_1(x_1^{(0)} + \Delta x_1^{(0)}, x_2^{(0)} + \Delta x_2^{(0)}, u) = 0 \quad (9.29)$$

$$g_2(x_1^*, x_2^*, u) = g_2(x_1^{(0)} + \Delta x_1^{(0)}, x_2^{(0)} + \Delta x_2^{(0)}, u) = 0 \quad (9.30)$$

Our problem now is to solve for $\Delta x_1^{(0)}$ and $\Delta x_2^{(0)}$, which we do by expanding Eqs. (9.29) and (9.30) in Taylor's series about the assumed solution to give

$$g_1(x_1^*, x_2^*, u) = g_1(x_1^{(0)}, x_2^{(0)}, u) + \Delta x_1^{(0)} \left. \frac{\partial g_1}{\partial x_1} \right|^{(0)} + \Delta x_2^{(0)} \left. \frac{\partial g_1}{\partial x_2} \right|^{(0)} + \dots = 0 \quad (9.31)$$

$$g_2(x_1^*, x_2^*, u) = g_2(x_1^{(0)}, x_2^{(0)}, u) + \Delta x_1^{(0)} \left. \frac{\partial g_2}{\partial x_1} \right|^{(0)} + \Delta x_2^{(0)} \left. \frac{\partial g_2}{\partial x_2} \right|^{(0)} + \dots = 0 \quad (9.32)$$

where the partial derivatives of order greater than 1 in the series of terms of the expansion have not been listed. The term $\partial g_1 / \partial x_1|^{(0)}$ indicates that the partial derivative is evaluated for the estimated values of $x_1^{(0)}$ and $x_2^{(0)}$. Other such terms are evaluated similarly.

If we neglect the partial derivatives of order greater than 1, we can rewrite Eqs. (9.31) and (9.32) in matrix form. We then have

$$\begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{bmatrix} \begin{bmatrix} \Delta x_1^{(0)} \\ \Delta x_2^{(0)} \end{bmatrix} = \begin{bmatrix} 0 - g_1(x_1^{(0)}, x_2^{(0)}, u) \\ 0 - g_2(x_1^{(0)}, x_2^{(0)}, u) \end{bmatrix} = \begin{bmatrix} b_1 - h_1(x_1^{(0)}, x_2^{(0)}, u) \\ b_2 - h_2(x_1^{(0)}, x_2^{(0)}, u) \end{bmatrix} \quad (9.33)$$

where the square matrix of partial derivatives is called the *Jacobian J* or in this case $J^{(0)}$ to indicate that the initial estimates $x_1^{(0)}$ and $x_2^{(0)}$ have been used to compute the numerical values of the partial derivatives. We note that $g_1(x_1^{(0)}, x_2^{(0)}, u)$ is the calculated value of g_1 based on the estimated values of $x_1^{(0)}$ and $x_2^{(0)}$, but this calculated value is not the zero value specified by Eq. (9.27) unless our estimated values $x_1^{(0)}$ and $x_2^{(0)}$ are correct. As before, we designate the specified value of g_1 minus the calculated value of g_1 as the mismatch $\Delta g_1^{(0)}$ and define the mismatch $\Delta g_2^{(0)}$ similarly. We then have the linear system of mismatch equations

$$J^{(0)} \begin{bmatrix} \Delta x_1^{(0)} \\ \Delta x_2^{(0)} \end{bmatrix} = \begin{bmatrix} \Delta g_1^{(0)} \\ \Delta g_2^{(0)} \end{bmatrix} \quad (9.34)$$

By solving the mismatch equations, either by triangular factorization of the Jacobian or (for very small problems) by finding its inverse, we can determine $\Delta x_1^{(0)}$ and $\Delta x_2^{(0)}$. However, since we truncated the series expansion, these values added to our initial guess do not determine for us the correct solution and we must try again by assuming new estimates $x_1^{(1)}$ and $x_2^{(1)}$, where

$$x_1^{(1)} = x_1^{(0)} + \Delta x_1^{(0)}, \quad x_2^{(1)} = x_2^{(0)} + \Delta x_2^{(0)} \quad (9.35)$$

We repeat the process until the corrections become so small in magnitude that they satisfy a chosen precision index $\epsilon > 0$; that is, until $|\Delta x_1|$ and $|\Delta x_2|$ are both less than ϵ . The concepts underlying the Newton-Raphson method are now exemplified numerically.

Example 9.4. Using the Newton-Raphson method, solve for x_1 and x_2 of the nonlinear equations

$$g_1(x_1, x_2, u) = h_1(x_1, x_2, u) - b_1 = 4ux_2 \sin x_1 + 0.6 = 0$$

$$g_2(x_1, x_2, u) = h_2(x_1, x_2, u) - b_2 = 4x_2^2 - 4ux_2 \cos x_1 + 0.3 = 0$$

Treat the parameter u as a fixed number equal to 1.0, and choose the initial conditions $x_1^{(0)} = 0$ rad and $x_2^{(0)} = 1.0$. The precision index ϵ is 10^{-5} .

Solution. Partial differentiation with respect to the x 's yields

$$J = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 4ux_2 \cos x_1 & 4u \sin x_1 \\ 4ux_2 \sin x_1 & 8x_2 - 4u \cos x_1 \end{bmatrix}$$

Here the parameter u has a fixed value equal to 1.0, but in some studies it could be treated as a specifiable or control variable.

First iteration. Setting $u = 1.0$ and using the initial estimates of x_1 and x_2 , we calculate the mismatches

$$\Delta g_1^{(0)} = 0 - g_{1, \text{calc}} = b_1 - h_1^{(0)} = -0.6 - 4 \sin(0) = -0.6$$

$$\Delta g_2^{(0)} = 0 - g_{2, \text{calc}} = b_2 - h_2^{(0)} = -0.3 - 4 \times (1.0)^2 + 4 \cos(0) = -0.3$$

which we use in Eq. (9.34) to yield the mismatch equations

$$\begin{bmatrix} 4 \cos(0) & 4 \sin(0) \\ 4 \sin(0) & 8x_2 - 4 \cos(0) \end{bmatrix} \begin{bmatrix} \Delta x_1^{(0)} \\ \Delta x_2^{(0)} \end{bmatrix} = \begin{bmatrix} -0.6 \\ -0.3 \end{bmatrix}$$

Inverting this simple 2×2 matrix, we determine the initial corrections

$$\begin{bmatrix} \Delta x_1^{(0)} \\ \Delta x_2^{(0)} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}^{-1} \begin{bmatrix} -0.6 \\ -0.3 \end{bmatrix} = \begin{bmatrix} -0.150 \\ -0.075 \end{bmatrix}$$

which provide first iteration values of x_1 and x_2 as follows:

$$x_1^{(1)} = x_1^{(0)} + \Delta x_1^{(0)} = 0.0 + (-0.150) = -0.150 \text{ rad}$$

$$x_2^{(1)} = x_2^{(0)} + \Delta x_2^{(0)} = 1.0 + (-0.075) = 0.925$$

The corrections exceed the specified tolerance, and so we continue.

Second iteration. The new mismatches are

$$\begin{bmatrix} \Delta g_1^{(1)} \\ \Delta g_2^{(1)} \end{bmatrix} = \begin{bmatrix} -0.6 - 4(0.925) \sin(-0.15) \\ -0.3 - 4(0.925)^2 + 4(0.925) \cos(-0.15) \end{bmatrix} = \begin{bmatrix} -0.047079 \\ -0.064047 \end{bmatrix}$$

and updating the jacobian, we compute the new corrections

$$\begin{bmatrix} \Delta x_1^{(1)} \\ \Delta x_2^{(1)} \end{bmatrix} = \begin{bmatrix} 3.658453 & -0.597753 \\ -0.552921 & 3.444916 \end{bmatrix}^{-1} \begin{bmatrix} -0.047079 \\ -0.064047 \end{bmatrix} = \begin{bmatrix} -0.016335 \\ -0.021214 \end{bmatrix}$$

These corrections also exceed the precision index, and so we move on to the next iteration with the new corrected values

$$x_1^{(2)} = -0.150 + (-0.016335) = -0.166335 \text{ rad}$$

$$x_2^{(2)} = 0.925 + (-0.021214) = 0.903786$$

Continuing on to the third iteration, we find that the corrections $\Delta x_1^{(2)}$ and $\Delta x_2^{(2)}$ are each smaller in magnitude than the stipulated tolerance of 10^{-4} . Accordingly, we calculate the solution

$$x_1^{(3)} = -0.166876 \text{ rad}; \quad x_2^{(3)} = 0.903057$$

The resultant mismatches are insignificant as may be easily checked.

In this example we have actually solved our first power-flow problem by the Newton-Raphson method. This is because the two nonlinear equations of the example are the power-flow equations for the simple system shown in Fig. 9.3.

$$\begin{aligned} g_1(x_1, x_2, u) &= P_2(x_1, x_2, u) - (P_{g2} - P_{d2}) \\ &= 4|V_1||V_2|\sin \delta_2 + 0.6 = 0 \end{aligned} \quad (9.36)$$

$$\begin{aligned} g_2(x_1, x_2, u) &= Q_2(x_1, x_2, u) - (Q_{g2} - Q_{d2}) \\ &= 4|V_1|^2 - 4|V_1||V_2|\cos \delta_2 + 0.3 = 0 \end{aligned} \quad (9.37)$$

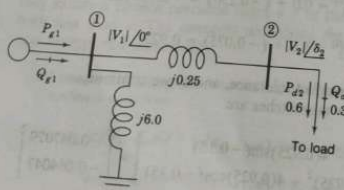


FIGURE 9.3
The system with power-flow equations corresponding to those of Example 9.4.

where x_1 represents the angle δ_2 and x_2 represents the voltage magnitude $|V_2|$ at bus ②. The control u denotes the voltage magnitude $|V_1|$ of the slack bus, and by changing its value from the specified value of 1.0 per unit, we may control the solution to the problem. In this textbook we do not investigate this control characteristic but, rather, concentrate on the application of the Newton-Raphson procedure in power-flow studies.

9.4 THE NEWTON-RAPHSON POWER-FLOW SOLUTION

To apply the Newton-Raphson method to the solution of the power-flow equations, we express bus voltages and line admittances in polar form. When n is set equal to i in Eqs. (9.6) and (9.7) and the corresponding terms are separated from the summations, we obtain

$$P_i = |V_i|^2 G_{ii} + \sum_{n=1, n \neq i}^N |V_i V_n Y_{in}| \cos(\theta_{in} + \delta_n - \delta_i) \quad (9.38)$$

$$Q_i = -|V_i|^2 B_{ii} - \sum_{n=1, n \neq i}^N |V_i V_n Y_{in}| \sin(\theta_{in} + \delta_n - \delta_i) \quad (9.39)$$

These equations can be readily differentiated with respect to voltage angles and magnitudes. The terms involving G_{ii} and B_{ii} come from the definition of Y_{ii} in Eq. (9.1) and the fact that the angle $(\delta_n - \delta_i)$ is zero when $n = i$.

Let us postpone consideration of voltage-controlled buses for now and regard all buses (except the slack bus) as load buses with known demands P_{di} and Q_{di} . The slack bus has specified values for δ_1 and $|V_1|$, and each of the other buses in the network has the two state variables δ_i and $|V_i|$ to be calculated in the power-flow solution. The known values of P_{di} and Q_{di} correspond to the negative of the b constants shown in Eqs. (9.27) and (9.28), as demonstrated in Example 9.4. At each of the nonslack buses estimated values of δ_i and $|V_i|$ correspond to the estimates $x_1^{(0)}$ and $x_2^{(0)}$ in the preceding section. Correspondence to the Δg mismatches of Eq. (9.34) follows from Eqs. (9.8) and (9.9) by writing the power mismatches for the typical load bus ①,

$$\Delta P_i = P_{i, \text{sch}} - P_{i, \text{calc}} \quad (9.40)$$

$$\Delta Q_i = Q_{i, \text{sch}} - Q_{i, \text{calc}} \quad (9.41)$$

For simplicity sake, we now write mismatch equations for a four-bus system, and it will become obvious how to extend those equations to systems with more than four buses.

Let us now bring together the results developed above in the following definitions:

Off-diagonal elements, $i \neq j$

$$M_{ij} \triangleq \frac{\partial P_i}{\partial \delta_j} = |V_i| \frac{\partial Q_i}{\partial |V_j|} \quad (9.64)$$

$$N_{ij} \triangleq \frac{\partial Q_i}{\partial \delta_j} = -|V_i| \frac{\partial P_i}{\partial |V_j|} \quad (9.65)$$

Diagonal elements, $i = j$

$$M_{ii} \triangleq \frac{\partial P_i}{\partial \delta_i} = |V_i| \frac{\partial Q_i}{\partial |V_i|} = -M_{ii} - 2|V_i|^2 B_{ii} \quad (9.66)$$

$$N_{ii} \triangleq \frac{\partial Q_i}{\partial \delta_i} = |V_i| \frac{\partial P_i}{\partial |V_i|} = N_{ii} + 2|V_i|^2 G_{ii} \quad (9.67)$$

Interrelationships among the elements in the four submatrices of the jacobian are more clearly seen if we use the definitions to rewrite Eq. (9.45) in the following form:

$$\begin{bmatrix} M_{22} & M_{23} & M_{24} \\ M_{32} & M_{33} & M_{34} \\ M_{42} & M_{43} & M_{44} \end{bmatrix} \begin{bmatrix} N_{22} + 2|V_2|^2 G_{22} & -N_{23} & -N_{24} \\ -N_{32} & N_{33} + 2|V_3|^2 G_{33} & -N_{34} \\ -N_{42} & -N_{43} & N_{44} + 2|V_4|^2 G_{44} \end{bmatrix} \begin{bmatrix} \Delta \delta_2 \\ \Delta \delta_3 \\ \Delta \delta_4 \end{bmatrix} = \begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \end{bmatrix} \quad (9.68)$$

$$\times \begin{bmatrix} \Delta |V_2|/|V_2| \\ \Delta |V_3|/|V_3| \\ \Delta |V_4|/|V_4| \end{bmatrix} = \begin{bmatrix} \Delta Q_2 \\ \Delta Q_3 \\ \Delta Q_4 \end{bmatrix}$$

So far we have regarded all nonslack buses as load buses. We now consider voltage-controlled buses also.

Voltage-controlled buses. In the polar form of the power-flow equations of the four-bus system is voltage controlled, then $|V_4|$ has a specified constant value and the voltage correction $\Delta |V_4|/|V_4|$ must always be zero. Consequently, the sixth column of the jacobian of Eq. (9.68) is always multiplied by zero, and so it may be removed. Furthermore, since Q_4 is not specified, the mismatch ΔQ_4 cannot be defined, and so we must omit the sixth row of Eq. (9.68) corresponding to Q_4 . Of course, Q_4 can be calculated after the power-flow solution becomes available.

In the general case if there are N_v voltage-controlled buses besides the slack bus, a row and column for each such bus is omitted from the polar form of the system jacobian, which then has $(2N - N_v - 2)$ rows and $(2N - N_v - 2)$ columns consistent with Table 9.1.

Example 9.5. The small power system of Example 9.2 has the line data and bus data given in Tables 9.2 and 9.3. A power-flow study of the system is to be made by the Newton-Raphson method using the polar form of the equations for P and Q . Determine the number of rows and columns in the jacobian. Calculate the initial mismatch $\Delta P_i^{(0)}$ and the initial values of the jacobian elements of the (second row, third column); of the (second row, second column); and of the (fifth row, fifth column). Use the specified values and initial voltage estimates shown in Table 9.3.

Solution. Since the slack bus has no rows or columns in the jacobian, a 6×6 matrix would be necessary if P and Q were specified for the remaining three buses. In fact, however, the voltage magnitude is specified (held constant) at bus 4, and thus the jacobian will be a 5×5 matrix. In order to calculate $P_{3,\text{calc}}$ based on the estimated and the specified voltages of Table 9.3, we need the polar form of the off-diagonal entries of Table 9.4.

$$Y_{31} = 26.359695 / 101.30993^\circ; \quad Y_{34} = 15.417934 / 101.30993^\circ$$

and the diagonal element $Y_{33} = 8.193267 - j40.863838$. Since Y_{32} and the initial values $\delta_3^{(0)}$ and $\delta_4^{(0)}$ are all zero, from Eq. (9.38) we obtain

$$\begin{aligned} P_{3,\text{calc}} &= |V_3|^2 G_{33} + |V_3 V_1 Y_{31}| \cos \theta_{31} + |V_3 V_4 Y_{34}| \cos \theta_{34} \\ &= (1.0)^2 8.193267 + (1.0 \times 1.0 \times 26.359695) \cos(101.30993^\circ) \\ &\quad + (1.0 \times 1.02 \times 15.417934) \cos(101.30993^\circ) \\ &= -0.06047 \text{ per unit} \end{aligned}$$

Scheduled real power into the network at bus ③ is -2.00 per unit, and so the initial mismatch which we are asked to calculate has the value

$$\Delta P_{3,\text{calc}} = -2.00 - (-0.06047) = -1.93953 \text{ per unit}$$

From Eq. (9.52) the (second row, third column) jacobian element is

$$\begin{aligned} \frac{\partial P_3}{\partial \delta_4} &= -|V_3 V_4 Y_{34}| \sin(\theta_{34} + \delta_4 - \delta_3) \\ &= -(1.0 \times 1.02 \times 15.417934) \sin(101.30993^\circ) \\ &= -15.420898 \text{ per unit} \end{aligned}$$

and from Eq. (9.53) the element of the (second row, second column) is

$$\begin{aligned} \frac{\partial P_3}{\partial \delta_3} &= -\frac{\partial P_3}{\partial \delta_1} - \frac{\partial P_3}{\partial \delta_2} - \frac{\partial P_3}{\partial \delta_4} \\ &= |V_3 V_1 Y_{31}| \sin(\theta_{31} + \delta_1 - \delta_3) - 0 - (-15.420898) \\ &= (1.0 \times 1.0 \times 26.359695) \sin(101.30993^\circ) + 15.420898 \\ &= 41.268707 \text{ per unit} \end{aligned}$$

For the element of the (fifth row, fifth column) Eq. (9.63) yields

$$\begin{aligned} |V_3| \frac{\partial Q_3}{\partial |V_3|} &= -\frac{\partial P_3}{\partial \delta_3} - 2|V_3|^2 B_{33} \\ &= -41.268707 - 2(1.0)^2(-40.863838) = 40.458969 \text{ per unit} \end{aligned}$$

Using the initial input data, we can similarly calculate initial values of the other elements of the jacobian and of the power mismatches at all the buses of the system.

For the system of the preceding example the numerical values for initialization of the mismatch equations are now shown, for convenience, to only three

decimal places as follows:

$$\begin{array}{c} \text{②} \quad \text{③} \quad \text{④} \quad \text{②} \quad \text{③} \\ \begin{array}{ccccc} \text{②} & 45.443 & 0 & -26.365 & 8.882 & 0 \\ \text{③} & 0 & 41.269 & -15.421 & 0 & 8.133 \\ \text{④} & -26.365 & -15.421 & 41.786 & -5.273 & -3.084 \\ \text{②} & -9.089 & 0 & 5.273 & 44.229 & 0 \\ \text{③} & 0 & -8.254 & 3.084 & 0 & 40.459 \end{array} \end{array} \begin{array}{c} \Delta \delta_2 \\ \Delta \delta_3 \\ \Delta \delta_4 \\ \Delta |V_2| \\ \Delta |V_3| \\ \Delta |V_4| \end{array}$$

$$= \begin{bmatrix} -1.597 \\ -1.940 \\ 2.213 \\ -0.447 \\ -0.835 \end{bmatrix}$$

This system of equations yields values for the voltage corrections of the first iteration which are needed to update the state variables according to Eqs. (9.49) and (9.50). At the end of the first iteration the set of updated voltages at the buses is:

Bus no. $i =$	①	②	③	④
δ_i (deg.)	0	-0.93094	-1.78790	-1.54383
$ V_i $ (per unit)	1.00	0.98335	0.97095	1.02

These updated voltages are then used to recalculate the jacobian and mismatches of the second iteration, and so on. The iterative procedure continues until either the mismatches ΔP_i and ΔQ_i become less than their stipulated allowable values or all $\Delta \delta_i$ and $\Delta |V_i|$ are less than the chosen precision index. When the solution is completed, we can use Eqs. (9.38) and (9.39) to calculate real and reactive power, P_i and Q_i , at the slack bus, and the reactive power Q_4 at voltage-controlled bus ④. Line flows can also be computed from the differences in bus voltages and the known parameters of the lines. Solved values of the bus voltages and line flows for the system of Example 9.5 are shown in Figs. 9.4 and 9.5.

The number of iterations required by the Newton-Raphson method using bus admittances is practically independent of the number of buses. The time for the Gauss-Seidel method (employing bus admittances) increases almost directly with the number of buses. On the other hand, computing the elements of the

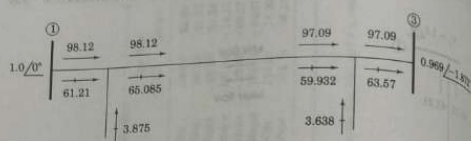


FIGURE 9.6 Single-line diagram showing flow of megawatts and megavars in the line connecting buses ① and ③ of the system of Examples 9.5 and 9.6.

Example 9.6. From the line flows shown in Fig. 9.6, calculate the current flowing in the equivalent circuit of the line from bus ① to bus ③ in the 230-kV system of Fig. 9.2. Using the calculated current and the line parameters given in Table 9.2, calculate the line I^2R loss and compare this value with the difference between the power in the line from bus ① and the power out at bus ③. Similarly, find I^2X in the line and compare with the value which could be found from the data in Fig. 9.6.

Solution. Figure 9.6 shows the megawatt and megavar flow in the per-phase equivalent circuit of line ①–③. The total megavoltampere flow through R and X of all three phases is

$$S = 98.12 + j65.085 = 117.744 \angle 33.56^\circ \text{ MVA}$$

$$\text{or } S = 97.09 + j59.932 = 114.098 \angle 31.69^\circ \text{ MVA}$$

$$\text{and } |I| = \frac{117.744}{\sqrt{3} \times 230 \times 1.0} = 295.56 \text{ A}$$

$$\text{or } |I| = \frac{114.098}{\sqrt{3} \times 230 \times 0.969} = 295.57 \text{ A}$$

The magnitude of current I in the series $R + jX$ of line ①–③ can also be calculated using $|I| = |V_1 - V_3| / |R + jX|$. The base impedance is

$$Z_{\text{base}} = \frac{(230)^2}{100} = 529 \Omega$$

and using the R and X parameters of Table 9.2, we have

$$I^2R \text{ loss} = 3 \times (295.56)^2 \times 0.00744 \times 529 \times 10^{-6} = 1.03 \text{ MW}$$

$$I^2X \text{ of line} = 3 \times (295.56)^2 \times 0.03720 \times 529 \times 10^{-6} = 5.157 \text{ Mvar}$$

These values compare with $(98.12 - 97.09) = 1.03 \text{ MW}$ and $(65.085 - 59.932) = 5.153 \text{ Mvar}$ from Fig. 9.6.

9.6 REGULATING TRANSFORMERS

As we have seen in Sec. 2.9, regulating transformers can be used to control the real and reactive power flows in a circuit. We now develop the bus admittance equations to include such transformers in power-flow studies.

Figure 9.7 is a more detailed representation of the regulating transformer of Fig. 2.24(b). The admittance Y in per unit is the reciprocal of the per-unit impedance of the transformer which has the transformation ratio $1:t$ as shown. The admittance Y is shown on the side of the ideal transformer nearest node ①, which is the tap-changing side. This designation is important in using the equations which are to be derived. If we are considering a transformer with off-nominal turns ratio, t may be real or imaginary, such as 1.02 for an approximate 2% boost in voltage magnitude or $e^{j\pi/60}$ for an approximate 3° shift per phase.

Figure 9.7 shows currents I_1 and I_2 entering the two buses, and the voltages are V_1 and V_2 referred to the reference node. The complex expressions for power into the ideal transformer from bus ① and bus ② are, respectively,

$$S_1 = V_1 I_1^* \quad S_2 = t V_2 I_2^* \quad (9.70)$$

Since we are assuming that we have an ideal transformer with no losses, the power S_1 into the ideal transformer from bus ① must equal the power $-S_2$ out of the ideal transformer on the bus ② side, and so from Eqs. (9.70) we obtain

$$I_1 = -t^* I_2 \quad (9.71)$$

The current I_2 can be expressed by

$$I_2 = (V_1 - t V_2) Y = -t V_1 Y + t V_2 Y \quad (9.72)$$

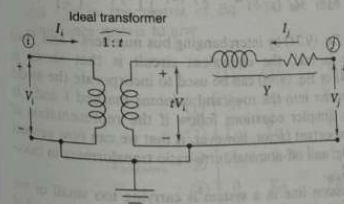


FIGURE 9.7 More detailed per-unit reactance diagram for the transformer of Fig. 2.24(b) whose turns ratio is $1:t$.

The Y_{ij} elements correspond to the parameters already in the network before the regulator is added. The vector of state variables depends on how bus (5) is treated within the power-flow model. There are two alternatives: either

- Tap t can be regarded as an independent parameter with a prescribed value before the power-flow solution begins. Bus (5) is then treated as a load bus with angle δ_5 and voltage magnitude $|V_5|$ to be determined along with the other five state variables represented in Eq. (9.68). In this case the state variable vector is

$$\mathbf{x} = [\delta_2, \delta_3, \delta_4, \delta_5, |V_2|, |V_3|, |V_4|, |V_5|]^T$$

or

- Voltage magnitude at bus (5) can be prescribed. Tap t then replaces $|V_5|$ as

$$\mathbf{x} = [\delta_2, \delta_3, \delta_4, \delta_5, |V_2|, |V_3|, |V_4|, t]^T$$

accordingly.

- In this case $\mathbf{x} = [\delta_2, \delta_3, \delta_4, \delta_5, |V_2|, |V_3|, |V_4|, t]^T$ and the Jacobian changes accordingly.

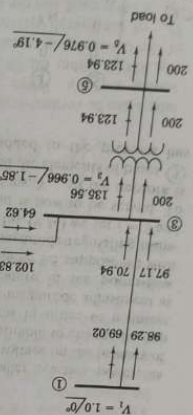


FIGURE 9.10 Flow of P and Q at bus (3) of the system of Fig. 9.5 when a regulating transformer is interposed between the bus and the load.

Example 9.7. Solve Example 2.13 using the Y_{bus} model of Eq. (9.74) for each of the two parallel transformers and compare the solution with the approximate results.

Solution. The admittance Y of each transformer is given by $1/j0.1 = -j10$ per unit. Hence, the currents in transformer T_1 of Fig. 9.11 can be determined from

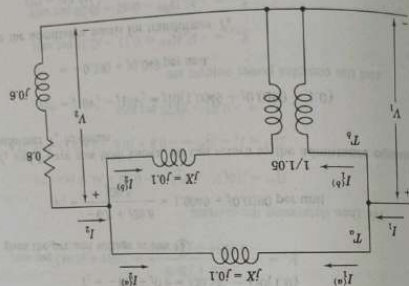


FIGURE 9.11 Circuit for Example 9.7. Values are in per unit.

the bus admittance equation

$$\begin{bmatrix} I_1^{(0)} \\ I_2^{(0)} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix} = \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix}$$

and the currents in transformer T_2 with $r = 1.05$, as shown in Fig. 9.11, are given by Eq. (9.74), which takes the numerical form

$$\begin{bmatrix} I_1^{(2)} \\ I_2^{(2)} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix} = \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix}$$

In Fig. 9.11 current $I_1 = (I_1^{(0)} + I_1^{(2)})$, and likewise, $I_2 = (I_2^{(0)} + I_2^{(2)})$, which means that the preceding two matrix equations can be directly added together

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix} = \begin{bmatrix} 1.0 \angle 0^\circ \\ -j10 \end{bmatrix}$$

In Example 2.13 V_2 is the reference voltage $1.0 \angle 0^\circ$ and the current I_2 is calculated to be $-0.8 + j0.6$. Therefore, from the second row of the preceding equation we have

$$I_2 = -0.8 + j0.6 = j20.5V_1 - j20(1.0)$$

which gives the per-unit voltage at bus ①

$$V_1 = \frac{-0.8 + j20.6}{j20.5} = 1.0049 + j0.0390 \text{ per unit}$$

Since V_1 and V_2 are now both known, we can return to the admittance equation for transformer T_2 to obtain

$$I_1^{(2)} = j10V_1 - j10V_2 = j10(1.0049 + j0.0390 - 1.0)$$

$$= -0.390 + j0.049 \text{ per unit}$$

and from the admittance matrix for transformer T_2

$$I_2^{(2)} = j10.5V_1 - j10V_2 = j10.5(1.0049 + j0.0390) - j10$$

$$= -0.41 + j0.551 \text{ per unit}$$

this example.

Again, the approximation values of Example 2.14 are close to the actual values of

$$S_{T2} = -V_2 I_2^{(2)*} = 0.67 + j0.29 \text{ per unit}$$

$$S_{T2} = -V_2 I_2^{(2)*} = 0.13 + j0.31 \text{ per unit}$$

and the complex power outputs are

$$= -0.67 + j0.29 \text{ per unit}$$

$$I_2^{(0)} = I_2 - I_2^{(2)} = -0.8 + j0.6 - (-0.13 + j0.31)$$

$$I_2^{(2)} = j10(V_1 - V_2) = -0.13 + j0.31 \text{ per unit}$$

We then determine the currents

$$V_1 = \frac{-0.5234 + j19.9863}{-0.8 + j20.6} = 1.031 + j0.013 \text{ per unit}$$

which yields the voltage at bus ①.

$$-0.8 + j0.6 = (-0.5234 + j19.9863)V_1 - j20(1.0)$$

Following the procedure of Example 9.7, we have

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -0.5234 + j19.9863 \\ -20.0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

in Example 9.7 to obtain the admittance equation for transformer T_2 , given

$$\begin{bmatrix} I_1^{(2)} \\ I_2^{(2)} \end{bmatrix} = \begin{bmatrix} -j10/1.0 & 10/87^\circ \\ 10/93^\circ & -j10 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

which can be added directly to the admittance equation for transformer T_2 , with

$$r = r_{T2}/60 = 1.0/\sqrt{3}^\circ \text{ is given by Eq. (9.74) as}$$

Example 9.8. Solve the phase-shifting problem of Example 2.14 by the exact Y_{bus} model of Eq. (9.74) and compare results.

Solution. The bus admittance equation for the phase-shifting transformer T_2 with

$$S_{T2} = -V_2 I_2^{(2)*} = 0.39 + j0.049 \text{ per unit}$$

$$S_{T2} = -V_2 I_2^{(2)*} = 0.41 + j0.551 \text{ per unit}$$

Hence, the complex power outputs of the transformers are

Using this decoupled version of the Newton-Raphson procedure, faster power-flow solutions may be found within a specified degree of precision.

Example 9.9. Using the decoupled form of the Newton-Raphson method, determine the first-iteration solution to the power-flow problem of Example 9.8. *Solution.* The B matrix can be read directly from Table 9.4 and the mismatches corresponding to the initial voltage estimates are already calculated in Example 9.8 so that Eq. (9.90) becomes

$$\begin{bmatrix} 44.835953 & 0 & -25.847809 \\ 0 & 40.863838 & -15.118528 \\ -25.847809 & -15.118528 & 40.863838 \end{bmatrix} \begin{bmatrix} \Delta \delta_1 \\ \Delta \delta_2 \\ \Delta \delta_3 \end{bmatrix} = \begin{bmatrix} -1.59661 \\ -1.93953 \\ 2.21286 \end{bmatrix}$$

Solving this equation gives the angle corrections in radians

$$\Delta \delta_1 = -0.02057; \quad \Delta \delta_2 = -0.03781; \quad \Delta \delta_3 = 0.02609$$

Adding these results to the flat-start estimates of Table 9.3 gives the updated values of δ_1 , δ_2 , and δ_3 , which we then use along with the elements of Y_{bus} to calculate the reactive mismatches

$$\frac{\Delta Q_1}{|V_1|} = \frac{1}{|V_1|} (Q_{1,kb} - Q_{1,cak})$$

$$= \frac{1}{|V_1|} \left[Q_{1,kb} - [-|V_1|^2 B_{11} - |Y_{12} V_1 V_2| \sin(\theta_{12} + \delta_1 - \delta_2)] \right]$$

$$= \frac{1}{|V_1|} \left[Q_{1,kb} - [-1.0535 + 1.0^2 (-44.835953) + 19.455965 \times 1.02 \sin(101.30993 \times \pi/180 + 0 + 0.02057) + 26.359695 \times 1.02 \sin(101.30993 \times \pi/180 + 0 + 0.03781) + 15.417934 \times 1.02 \sin(101.30993 \times \pi/180 + 0.02609 + 0.03781)] \right]$$

$$= -0.80370 \text{ per unit}$$

$$\frac{\Delta Q_2}{|V_2|} = \frac{1}{|V_2|} (Q_{2,kb} - Q_{2,cak})$$

$$= \frac{1}{|V_2|} \left[Q_{2,kb} - [-|V_2|^2 B_{22} - |Y_{21} V_1 V_2| \sin(\theta_{21} + \delta_2 - \delta_1) - |Y_{23} V_2 V_3| \sin(\theta_{23} + \delta_2 - \delta_3)] \right]$$

$$= -1.27684 \text{ per unit}$$

A reactive mismatch calculation is not required for bus 3, which is voltage controlled. Accordingly, in this example Eq. (9.91) becomes

$$\begin{bmatrix} 44.835953 & 0 & 0 \\ 0 & 40.863838 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_3 \end{bmatrix} = \begin{bmatrix} -0.80370 \\ -1.27684 \\ 0 \end{bmatrix}$$

which yields the solutions $\Delta V_1 = -0.01793$ and $\Delta V_2 = -0.03125$. The new voltage magnitudes at buses 1 and 2 are $|V_1| = 0.98207$ and $|V_2| = 0.96875$, which completes the first iteration. Updated mismatches for the second iteration of Eq. (9.90) are calculated using the new voltage values. Repeating the procedure over a number of iterations yields the same solution as tabulated in Fig. 9.4.

Often in industry-based programs certain modifications are made in Eqs. (9.90) and (9.91). The modifications to B in Eq. (9.91) are generally as follows:

• Omit the angle-shifting effects of phase shifters from B by setting $t = 1.0/0^\circ$. When rows and columns for voltage-controlled buses are also omitted as previously indicated, the resulting matrix is called B' .

The coefficient matrix in Eq. (9.90) is generally modified as follows:

• Omit from B those elements that mainly affect megawatt flows such as shunt capacitors and reactors, and set taps t of off-nominal transformers equal to 1. Also, ignore series resistances in the equivalent- π circuits of the transmission lines in forming Y_{bus} from which B in Eq. (9.90) is obtained. The resulting matrix is called B'' .

When B in Eq. (9.90) is replaced by B' , the model becomes a lossless network. If, in addition, all bus voltages are assumed to remain constant at nominal values of 1.0 per unit, the so-called dc power-flow model results. Under these additional assumptions, Eq. (9.91) is not necessary (since $\Delta V_i = 0$ at each bus ①) and Eq. (9.90) for the dc power flow becomes

$$\begin{bmatrix} -B_{12} & -B_{13} & -B_{14} \\ -B_{22} & -B_{23} & -B_{24} \\ -B_{32} & -B_{33} & -B_{34} \\ -B_{42} & -B_{43} & -B_{44} \end{bmatrix} \begin{bmatrix} \Delta \delta_1 \\ \Delta \delta_2 \\ \Delta \delta_3 \end{bmatrix} = \begin{bmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \end{bmatrix} \quad (9.92)$$

where it is understood that the elements of B' are calculated assuming all lines are lossless. Dc power-flow analysis can be used where approximate solutions are acceptable, as in the contingency studies discussed in Chap. 14.

and I_L in the equations

$$E_g'' = V_f + (Z_{ext} + jX_{dg}'')I_L \quad (10.7)$$

$$E_m'' = V_f - jX_{dm}''I_L \quad (10.8)$$

When the fault is on the system, as shown in Fig. 10.3(b), the subtransient currents I_g'' out of the generator and I_m'' out of the motor are found from the relations

$$I_g'' = \frac{E_g''}{Z_{ext} + jX_{dg}''} = \frac{V_f}{Z_{ext} + jX_{dg}''} + I_L \quad (10.9)$$

$$I_m'' = \frac{E_m''}{jX_{dm}''} = \frac{V_f}{jX_{dm}''} - I_L \quad (10.10)$$

These two currents add together to give the total symmetrical fault current I_f'' shown in Fig. 10.3(b). That is,

$$I_f'' = I_g'' + I_m'' = \frac{V_f}{Z_{ext} + jX_{dg}''} + \frac{V_f}{jX_{dm}''} \quad (10.11)$$

where I_g'' and I_m'' are the respective contributions of the generator and motor to the fault current I_f'' . Note that the fault current does not include the prefault (load) current.

The alternative approach using Thévenin's theorem is based on the observation that Eq. (10.11) requires a knowledge of only V_f , the prefault voltage of the fault point, and the parameters of the network with the subtransient reactances representing the machines. Therefore, I_f'' and the additional currents produced throughout the network by the fault can be found simply by applying voltage V_f to the fault point P in the dead subtransient network of the system, as shown in Fig. 10.4(a). If we redraw that network as shown in Fig. 10.4(b), it becomes clear that the symmetrical values of the subtransient fault currents can be found from the Thévenin equivalent circuit of the subtransient network at the fault point. The Thévenin equivalent circuit is a single generator and a single impedance terminating at the point of application of the fault. The equivalent generator has an internal voltage equal to V_f , the voltage at the fault point before the fault occurs. The impedance is that measured at the point of application of the fault looking back into the circuit with all the generated voltages short-circuited. Subtransient reactances are used since the initial sym-

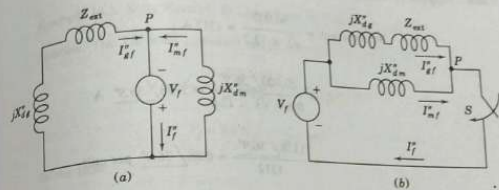


FIGURE 10.4 Circuits illustrating the additional current flows due to the three-phase fault at P: (a) applying V_f to dead network to simulate the fault; (b) Thévenin equivalent looking into the circuit at point P.

metrical fault current is desired. In Fig. 10.4(b) the Thévenin impedance Z_{th} is

$$Z_{th} = \frac{jX_{dg}''(Z_{ext} + jX_{dm}'')}{Z_{ext} + j(X_{dg}'' + X_{dm}'')} \quad (10.12)$$

Upon the occurrence of a three-phase short circuit at P, simulated by closing switch S, the subtransient current in the fault is

$$I_f'' = \frac{V_f}{Z_{th}} = \frac{V_f [Z_{ext} + j(X_{dg}'' + X_{dm}'')]}{jX_{dg}''(Z_{ext} + jX_{dm}'')} \quad (10.13)$$

Thus, three-phase symmetrical faults on systems containing generators and motors under load may be analyzed by either the use of subtransient internal voltages or Thévenin's theorem, as illustrated in the following examples.

Example 10.1. A synchronous generator and motor are rated 30,000 kVA, 13.2 kV, and both have subtransient reactances of 20%. The line connecting them has a reactance of 10% on the base of the machine ratings. The motor is drawing 20,000 kW at 0.8 power-factor leading and a terminal voltage of 12.8 kV when a symmetrical three-phase fault occurs at the motor terminals. Find the subtransient currents in the generator, the motor, and the fault by using the internal voltages of the machines.

Solution. The prefault equivalent circuit of the system corresponds to Fig. 10.3(a). Choosing a base of 30,000 kVA, 13.2 kV and using the voltage V_f at the fault point as the reference phasor, we obtain

$$V_f = \frac{12.8}{13.2} = 0.970 \angle 0^\circ \text{ per unit}$$

$$\text{Base current} = \frac{30,000}{\sqrt{3} \times 13.2} = 1312 \text{ A}$$

$$I_L = \frac{20,000 / 36.9^\circ}{0.8 \times \sqrt{3} \times 12.8} = 1128 \angle 36.9^\circ \text{ A}$$

$$= \frac{1128 \angle 36.9^\circ}{1312} = 0.86 \angle 36.9^\circ \text{ per unit}$$

$$= 0.86(0.8 + j0.6) = 0.69 + j0.52 \text{ per unit}$$

For the generator

$$V_f = 0.970 + j0.1(0.69 + j0.52) = 0.918 + j0.069 \text{ per unit}$$

$$E_g'' = 0.918 + j0.069 + j0.2(0.69 + j0.52) = 0.814 + j0.207 \text{ per unit}$$

$$I_g'' = \frac{0.814 + j0.207}{j0.3} = 0.69 - j2.71 \text{ per unit}$$

$$= 1312(0.69 - j2.71) = 905 - j3550 \text{ A}$$

For the motor

$$V_f = V_g = 0.970 \angle 0^\circ \text{ per unit}$$

$$E_m'' = 0.970 + j0 - j0.2(0.69 + j0.52) = 0.970 - j0.138 + 0.104 \text{ per unit}$$

$$= 1.074 - j0.138 \text{ per unit}$$

$$I_m'' = \frac{1.074 - j0.138}{j0.2} = -0.69 - j5.37 \text{ per unit}$$

$$= 1312(-0.69 - j5.37) = -905 - j7050 \text{ A}$$

In the fault

$$I_f'' = I_g'' + I_m'' = 0.69 - j2.71 - 0.69 - j5.37 = -j8.08 \text{ per unit}$$

$$= -j8.08 \times 1312 = -j10,600 \text{ A}$$

Figure 10.3(b) shows the paths of I_g'' , I_m'' , and I_f'' .

Example 10.2. Solve Example 10.1 by the use of Thévenin's theorem.

Solution. The Thévenin equivalent circuit corresponds to Fig. 10.4.

$$Z_m = \frac{j0.3 \times j0.2}{j0.3 + j0.2} = j0.12 \text{ per unit}$$

$$V_f = 0.970 \angle 0^\circ \text{ per unit}$$

In the fault

$$I_f'' = \frac{V_f}{Z_m} = \frac{0.97 + j0}{j0.12} = -j8.08 \text{ per unit}$$

This fault current is divided between the parallel circuits of the machines inversely as their impedances. By simple current division we obtain the fault currents

$$\text{From generator: } I_{gf}'' = -j8.08 \times \frac{j0.2}{j0.5} = -j3.23 \text{ per unit}$$

$$\text{From motor: } I_{mf}'' = -j8.08 \times \frac{j0.3}{j0.5} = -j4.85 \text{ per unit}$$

Neglecting load current gives

$$\text{Fault current from generator} = 3.23 \times 1312 = 4240 \text{ A}$$

$$\text{Fault current from motor} = 4.85 \times 1312 = 6360 \text{ A}$$

$$\text{Current in fault} = 8.08 \times 1312 = 10,600 \text{ A}$$

The current in the fault is the same whether or not load current is considered, but the currents in the lines differ. When load current I_L is included, we find from Example 10.1 that

$$I_g'' = I_{gf}'' + I_L = -j3.23 + 0.69 + j0.52 = 0.69 - j2.71 \text{ per unit}$$

$$I_m'' = I_{mf}'' - I_L = -j4.85 - 0.69 - j0.52 = -0.69 - j5.37 \text{ per unit}$$

Note that I_L is in the same direction as I_g'' but opposite to I_m'' . The per-unit values found for I_g'' , I_{gf}'' , and I_m'' are the same as in Example 10.1, and so the ampere values will also be the same.

$$\text{Fault current from generator} = |905 - j3550| = 3600 \text{ A}$$

$$\text{Fault current from motor} = |-905 - j7050| = 7200 \text{ A}$$

The sum of the magnitudes of the generator and motor currents does not equal the fault current because the currents from the generator and motor are not in phase when load current is included.

The discussion of this section shows that only column k of Z_{bus} , which is now denote by Z_{bk} , is required to evaluate the impact on the system of a symmetrical three-phase fault at bus (k) . If necessary, the elements of Z_{bk} be generated from the triangular factors of Y_{bus} as demonstrated in Sec. 8.1.

Example 10.3. A three-phase fault occurs at bus (2) of the network of Fig. 10. Determine the initial symmetrical rms current (that is, the subtransient current) in the fault, the voltages at buses (1), (3), and (4) during the fault, the current in the line from bus (3) to bus (1); and the current contributions to the fault in lines (3)-(2), (1)-(2), and (4)-(2). Take the prefault voltage V_f at bus (1) equal to $1.0/\angle 0^\circ$ per unit and neglect all prefault currents.

Solution. Applying the Z_{bus} building algorithm to Fig. 10.5, we find that

$$Z_{bus} = \begin{matrix} & \begin{matrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \end{matrix} \\ \begin{matrix} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{matrix} & \begin{bmatrix} j0.2436 & j0.1938 & j0.1544 & j0.1456 \\ j0.1938 & j0.2295 & j0.1494 & j0.1506 \\ j0.1544 & j0.1494 & j0.1954 & j0.1046 \\ j0.1456 & j0.1506 & j0.1046 & j0.1954 \end{bmatrix} \end{matrix}$$

Since load currents are neglected, the prefault voltage at each bus is $1.0/\angle 0^\circ$ per unit, the same as V_f at bus (2). When the fault occurs,

$$I_f' = \frac{1.0}{Z_{22}} = \frac{1.0}{j0.2295} = -j4.3573 \text{ per unit}$$

and from Eq. (10.18) the voltages during the fault are

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 1 - j0.1938 \\ 0 \\ 1 - j0.1494 \\ 1 - j0.2295 \end{bmatrix} \begin{bmatrix} 0.1556 \\ 0 \\ 0.3490 \\ 0.3438 \end{bmatrix} \text{ per unit}$$

The current flow in line (3)-(1) is

$$I_{31} = \frac{V_3 - V_1}{Z_{b6}} = \frac{0.3490 - 0.1556}{j0.25} = -j0.7736 \text{ per unit}$$

Fault currents contributed to bus (2) by the adjacent unfaulted buses are

$$\text{From bus (1): } \frac{V_1}{Z_{k1}} = \frac{0.1556}{j0.125} = -j1.2448 \text{ per unit}$$

$$\text{From bus (3): } \frac{V_3}{Z_{k3}} = \frac{0.3490}{j0.25} = -j1.3960 \text{ per unit}$$

$$\text{From bus (4): } \frac{V_4}{Z_{k4}} = \frac{0.3438}{j0.20} = -j1.7190 \text{ per unit}$$

Except for round-off errors, the sum of these current contributions equals I_f' .

10.4 FAULT CALCULATIONS USING Z_{bus} EQUIVALENT CIRCUITS

We cannot devise a physically realizable network which directly incorporates all the individual elements of the bus impedance matrix. However, Fig. 8.4 shows that we can use the matrix elements to construct the Thevenin equivalent circuit between any pair of buses in the network that may be of interest. The Thevenin equivalent circuit is very helpful for illustrating the symmetrical fault equations, which are developed in Sec. 10.3.

In the Thevenin equivalent circuit of Fig. 10.7(a) bus (k) is assumed to be the fault bus and bus (i) is unfaulted. The impedances shown correspond directly to the elements of the network Z_{bus} and all the prefault bus voltages are the same as V_f of the fault bus if load currents are neglected. The two points

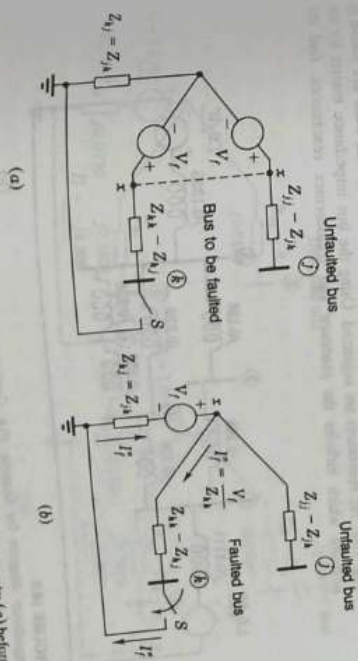


FIGURE 10.7 Thevenin equivalent between buses (i) and (k) of system with no prefault load currents: (a) before the fault (S open); (b) during the fault (S closed).

marked x have the same potential, and so they can be joined together to give the equivalent circuit of Fig. 10.7(b) with a single voltage source V_f as shown. If the switch S is open between bus (k) and the reference node, there is no short circuit and no current flows in any branch of the network. When S is closed, it represents the fault on bus (k), current flows in the circuit toward bus (k). The current is $I_f = V_f / Z_{kk}$, which agrees with Eq. (10.19), and it induces a voltage drop $(Z_{kf} / Z_{kk}) V_f$ in the direction from the reference node toward bus (k). The voltage from bus (k) to the reference changes therefore by the amount $-(Z_{kf} / Z_{kk}) V_f$, so that the voltage at bus (k) during the fault is $V_f - (Z_{kf} / Z_{kk}) V_f$, which is consistent with Eq. (10.20).

Thus, by substituting appropriate numerical values for the impedances in the simple equivalent circuit of Fig. 10.7(b), we can calculate the bus voltages of the system before and after the fault occurs. With switch S open in the circuit the voltages at bus (k) and the representative bus (l) are equal to V_f . The same uniform voltage profile occurs in Fig. 10.6 if there are no prefault currents so that E_g and E_g equal V_f . If S is closed in Fig. 10.7(b), the circuit reflects the voltage of representative bus (l) with respect to reference while the fault is on bus (k). Therefore, if a three-phase short-circuit fault occurs at bus (k) of a large-scale network, we can calculate the current in the fault and the voltage at any of the unfaulted buses simply by inserting the proper impedance values into elementary circuits like those in Fig. 10.7. The following example illustrates the procedure.

Example 10.4. A five-bus network has generators at buses (1) and (3) rated 270 and 225 MVA, respectively. The generator subtransient reactances plus the reactances of the transformers connecting them to the buses are each 0.30 per unit on the generator rating as base. The turns ratios of the transformers are such that the voltage base in each generator circuit is equal to the voltage rating of the generator. Line impedances in per unit on a 100-MVA system base are shown in Fig. 10.8. All resistances are neglected. Using the bus impedance matrix for the network which includes the generator and transformer reactances, find the

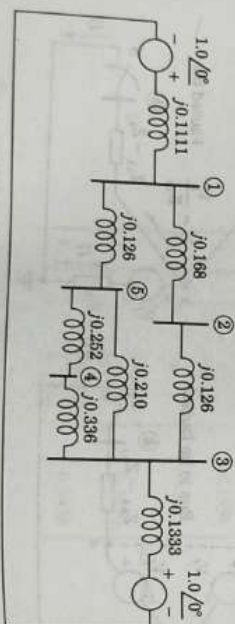


FIGURE 10.8 Impedance diagram for Example 10.4. Generator reactances include subtransient values plus reactances of set-up transformers. All values in per unit on a 100-MVA base.

subtransient current in a three-phase fault at bus (4) and the current coming to the faulted bus over each line. Prefault current is to be neglected and all voltages are assumed to be 1.0 per unit before the fault occurs.

Solution. Converted to the 100-MVA base, the combined generator and transformer reactances are

$$\text{Generator at bus (1): } X = 0.30 \times \frac{100}{270} = 0.1111 \text{ per unit}$$

$$\text{Generator at bus (3): } X = 0.30 \times \frac{100}{225} = 0.1333 \text{ per unit}$$

These values, along with the line impedances, are marked in per unit in Fig. 10.8 from which the bus impedance matrix can be determined by the Z_{bus} building algorithm to yield

$$Z_{bus} = \begin{bmatrix} \text{(1)} & \text{(2)} & \text{(3)} & \text{(4)} & \text{(5)} \\ \text{(2)} & j0.0793 & j0.0558 & j0.0382 & j0.0511 & j0.0608 \\ \text{(3)} & j0.0558 & j0.1338 & j0.0664 & j0.0630 & j0.0605 \\ \text{(4)} & j0.0382 & j0.0664 & j0.0875 & j0.0720 & j0.0603 \\ \text{(5)} & j0.0511 & j0.0630 & j0.0720 & j0.2321 & j0.1002 \\ \text{(6)} & j0.0608 & j0.0605 & j0.0603 & j0.1002 & j0.1301 \end{bmatrix}$$

Since we are to calculate the currents from buses (3) and (5) into the fault at bus

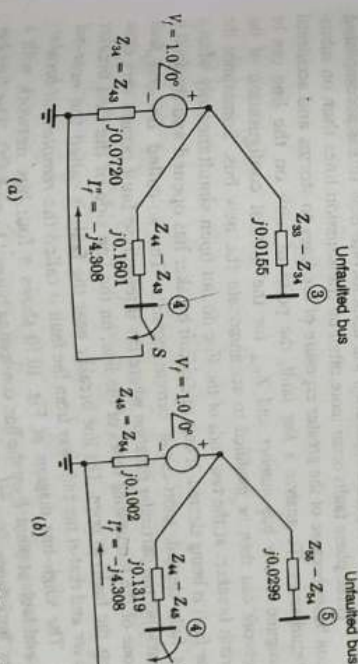


FIGURE 10.9 Use of Thevenin equivalent circuits to calculate voltages at (a) bus (3) and (b) bus (4) due to fault at bus (5).

reducing the matrix Z to obtain the new 5×5 matrix $Z_{bus, new}$ including bus ① as explained for Case 4 of Table 8.1. However, since $Z_{kk, new}$ is the only element required to calculate the current in the fault at bus ② (point P of Fig. 10.10), we can save work by observing from Eq. (10.22) that the Kron reduction line gives

$$Z_{kk, new} = Z_{11} + Z_b - \frac{(Z_{11} - Z_{21})^2}{Z_{11} - Z_{21} - Z_b} \quad (10.23)$$

Again, we note that $Z_{12} = Z_{21}$ and $Z_{1b,12} = Z_{11} + Z_{22} - 2Z_{12}$. By neglecting prefault currents and assigning prefault voltage $V_f = 1.0/\angle 0^\circ$ per unit to the fault point P , we find the line-end fault current I_f' out of bus ② as follows:

$$I_f' = \frac{1.0}{Z_{kk, new}} = \frac{1.0}{Z_{11} + Z_b - (Z_{11} - Z_{21})^2 / (Z_{11} - Z_{21} - Z_b)} \quad (10.24)$$

Thus, the only elements of Z_{orig} entering into the calculation of I_f' are $Z_{11} = Z_{21}$, and Z_{22} .

It is worthwhile observing that the same equation for the line-end fault current can be found directly by inspection of Fig. 10.11(a), which shows the Thevenin equivalent circuit between buses ① and ② of the prefault network. The impedances Z_b and $-Z_b$ are connected as shown in accordance with steps 1 and 2 above. Circuit analysis then shows in a straightforward manner that the impedance looking back into the circuit from the terminals of the open switch S is

$$Z_{kk, new} = Z_b + \frac{(Z_{11} - Z_{12})(Z_{22} - Z_{21} - Z_b)}{Z_{11} - Z_{12} + Z_{22} - Z_{21} - Z_b} + Z_{12} \quad (10.25)$$

Since $Z_{12} = Z_{21}$ and $Z_{1b,12} = Z_{11} + Z_{22} - 2Z_{12}$, Eq. (10.25) can be reduced to give

$$\begin{aligned} Z_{kk, new} &= Z_b + \frac{(Z_{11} - Z_{12})(Z_{1b,12} - Z_b) - (Z_{11} - Z_{12})^2}{Z_{1b,12} - Z_b} + Z_{12} \\ &= Z_{11} + Z_b - \frac{(Z_{11} - Z_{21})^2}{Z_{1b,12} - Z_b} \end{aligned} \quad (10.26)$$

Thus, simply by closing switch S as shown in Fig. 10.11(b) and using elementary circuit analysis, we can calculate the line-end fault current I_f' in agreement with Eq. (10.24). Of course, the circuit approach using the Thevenin equivalent must yield the same results as the matrix manipulations of Eq. (10.22)—for the same

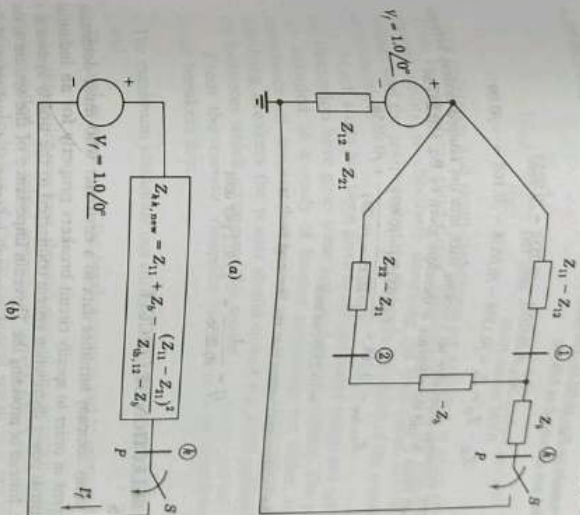


FIGURE 10.11 Simulating the line-end fault of Fig. 10.10 by Thevenin equivalent circuit: (a) with line ①–② open before the fault; (b) during the fault (S closed).

external connections are being made to the larger system model as to its Thevenin equivalent.

Other uses of the equivalent circuits based on the bus impedance matrix are possible.

Example 10.5. In the five-bus system of Fig. 10.8 a line-end, short-circuit fault occurs on line ①–②, on the line side of the breaker at bus ②. Neglecting prefault currents and assuming rated system voltage at the fault point, calculate the subtransient current into the fault when only the near-end breaker at bus ② opens.

Solution. Figure 10.8 shows that the impedance of line ①–② is $Z_{12} = j0.168$ per unit and the required elements of Z_{bus} are given in Example 10.4. The Thevenin equivalent circuit looking into the *inter* system between buses ① and ②

corresponds to Fig. 10.11(a). The numerical values of the impedances shown in parallel are calculated as follows:

$$Z_{11} = j0.0793 - j0.0558 = j0.0235$$

$$Z_{22} - Z_{21} - Z_0 = j0.1338 - j0.0558 - j0.168 = -j0.09$$

The new Thévenin impedance seen looking back into the faulted system from the fault point P and the reference is therefore given by Eq. (10.25) as

$$Z_{th, new} = j0.168 + \frac{(j0.0235)(-j0.09)}{(j0.0235 - j0.09)} + j0.0558 \\ = j0.2556 \text{ per unit}$$

Thus, the subtransient current into the line-end fault is

$$I_f = \frac{1}{j0.2556} = -j3.912 \text{ per unit}$$

10.5 THE SELECTION OF CIRCUIT BREAKERS

The electric utility company furnishes data to a customer who must determine the fault current in order to specify circuit breakers properly for an industrial plant or industrial power distribution system connected to the utility system at a certain point. Instead of providing the Thévenin impedance of the system at the point of connection, usually the power company informs the customer of the short-circuit megavoltamperes which can be expected at nominal voltage; that is,

$$\text{Short-circuit MVA} = \sqrt{3} \times (\text{nominal kV}) \times |I_{sc}| \times 10^{-3} \quad (10.27)$$

where $|I_{sc}|$ in amperes is the rms magnitude of the short-circuit current in a three-phase fault at the connection point. Base megavoltamperes are related to base kilovolts and base amperes $|I_{base}|$ by

$$\text{Base MVA} = \sqrt{3} \times (\text{base kV}) \times |I_{base}| \times 10^{-3} \quad (10.28)$$

If base kilovolts equal nominal kilovolts, then dividing Eq. (10.27) by Eq. (10.28) converts the former to per unit, and we obtain

$$\text{Short-circuit MVA in per unit} = |I_{sc}| \text{ in per unit} \quad (10.29)$$

At nominal voltage the Thévenin equivalent circuit looking back into the system from the point of connection is an emf of $1.0 \angle 0^\circ$ per unit in series with the

per-unit impedance Z_{th} . Therefore, under short-circuit conditions,

$$|Z_{th}| = \frac{1.0}{|I_{sc}|} \text{ per unit} = \frac{1.0}{\text{short-circuit MVA}} \text{ per unit} \quad (10.30)$$

Often resistance and shunt capacitance are neglected, in which case $Z_{th} = X_{th}$. Thus, by specifying short-circuit megavoltamperes at the customer's bus, the electric utility is effectively describing the short-circuit current at nominal voltage and the reciprocal of the Thévenin impedance of the system at the point of connection.

Much study has been given to circuit-breaker ratings and applications, and our discussion here gives some introduction to the subject. The presentation is not intended as a study of breaker applications but, rather, to indicate the importance of understanding fault calculations. For additional guidance in specifying breakers the reader should consult the ANSI publications listed in the footnotes which accompany this section.

From the current viewpoint two factors to be considered in selecting circuit breakers are:

- The maximum instantaneous current which the breaker must carry (*withstand*) and
- The total current when the breaker contacts part to *interrupt* the circuit.

Up to this point we have devoted most of our attention to the subtransient current called the *initial symmetrical current*, which does not include the dc component. Inclusion of the dc component results in a rms value of current immediately after the fault, which is higher than the subtransient current. For oil circuit breakers above 5 kV the subtransient current multiplied by 1.6 is considered to be the rms value of the current whose disruptive forces the breaker must withstand during the first half cycle after the fault occurs. This current is called the *momentary current*, and for many years circuit breakers were rated in terms of their momentary current as well as other criteria.¹

The *interrupting rating* of a circuit breaker was specified in kilovoltamperes or megavoltamperes. The interrupting kilovoltamperes equal $\sqrt{3} \times$ (the kilovolts of the bus to which the breaker is connected) $\times$ (the current which the breaker must be capable of interrupting when its contacts part). This *interrupting current* is, of course, lower than the momentary current and depends on the speed of the breaker, such as 8, 5, 3, or 2 cycles, which is a measure of the time from the occurrence of the fault to the extinction of the arc. Breakers of different speeds are classified by their *rated interrupting times*. The rated

¹See G. N. Lester, "High Voltage Circuit Breaker Standards in the USA: Past, Present, and Future," *IEEE Transactions on Power Apparatus and Systems*, vol. 93, 1974, pp. 590-600.

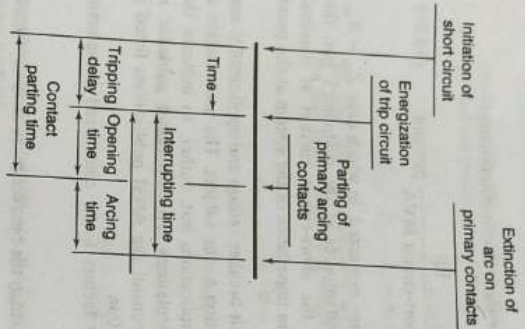


FIGURE 10.12
Definition of interrupting time given in ANSI/IEEE Standard C37.101-1979, *Application Guide for AC High Voltage Circuit Breakers Rated on a Symmetrical Current Basis*.

interrupting time of a circuit breaker is the period between the instant of energizing the trip circuit and the arc extinction on an opening operation, Fig. 10.12. Preceding this period is the *tripping delay time*, which is usually assumed to be $\frac{1}{2}$ cycle for relays to pick up.

The current which a breaker must interrupt is usually asymmetrical since it still contains some of the decaying dc component. A schedule of preferred ratings for ac high-voltage oil circuit breakers specifies the interrupting current ratings of breakers in terms of the component of the asymmetrical current which is symmetrical about the zero axis. This current is properly called the *required symmetrical interrupting capability* or simply the *rated symmetrical short-circuit current*. Often the adjective *symmetrical* is omitted. Selection of circuit breakers may also be made on the basis of total current (dc component included).² We shall limit our discussion to a brief treatment of the symmetrical basis of breaker selection.

²See *Preferred Ratings and Related Required Capabilities for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis*, ANSI C37.106-1987, and *Guide for Calculation of Fault Currents for Application of AC High-Voltage Circuit Breakers Rated on a Total Current Basis*, ANSI C37.5-1978, American National Standards Institute, New York.

Breakers are identified by nominal-voltage class, such as 69 kV. Among other factors specified are rated continuous current, rated maximum voltage, voltage range factor K , and rated short-circuit current at rated maximum voltages. The *rated maximum voltage* of a circuit breaker is the highest rms voltage for which the circuit breaker is designed. The *rated voltage range factor* K is the ratio (rated maximum voltage $\div$ the lower limit of the range of operating voltage). K determines the range of voltage over which the product (rated short-circuit current $\times$ operating voltage) is constant. In the application of circuit breakers it is important not to exceed the short-circuit capabilities of breakers. A breaker is required to have a *maximum symmetrical interrupting capability* equal to $K \times$ rated short-circuit current. Between the rated maximum voltage and $1/K$ times the rated maximum voltage the *symmetrical interrupting capability* is defined as the product [rated short-circuit current $\times$ (rated maximum voltage/operating voltage)].

Example 10.6. A 69-kV circuit breaker having a voltage range factor K of 1.21 and a continuous current rating of 1200 A has a rated short-circuit current of 19,000 A at the maximum rated voltage of 72.5 kV. Determine the maximum symmetrical interrupting capability of the breaker and explain its significance at lower operating voltages.

Solution. The maximum symmetrical interrupting capability is given by

$$K \times \text{rated short-circuit current} = 1.21 \times 19,000 = 22,990 \text{ A}$$

This value of symmetrical interrupting current must not be exceeded. From the definition of K we have

$$\text{Lower limit of operating voltage} = \frac{\text{rated maximum voltage}}{K} = \frac{72.5}{1.21} \approx 60 \text{ kV}$$

Hence, in the operating voltage range 72.5–60 kV, the symmetrical interrupting current may exceed the rated short-circuit current of 19,000 A, but it is limited to 22,990 A. For example, at 66 kV the interrupting current can be

$$\frac{72.5}{66} \times 19,000 = 20,871 \text{ A}$$

Breakers of the 115-kV class and higher have a K of 1.0.

A simplified procedure for calculating the symmetrical short-circuit current, called the *E/X method*,³ disregards all resistance, all static load, and all

³See *Application Guide for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis*, ANSI C37.101-1979, American National Standards Institute, New York. This publication is also IEEE Std. 320-1979.

prefault current. Subtransient reactance is used for generators in the method, and for synchronous motors the recommended reactance in the motor times 1.5, which is the approximate value of the transient reactance X_d' of the motor. Induction motors below 50 hp are neglected, and multiplying factors are applied to the X_d'' of larger induction motors depending on their size. If no motors are present, symmetrical short-circuit current equals subtransient current.

The impedance by which the voltage V_f at the fault is divided to find short-circuit current must be examined when the E/X method is used to find specifying a breaker for bus (1), this impedance is Z_{1f} of the bus impedance matrix with the proper machine reactances since the short-circuit current is expressed by Eq. (10.19). If the ratio of X/R of this impedance is 15 or less, a breaker of the correct voltage and kilovoltamperes may be used if its interrupting current rating is equal to or exceeds the calculated current. If the X/R ratio is unknown, the calculated current should be no more than 80% of the allowed value for the breaker at the existing bus voltage. The ANSI application guide specifies a corrected method to account for ac and dc time constants for the decay of the current amplitude if the X/R ratio exceeds 15. The corrected method also considers breaker speed.

Example 10.7. A 25,000-kVA 13.8-kV generator with $X_d'' = 15\%$ is connected through a transformer to a bus which supplies four identical motors, as shown in Fig. 10.13. The subtransient reactance X_d'' of each motor is 20% on a base of 500 kVA, 6.9 kV. The three-phase rating of the transformer is 25,000 kVA, 13.8/6.9 kV, with a leakage reactance of 10%. The bus voltage at the motors is 6.9 kV when a three-phase fault occurs at point P. For the fault specified, determine (a) the subtransient current in the fault, (b) the subtransient current in breaker A, and (c) the symmetrical short-circuit interrupting current (as defined for circuit-breaker applications) in the fault and in breaker A.

Solution. (a) For a base of 25,000 kVA, 13.8 kV in the generator circuit the base for the motors is 25,000 kVA, 6.9 kV. The subtransient reactance of each motor's

$$X_d'' = 0.20 \frac{25,000}{5000} = 1.0 \text{ per unit}$$

Figure 10.14 is the diagram with subtransient values of reactance marked. For a

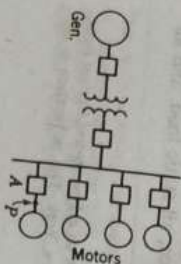


FIGURE 10.13
One-line diagram for Example 10.7.

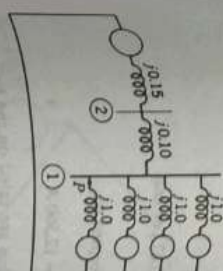


FIGURE 10.14
Reactance diagram for Example 10.7.

$$V_f = 1.0 \angle 0^\circ \text{ per unit} \quad Z_a = j0.125 \text{ per unit}$$

$$I_f = \frac{1.0 \angle 0^\circ}{j0.125} = -j8.0 \text{ per unit}$$

The base current in the 6.9 kV circuit is

$$|I_{\text{base}}| = \frac{25,000}{\sqrt{3} \times 6.9} = 2090 \text{ A}$$

and so

$$|I_f| = 8 \times 2090 = 16,720 \text{ A}$$

(b) Through breaker A comes the contribution from the generator and three of the four motors. The generator contributes a current of

$$-j8.0 \times \frac{0.25}{0.50} = -j4.0 \text{ per unit}$$

Each motor contributes 25% of the remaining fault current, or $-j1.0$ per unit amperes each. breaker A

$$I'' = -j4.0 + 3(-j1.0) = -j7.0 \text{ per unit} \quad \text{or } 7 \times 2090 = 14,630 \text{ A}$$

(c) To compute the current to be interrupted by breaker A, replace the subtransient reactance of $j1.0$ by the transient reactance of $j1.5$ in the motor circuits of Fig. 10.14. Then,

$$Z_{\text{th}} = j \frac{0.375 \times 0.25}{0.375 + 0.25} = j0.15 \text{ per unit}$$

The generator contributes a current of

$$\frac{1.0}{j0.15} \times \frac{0.375}{0.625} = -j4.0 \text{ per unit}$$

Each motor contributes a current of

$$\frac{1}{4} \times \frac{1.0}{j0.15} \times 0.25 = -j0.67 \text{ per unit}$$

The symmetrical short-circuit current to be interrupted is

$$(4.0 + 3 \times 0.67) \times 2090 = 12,560 \text{ A}$$

Suppose that all the breakers connected to the bus are rated on the basis of the current into a fault on the bus. In that case the short-circuit current interrupting rating of the breakers connected to the 6.9 kV bus must be at least

$$4 + 4 \times 0.67 = 6.67 \text{ per unit}$$

$$\text{or } 6.67 \times 2090 = 13,940 \text{ A}$$

A 14.4-kV circuit breaker has a rated maximum voltage of 15.5 kV and a breaking current of 15.5 kV. At 15.5 kV its rated short-circuit interrupting current is 8900 A. The breaker is rated for a symmetrical short-circuit interrupting current of 2.67 kA = 23,760 A, at a voltage of 15.5/2.67 = 5.8 kV. This current is the maximum that can be interrupted even though the breaker may be in a circuit of lower voltage. The short-circuit interrupting current rating at 6.9 kV is

$$\frac{15.5}{6.9} \times 8900 = 20,000 \text{ A}$$

The required capability of 13,940 A is well below 80% of 20,000 A, and the breaker is suitable with respect to short-circuit current.

The short-circuit current could have been found by using the bus impedance matrix. For this purpose two buses ① and ② are identified in Fig. 10.14. Bus ① is on the low-voltage side of the transformer and bus ② is on the high-voltage side. For motor reactance of 1.5 per unit

$$Y_{11} = -j10 + \frac{1}{j1.5/4} = -j12.67$$

$$Y_{12} = j10 \quad Y_{22} = -j10 - j6.67 = -j16.67$$

The node admittance matrix and its inverse are

$$Y_{\text{bus}} = \begin{bmatrix} \text{①} & \text{②} \\ \text{②} & \end{bmatrix} \begin{bmatrix} -j12.67 & j10.00 \\ j10.00 & -j16.67 \end{bmatrix} \quad Z_{\text{bus}} = \begin{bmatrix} \text{①} & \text{②} \\ \text{②} & \end{bmatrix} \begin{bmatrix} j0.150 & j0.090 \\ j0.090 & j0.114 \end{bmatrix}$$

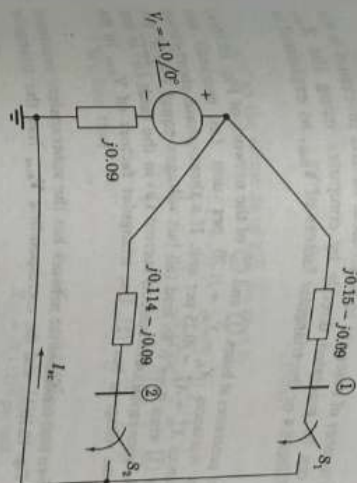


FIGURE 10.15 Bus impedance equivalent circuit for the Z_{bus} of Fig. 10.14.

Figure 10.15 is the network corresponding to Z_{bus} and $V_f = 1.0$ per unit. Closing S_1 with S_2 open represents a fault on bus ①.

The symmetrical short-circuit interrupting current in a three-phase fault at bus ① is

$$I_{sc} = \frac{1.0}{j0.15} = -j6.67 \text{ per unit}$$

which agrees with our previous calculations. The bus impedance matrix also gives us the voltage at bus ② with the fault on bus ①.

$$V_2 = 1.0 - I_{sc} Z_{21} = 1.0 - (-j6.67)(j0.09) = 0.4$$

and since the admittance between buses ① and ② is $-j10$, the current into the fault from the transformer is

$$(0.4 - 0.0)(-j10) = -j4.0 \text{ per unit}$$

which also agrees with our previous result.

We also know immediately the short-circuit current in a three-phase fault at bus ②, which, by referring to Fig. 10.15 with S_1 open and S_2 closed, is

$$I_{sc} = \frac{1.0}{j0.114} = -j8.77 \text{ per unit}$$

This simple example illustrates the value of the bus impedance matrix where the

effects of a fault at a number of buses are to be studied. Matrix inversion is necessary, for Z_{bus} can be generated directly by computer using the building algorithm of Sec. 8.4 or the triangular factors of Y_{bus} as explained in Sec. 8.5.

Example 10.8. The generators at buses ① and ② of the network of Fig. 10.16 have synchronous reactances $X_{d1} = X_{d2} = j1.70$ per unit (as marked) and subtransient reactances $X'_{d1} = X'_{d2} = j0.25$ per unit. If a three-phase short-circuit fault occurs at bus ③ when there is no load (all bus voltages equal $1.0 \angle 0^\circ$ per unit), find the initial symmetrical (subtransient) current (a) in the fault (b) in line ①–③, and (c) the voltage at bus ②. Use triangular factors of Y_{bus} in the calculations.

Solution. For the given fault conditions the network has the subtransient reactance diagram shown in Fig. 10.16(b), and the corresponding Y_{bus} has the triangular factors

$$Y_{bus} = \begin{bmatrix} -j10 & & \\ j1 & -j7.9 & \\ j5 & j3.5 & -j3.94937 \end{bmatrix} \begin{bmatrix} 1 & -0.1 & -0.5 \\ & 1 & -0.44304 \\ & & 1 \end{bmatrix}$$

Since the fault is at bus ③, Eqs. (10.19) through (10.21) show that the calculations involve column 3, $Z_{bus}^{(3)}$ of the subtransient Z_{bus} , which we now generate as follows:

$$\begin{bmatrix} -j10 & & \\ j1 & -j7.9 & \\ j5 & j3.5 & -j3.94937 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

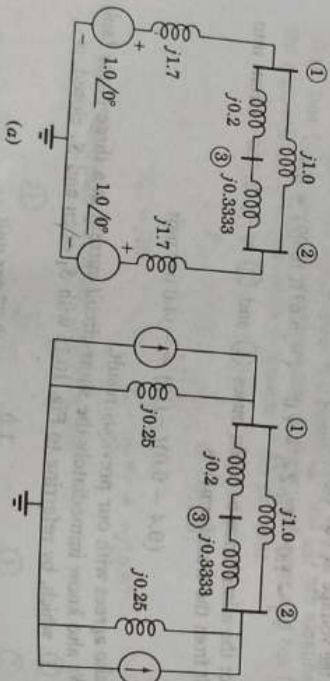


FIGURE 10.16

The reactance diagram for Example 10.8 with generators represented by: (a) series voltage source behind X'_d ; (b) equivalent current source in parallel with X'_d .

so that, we obtain

$$x_1 = x_2 = 0; \quad x_3 = \frac{1}{-j3.94937} = j0.25320 \text{ per unit}$$

and so the elements of $Z_{bus}^{(3)}$ are given by

$$\begin{bmatrix} 1 & -0.1 & -0.5 \\ & 1 & -0.44304 \\ & & 1 \end{bmatrix} \begin{bmatrix} Z_{13} \\ Z_{23} \\ Z_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ j0.25320 \end{bmatrix}$$

We find that

$$Z_{33} = j0.25320 \text{ per unit}$$

$$Z_{23} = j0.11218 \text{ per unit}$$

$$Z_{13} = j0.13782 \text{ per unit}$$

(a) According to Eq. (10.19), the subtransient current in the fault is

$$I_f' = \frac{V_f}{Z_{33}} = \frac{1}{j0.25320} = -j3.94937 \text{ per unit}$$

(b) From Eq. (10.21) we can write for the current in line ①–③

$$I_{13}' = -\frac{V_f}{Z_b} \left(\frac{Z_{13} - Z_{33}}{Z_{33}} \right) = -\frac{1}{j0.2} \left(\frac{j0.13782 - j0.25320}{j0.25320} \right) = -j2.27844 \text{ per unit}$$

(c) During the fault the voltage at bus ② is given by Eq. (10.20) as follows:

$$V_2 = V_f \left(1 - \frac{Z_{23}}{Z_{33}} \right) = 1.0 \left(\frac{j0.11218}{j0.25320} \right) = 0.55695 \text{ per unit}$$

10.6 SUMMARY

The current flowing immediately after a fault occurs in a power network is determined by the impedances of the network components and the synchronous machines. The initial symmetrical rms fault current can be determined by representing each machine by its subtransient reactance in series with its subtransient internal voltage. Subtransient currents are larger than the transient

where, for convenience, we let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \quad (11.9)$$

Then, as may be verified easily,

$$\mathbf{A}^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \quad (11.10)$$

and premultiplying both sides of Eq. (11.8) by $\mathbf{A}^{-1}$ yields

$$\begin{bmatrix} V_a^{(0)} \\ V_a^{(1)} \\ V_a^{(2)} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad (11.11)$$

which shows us how to resolve three unsymmetrical phasors into their symmetrical components. These relations are so important that we write the separate equations in the expanded form

$$V_a^{(0)} = \frac{1}{3}(V_a + V_b + V_c) \quad (11.12)$$

$$V_a^{(1)} = \frac{1}{3}(V_a + aV_b + a^2V_c) \quad (11.13)$$

$$V_a^{(2)} = \frac{1}{3}(V_a + a^2V_b + aV_c) \quad (11.14)$$

If required, the components $V_b^{(0)}$, $V_b^{(1)}$, $V_b^{(2)}$, $V_c^{(0)}$, $V_c^{(1)}$, and $V_c^{(2)}$ can be found by Eqs. (11.4). Similar results apply to line-to-line voltages simply by replacing V_a , V_b , and V_c in above equations by V_{ab} , V_{bc} , and V_{ca} , respectively.

Equation (11.12) shows that no zero-sequence components exist if the sum of the unbalanced phasors is zero. Since the sum of the line-to-line voltage phasors in a three-phase system is always zero, zero-sequence components are never present in the line voltages regardless of the degree of unbalance. The sum of the three line-to-line neutral voltage phasors is not necessarily zero, and voltages to neutral may contain zero-sequence components.

The preceding equations could have been written for any set of related phasors, and we might have written them for currents instead of for voltages. They may be solved either analytically or graphically. Because some of the

preceding equations are so fundamental, they are summarized for currents:

$$I_a = I_a^{(0)} + I_a^{(1)} + I_a^{(2)}$$

$$I_b = I_a^{(0)} + a^2 I_a^{(1)} + a I_a^{(2)} \quad (11.15)$$

$$I_c = I_a^{(0)} + a I_a^{(1)} + a^2 I_a^{(2)}$$

$$I_a^{(0)} = \frac{1}{3}(I_a + I_b + I_c)$$

$$I_a^{(1)} = \frac{1}{3}(I_a + aI_b + a^2I_c) \quad (11.16)$$

$$I_a^{(2)} = \frac{1}{3}(I_a + a^2I_b + aI_c)$$

Finally, these results can be extended to phase currents of a Δ circuit [such as that of Fig. 11.4(a)] by replacing I_a , I_b , and I_c by I_{ab} , I_{bc} , and I_{ca} , respectively.

Example 11.1. One conductor of a three-phase line is open. The current flowing to the Δ -connected load through line a is 10 A. With the current in line a as reference and assuming that line c is open, find the symmetrical components of the line currents.

Solution. Figure 11.3 is a diagram of the circuit. The line currents are

$$I_a = 10 \angle 0^\circ \text{ A} \quad I_b = 10 \angle 180^\circ \text{ A} \quad I_c = 0 \text{ A}$$

From Eqs. (11.16)

$$I_a^{(0)} = \frac{1}{3}(10 \angle 0^\circ + 10 \angle 180^\circ + 0) = 0$$

$$I_a^{(1)} = \frac{1}{3}(10 \angle 0^\circ + 10 \angle 180^\circ + 120^\circ + 0)$$

$$= 5 - j2.89 = 5.78 \angle -30^\circ \text{ A}$$

$$I_a^{(2)} = \frac{1}{3}(10 \angle 0^\circ + 10 \angle 180^\circ + 240^\circ + 0)$$

$$= 5 + j2.89 = 5.78 \angle 30^\circ \text{ A}$$

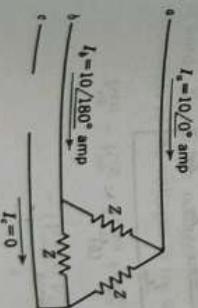


FIGURE 11.3
Circuit for Example 11.1.

From Eqs. (11.4)

$$\begin{aligned}
 I_b^{(0)} &= 0 & I_c^{(0)} &= 0 \\
 I_b^{(1)} &= 5.78 \angle -150^\circ \text{ A} & I_c^{(1)} &= 5.78 \angle 90^\circ \text{ A} \\
 I_b^{(2)} &= 5.78 \angle 150^\circ \text{ A} & I_c^{(2)} &= 5.78 \angle -90^\circ \text{ A}
 \end{aligned}$$

The result $I_a^{(0)} = I_b^{(0)} = I_c^{(0)} = 0$ holds for any three-wire system.

In Example 11.1 we note that components $I_a^{(1)}$ and $I_c^{(2)}$ have the same values although line c is open and can carry no net current. As is expected, therefore, the sum of the components in line c is zero. Of course, the sum of the components in line a is $10 \angle 0^\circ \text{ A}$, and the sum of the components in line b is $10 \angle 180^\circ \text{ A}$.

11.3 SYMMETRICAL Y AND Δ CIRCUITS

In three-phase systems circuit elements are connected between lines a, b , and c in either Y or Δ configuration. Relationships between the symmetrical components of Y and Δ currents and voltages can be established by referring to Fig. 11.4, which shows symmetrical impedances connected in Y and Δ. Let us agree that the reference phase for Δ quantities is branch $a-b$. The particular choice of reference phase is arbitrary and does not affect the results for currents we have

$$\begin{aligned}
 I_a &= I_{ab} - I_{ca} \\
 I_b &= I_{bc} - I_{ab} \\
 I_c &= I_{ca} - I_{bc}
 \end{aligned} \quad (11.1)$$

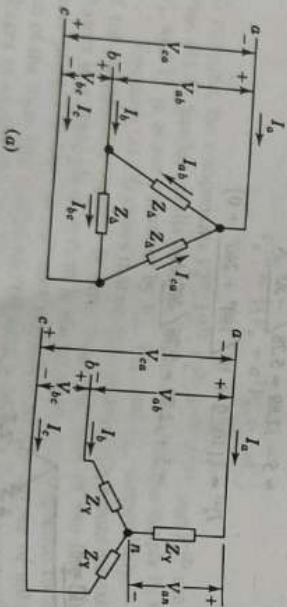


FIGURE 11.4
Symmetrical impedances: (a) Δ-connected; (b) Y-connected.

Adding all three equations together and invoking the definition of zero-sequence current, we obtain $I_a^{(0)} = (I_a + I_b + I_c)/3 = 0$, which means that line currents into a Δ-connected circuit have no zero-sequence currents. Substituting components of current in the equation for I_a yields

$$\begin{aligned}
 I_a^{(1)} + I_a^{(2)} &= (I_{ab}^{(0)} + I_{ab}^{(1)} + I_{ab}^{(2)}) - (I_{ca}^{(0)} + I_{ca}^{(1)} + I_{ca}^{(2)}) \\
 &= \frac{(I_{ab}^{(0)} - I_{ca}^{(0)}) + (I_{ab}^{(1)} - I_{ca}^{(1)}) + (I_{ab}^{(2)} - I_{ca}^{(2)})}{0} \quad (11.18)
 \end{aligned}$$

Evidently, if a nonzero value of circulating current $I_{ab}^{(0)}$ exists in the Δ circuit, it cannot be determined from the line currents alone. Noting that $I_{ca}^{(1)} = aI_{ab}^{(1)}$ and $I_{ca}^{(2)} = a^2I_{ab}^{(2)}$, we now write Eq. (11.18) as follows:

$$I_a^{(1)} + I_a^{(2)} = (1 - a)I_{ab}^{(1)} + (1 - a^2)I_{ab}^{(2)} \quad (11.19)$$

A similar equation for phase b is $I_b^{(1)} + I_b^{(2)} = (1 - a)I_{bc}^{(1)} + (1 - a^2)I_{bc}^{(2)}$, and expressing $I_{bc}^{(1)}$, $I_{bc}^{(2)}$, and $I_{bc}^{(2)}$ in terms of $I_{ab}^{(1)}$, $I_{ab}^{(2)}$, and $I_{ab}^{(2)}$, we obtain a resultant equation which can be solved along with Eq. (11.19) to yield the important results

$$I_a^{(1)} = \sqrt{3} \angle -30^\circ \times I_{ab}^{(1)} \quad I_a^{(2)} = \sqrt{3} \angle 30^\circ \times I_{ab}^{(2)} \quad (11.20)$$

These results amount to equating currents of the same sequence in Eq. (11.19). Complete sets of positive- and negative-sequence components of currents are shown in the phasor diagram of Fig. 11.5(a).

In a similar manner, the line-to-line voltages can be written in terms of line-to-neutral voltages of a Y-connected system.

$$\begin{aligned}
 V_{ab} &= V_{an} - V_{bn} \\
 V_{bc} &= V_{bn} - V_{cn} \\
 V_{ca} &= V_{cn} - V_{an}
 \end{aligned} \quad (11.21)$$

Adding together all three equations shows that $V_{ab}^{(0)} = (V_{an} + V_{bn} + V_{cn})/3 = 0$. In words, line-to-line voltages have no zero-sequence components. Substituting components of the voltages in the equation for V_{ab} yields

$$\begin{aligned}
 V_{ab}^{(1)} + V_{ab}^{(2)} &= (V_{an}^{(0)} + V_{an}^{(1)} + V_{an}^{(2)}) - (V_{bn}^{(0)} + V_{bn}^{(1)} + V_{bn}^{(2)}) \\
 &= \frac{(V_{an}^{(0)} - V_{bn}^{(0)}) + (V_{an}^{(1)} - V_{bn}^{(1)}) + (V_{an}^{(2)} - V_{bn}^{(2)})}{0} \quad (11.22)
 \end{aligned}$$

tive- or negative-sequence currents are concerned. Of course, this result could have been anticipated from the usual Δ -Y transformations of Table 12. The relation $Z_Y = Z_\Delta/3$ is correct when the impedances Z_a and Z_Y are both expressed in ohms or in per unit on the same kilovoltampere and voltage base.

Example 11.2. Three identical Y-connected resistors form a load bank with a three-phase rating of 2300 V and 500 kVA. If the load bank has applied voltages

$$|V_{ab}| = 1840 \text{ V} \quad |V_{bc}| = 2760 \text{ V} \quad |V_{ca}| = 2300 \text{ V}$$

find the line voltages and currents in per unit into the load. Assume that the neutral of the load is not connected to the neutral of the system and select a base of 2300 V, 500 kVA.

Solution. The rating of the load bank coincides with the specified base, and so the resistance values are 1.0 per unit. On the same base the given line voltages in per unit are

$$|V_{ab}| = 0.8 \quad |V_{bc}| = 1.2 \quad |V_{ca}| = 1.0$$

Assuming an angle of 180° for V_{ca} and using the law of cosines to find the angles of the other line voltages, we find the per-unit values

$$V_{ab} = 0.8 \angle 82.8^\circ \quad V_{bc} = 1.2 \angle -41.4^\circ \quad V_{ca} = 1.0 \angle 180^\circ$$

The symmetrical components of the line voltages are

$$\begin{aligned} V_{ab}^{(0)} &= \frac{1}{3} \left(0.8 \angle 82.8^\circ + 1.2 \angle 120^\circ - 41.4^\circ + 1.0 \angle 240^\circ + 180^\circ \right) \\ &= \frac{1}{3} (0.1003 + j0.7937 + 0.2372 + j1.1763 + 0.5 + j0.8660) \\ &= 0.2792 + j0.9453 = 0.9857 \angle 73.6^\circ \text{ per unit (line-to-line voltage base)} \end{aligned}$$

$$\begin{aligned} V_{ab}^{(2)} &= \frac{1}{3} \left(0.8 \angle 82.8^\circ + 1.2 \angle 240^\circ - 41.4^\circ + 1.0 \angle 120^\circ + 180^\circ \right) \\ &= \frac{1}{3} (0.1003 + j0.7937 - 1.1373 - j0.3828 + 0.5 - j0.8660) \\ &= -0.1790 - j0.1517 = 0.2346 \angle 220.3^\circ \text{ per unit (line-to-line voltage base)} \end{aligned}$$

The absence of a neutral connection means that zero-sequence currents are not present. Therefore, the phase voltages at the load contain positive- and negative-sequence components only. The phase voltages are found from Eqs. (11.23) with the $\sqrt{3}$ factor omitted since the line voltages are expressed in terms of the base voltage from line to line and the phase voltages are desired in per unit of the base

voltage to neutral. Thus,

$$\begin{aligned} V_{an}^{(1)} &= 0.9857 \angle 73.6^\circ - 30^\circ \\ &= 0.9857 \angle 43.6^\circ \text{ per unit (line-to-neutral voltage base)} \\ V_{an}^{(2)} &= 0.2346 \angle 220.3^\circ + 30^\circ \\ &= 0.2346 \angle 250.3^\circ \text{ per unit (line-to-neutral voltage base)} \end{aligned}$$

Since each resistor has an impedance of $1.0 \angle 0^\circ$ per unit,

$$\begin{aligned} I_a^{(1)} &= \frac{V_a^{(1)}}{1.0 \angle 0^\circ} = 0.9857 \angle 43.6^\circ \text{ per unit} \\ I_a^{(2)} &= \frac{V_a^{(2)}}{1.0 \angle 0^\circ} = 0.2346 \angle 250.3^\circ \text{ per unit} \end{aligned}$$

The positive direction of current is chosen to be from the supply toward the load.

11.4 POWER IN TERMS OF SYMMETRICAL COMPONENTS

If the symmetrical components of current and voltage are known, the power expended in a three-phase circuit can be computed directly from the components. Demonstration of this statement is a good example of the matrix manipulation of symmetrical components.

The total complex power flowing into a three-phase circuit through three lines a , b , and c is

$$S_{3\phi} = P + jQ = V_a I_a^* + V_b I_b^* + V_c I_c^* \quad (11.26)$$

where V_a , V_b , and V_c are the voltages to reference at the terminals and I_a , I_b , and I_c are the currents flowing into the circuit in the three lines. A neutral connection may or may not be present. If there is impedance in the neutral connection to ground, then the voltages V_a , V_b , and V_c must be interpreted as voltages from the line to ground rather than to neutral. In matrix notation

$$S_{3\phi} = \begin{bmatrix} V_a & V_b & V_c \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}^* = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}^* \quad (11.27)$$

where the conjugate of a matrix is understood to be composed of elements that are the conjugates of the corresponding elements of the original matrix. To introduce the symmetrical components of the voltages and currents, we make use of Eq. (11.8) to obtain

$$S_{3\phi} = [AV_{012}]^T [AI_{012}]^* \quad (11.26)$$

$$\text{where } V_{012} = \begin{bmatrix} V_a^{(0)} \\ V_a^{(1)} \\ V_a^{(2)} \end{bmatrix} \quad \text{and} \quad I_{012} = \begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} \quad (11.27)$$

The *reversal rule* of matrix algebra states that the transpose of the product of two matrices is equal to the product of the transposes of the matrices in reverse order. According to this rule,

$$[AV_{012}]^T = V_{012}^T A^T$$

and so

$$S_{3\phi} = V_{012}^T A^T [AI_{012}]^* = V_{012}^T A^T A^* I_{012}^* \quad (11.28)$$

Noting that $A^T = A$ and that a and a^2 are conjugates, we obtain

$$S_{3\phi} = \begin{bmatrix} V_a^{(0)} & V_a^{(1)} & V_a^{(2)} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a^{(0)*} \\ I_a^{(1)*} \\ I_a^{(2)*} \end{bmatrix} \quad (11.29)$$

or since

$$A^T A^* = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S_{3\phi} = 3 \begin{bmatrix} V_a^{(0)} & V_a^{(1)} & V_a^{(2)} \end{bmatrix} \begin{bmatrix} I_a^{(0)*} \\ I_a^{(1)*} \\ I_a^{(2)*} \end{bmatrix} \quad (11.30)$$

So, complex power is

$$S_{3\phi} = V_a I_a^* + V_b I_b^* + V_c I_c^* = 3V_a^{(0)} I_a^{(0)*} + 3V_a^{(1)} I_a^{(1)*} + 3V_a^{(2)} I_a^{(2)*} \quad (11.31)$$

which shows how complex power (in voltamperes) can be computed from the symmetrical components of the voltages to reference (in volts) and line currents

(in amperes) of an unbalanced three-phase circuit. It is important to note that the transformation of a - b - c voltages and currents to symmetrical components is the power-invariant in the sense discussed in Sec. 8.6, only if each product of sequence voltage (in volts) times the complex conjugate of the corresponding sequence current (in amperes) is multiplied by 3, as shown in Eq. (11.34). When, however, the multiplier 3 disappears.

Example 11.3. Using symmetrical components, calculate the power absorbed in the load of Example 11.2 and check the answer.

Solution. In per unit of the three-phase 500-kVA base, Eq. (11.34) becomes

$$S_{3\phi} = V_a^{(0)} I_a^{(0)*} + V_a^{(1)} I_a^{(1)*} + V_a^{(2)} I_a^{(2)*}$$

Substituting the components of voltages and currents from Example 11.2, we obtain

$$\begin{aligned} S_{3\phi} &= 0 + 0.9857 \angle 43.6^\circ \times 0.9857 \angle -43.6^\circ + 0.2346 \angle 250.3^\circ \times 0.2346 \angle -250.3^\circ \\ &= (0.9857)^2 + (0.2346)^2 = 1.02664 \text{ per unit} \\ &= 513.32 \text{ kW} \end{aligned}$$

The per-unit value of the resistors in each phase of the Y-connected load bank is 1.0 per unit. In ohms, therefore,

$$R_Y = \frac{(2300)^2}{500,000} = 10.58 \, \Omega$$

and the equivalent Δ -connected resistors are

$$R_\Delta = 3R_Y = 31.74 \, \Omega$$

From the given line-to-line voltages we calculate directly

$$\begin{aligned} S_{3\phi} &= \frac{|V_{ab}|^2}{R_\Delta} + \frac{|V_{bc}|^2}{R_\Delta} + \frac{|V_{ca}|^2}{R_\Delta} \\ &= \frac{(1840)^2 + (2760)^2 + (2300)^2}{31.74} = 513.33 \text{ kW} \end{aligned}$$

11.5 SEQUENCE CIRCUITS OF Y AND Δ IMPEDANCES

If impedance Z_n is inserted between the neutral and ground of the Y-connected impedances shown in Fig. 11.6(b), then the sum of the line currents is equal to

Likewise, Fig. 11.7(b) is called the *positive-sequence circuit* and Z_1 is called the *impedance to positive-sequence current*, whereas Fig. 11.7(c) is the *negative-sequence circuit* and Z_2 is the *impedance to negative-sequence current*. The names of the impedances to currents of the different sequences are shortened to the less descriptive terms *zero-sequence impedance*, *positive-sequence impedance*, and *negative-sequence impedance*. Here the positive-sequence impedance Z_1 and negative-sequence impedance Z_2 are equal to the usual per-phase impedance Z_Y , which is generally the case for stationary symmetrical circuits. Each of the three sequence circuits has only one phase of the actual three-phase circuit when the latter carries current in only that sequence. When the three sequence currents are simultaneously present, *all three* sequence circuits are needed to fully represent the original circuit.

Voltages in the positive-sequence and negative-sequence circuits can be regarded as voltages measured with respect to either neutral or ground, whether or not there is a connection of some finite value of impedance Z_n between neutral and ground. Accordingly, in the positive-sequence circuit there is a difference between $V_n^{(0)}$ and $V_n^{(1)}$, and a similar statement applies to $V_n^{(2)}$ in the negative-sequence circuit. However, a voltage difference can exist between the neutral and the reference of the zero-sequence circuit. In the circuit of Fig. 11.7(a) the current $I_n^{(0)}$ flowing through impedance Z_n produces the same voltage drop from neutral to ground as the current $3I_n^{(0)}$ flowing through impedance Z_n in the actual circuit of Fig. 11.6(b).

If the neutral of the Y-connected circuit is grounded through an impedance, we set $Z_n = 0$ and a zero-impedance connection then joins the neutral point to the reference node of the zero-sequence circuit. If there is no connection between neutral and ground, there cannot be any zero-sequence current flow, for then $Z_n = \infty$, which is indicated by the open circuit between neutral and the reference node in the zero-sequence circuit of Fig. 11.8(a).

Obviously, a Δ -connected circuit cannot provide a path through neutral and so line currents flowing into a Δ -connected load or its equivalent Y circuit cannot contain any zero-sequence components. Consider the symmetrical Δ -connected circuit of Fig. 11.4 with

$$V_{ab} = Z_{\Delta} I_{ab} \quad V_{bc} = Z_{\Delta} I_{bc} \quad V_{ca} = Z_{\Delta} I_{ca} \quad (11.4)$$

Adding the three preceding equations together, we obtain

$$V_{ab} + V_{bc} + V_{ca} = 3V_{ab}^{(0)} = 3Z_{\Delta} I_{ab}^{(0)} \quad (11.5)$$

and since the sum of the line-to-line voltages is always zero, we therefore have

$$V_{ab}^{(0)} = I_{ab}^{(0)} = 0 \quad (11.6)$$

Thus, in Δ -connected circuits with impedances only and no sources or mutual coupling there cannot be any circulating currents. Sometimes single-phase

11.5 SEQUENCE CIRCUITS OF Y AND Δ IMPEDANCES 433

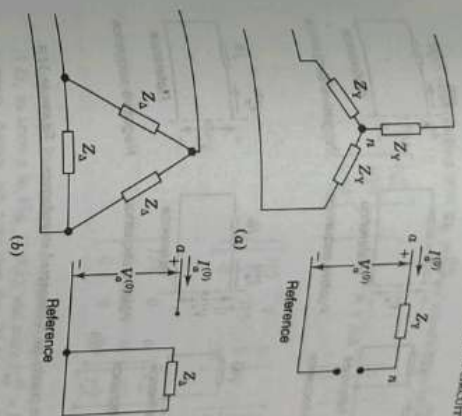


FIGURE 11.5
(a) Ungrounded Y-connected and
(b) Δ -connected circuits and their
zero-sequence circuits.

circulating currents can be produced in the Δ circuits of transformers and generators by either induction or zero-sequence generated voltages. A Δ circuit and its zero-sequence circuit are shown in Fig. 11.8(b). Note, however, that even if zero-sequence voltages were generated in the phases of the Δ , no zero-sequence voltage could exist between the Δ terminals, for the rise in voltage in each phase would then be matched by the voltage drop in the zero-sequence impedance of each phase.

Example 11.4. Three equal impedances of $j21 \Omega$ are connected in Δ . Determine the sequence impedances and circuits of the combination. Repeat the solution for the case where a mutual impedance of $j6 \Omega$ exists between each pair of adjacent branches in the Δ .

Solution. The line-to-line voltages are related to the Δ currents by

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} = \begin{bmatrix} j21 & 0 & 0 \\ 0 & j21 & 0 \\ 0 & 0 & j21 \end{bmatrix} \begin{bmatrix} I_{ab} \\ I_{bc} \\ I_{ca} \end{bmatrix}$$

Transforming to symmetrical components of voltages and currents gives

$$\mathbf{A} \begin{bmatrix} V_{ab}^{(0)} \\ V_{ab}^{(1)} \\ V_{ab}^{(2)} \end{bmatrix} = \begin{bmatrix} j21 & 0 & 0 \\ 0 & j21 & 0 \\ 0 & 0 & j21 \end{bmatrix} \mathbf{A} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

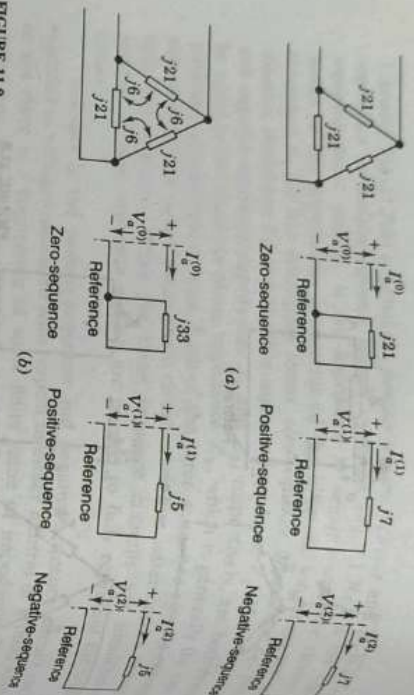


FIGURE 11.9 Zero-, positive-, and negative-sequence circuits for Δ -connected impedances of Example 11.4

and premultiplying each side by $\mathbf{A}^{-1}$, we obtain

$$\begin{bmatrix} V_{ab}^{(0)} \\ V_{ab}^{(1)} \\ V_{ab}^{(2)} \end{bmatrix} = j21\mathbf{A}^{-1}\mathbf{A} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix} = \begin{bmatrix} j21 & 0 & 0 \\ 0 & j21 & 0 \\ 0 & 0 & j21 \end{bmatrix} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

The positive- and negative-sequence circuits have per-phase impedances $Z_1 = Z_2 = j7 \Omega$, as shown in Fig. 11.9(a), and since $V_{ab}^{(0)} = 0$, the zero-sequence current in the zero-sequence circuit is an open circuit. The $j21\text{-}\Omega$ resistance source in the original Δ circuit.

When there is mutual inductance $j6 \Omega$ between phases,

$$\mathbf{A} \begin{bmatrix} V_{ab}^{(0)} \\ V_{ab}^{(1)} \\ V_{ab}^{(2)} \end{bmatrix} = \begin{bmatrix} j21 & j6 & j6 \\ j6 & j21 & j6 \\ j6 & j6 & j21 \end{bmatrix} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

The coefficient matrix can be separated into two parts as follows:

$$\begin{bmatrix} j21 & j6 & j6 \\ j6 & j21 & j6 \\ j6 & j6 & j21 \end{bmatrix} = j15 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + j6 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

and substituting into the previous equation, we obtain

$$\begin{bmatrix} V_{ab}^{(0)} \\ V_{ab}^{(1)} \\ V_{ab}^{(2)} \end{bmatrix} = \left\{ j15\mathbf{A}^{-1}\mathbf{A} + j6\mathbf{A}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \mathbf{A} \right\} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

$$= \begin{bmatrix} j15 & 0 & 0 \\ 0 & j15 & 0 \\ 0 & 0 & j15 \end{bmatrix} + j6 \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

$$= \begin{bmatrix} j33 & 0 & 0 \\ 0 & j15 & 0 \\ 0 & 0 & j15 \end{bmatrix} \begin{bmatrix} I_{ab}^{(0)} \\ I_{ab}^{(1)} \\ I_{ab}^{(2)} \end{bmatrix}$$

The positive- and negative-sequence impedances Z_1 and Z_2 now take on the value $j5 \Omega$, as shown in Fig. 11.9(b), and since $V_{ab}^{(0)} = 0$, the zero-sequence circuit is open. Again, we note that the $j33\text{-}\Omega$ resistance in the zero-sequence network has no significance because there is no internal source in the original Δ circuit.

The matrix manipulations of this example are useful in the sections which follow.

11.6 SEQUENCE CIRCUITS OF A SYMMETRICAL TRANSMISSION LINE

We are concerned primarily with systems that are essentially symmetrically balanced and which become unbalanced only upon the occurrence of an unsymmetrical fault. In practical transmission systems such complete symmetry is usually small, perfect balance between phases is often assumed especially if the lines are transposed along their lengths. Let us consider Fig. 11.10, for instance, which shows one section of a three-phase transmission line with a neutral conductor. The self-impedance Z_{aa} is the same for each phase conductor, and the neutral conductor has self-impedance Z_{nn} . When currents I_a , I_b , and I_c in the phase conductors are unbalanced, the neutral conductor serves as a return path. All the currents are assumed positive in the directions shown even though some of their numerical values may be negative under unbalanced conditions caused by faults. Because of mutual coupling, current flow in any one of the phases induces voltages in each of the other adjacent phases and in the neutral conductor. Similarly, I_n in the neutral conductor induces voltages in each of the phases. The coupling between all three phase conductors is regarded as being symmetrical and mutual impedance Z_{ab} is assumed between

to perfectly conducting ground, and we can write

$$\begin{aligned} V_{aa}^{(0)} &= V_a^{(0)} - V_d^{(0)} = Z_0 I_a^{(0)} \\ V_{aa}^{(1)} &= V_a^{(1)} - V_d^{(1)} = Z_1 I_a^{(1)} \\ V_{aa}^{(2)} &= V_a^{(2)} - V_d^{(2)} = Z_2 I_a^{(2)} \end{aligned} \quad (11.46)$$

where the sequence components of the voltages V_a and V_d are now with respect to ideal ground.

In deriving the equations for inductance and capacitance of transmission lines, we assumed balanced three-phase currents and did not specify phase order. The resulting parameters are therefore valid for both positive- and negative-sequence impedances. When only zero-sequence current flows in a transmission line, the current in each phase is identical. The current returns through the ground, through overhead ground wires, or through both. Because zero-sequence current is identical in each phase conductor (rather than equal only in magnitude and displaced in phase by 120° from other phase currents), the magnetic field due to zero-sequence current is very different from the magnetic field caused by either positive- or negative-sequence current. The difference in magnetic field results in the zero-sequence inductive reactance of overhead transmission lines being 2 to 3.5 times as large as the positive-sequence reactance. The ratio is toward the higher portion of the specified range for double-circuit lines and lines without ground wires.

Example 11.5. In Fig. 11.10 the terminal voltages at the left-hand and right-hand ends of the line are given by

$$\begin{aligned} V_{an} &= 182.0 + j70.0 \text{ kV} & V_{a'n'} &= 154.0 + j28.0 \text{ kV} \\ V_{bn} &= 72.24 - j32.62 \text{ kV} & V_{b'n'} &= 44.24 - j74.62 \text{ kV} \\ V_{cn} &= -170.24 + j88.62 \text{ kV} & V_{c'n'} &= -198.24 + j46.62 \text{ kV} \end{aligned}$$

The line impedances in ohms are

$$Z_{aa} = j60 \quad Z_{ab} = j20 \quad Z_{an} = j80 \quad Z_{nn} = 0$$

Determine the line currents I_a , I_b , and I_c using symmetrical components. Repeat the solution without using symmetrical components.

Solution. The sequence impedances have calculated values

$$\begin{aligned} Z_0 &= Z_{aa} + 2Z_{ab} + 3Z_{nn} = j60 + j40 + j240 = j340 \Omega \\ Z_1 &= Z_2 = Z_{aa} - Z_{ab} = j60 - j20 = j40 \Omega \end{aligned}$$

The sequence components of the voltage drops in the line are

$$\begin{aligned} \begin{bmatrix} V_{aa}^{(0)} \\ V_{aa}^{(1)} \\ V_{aa}^{(2)} \end{bmatrix} &= \mathbf{A}^{-1} \begin{bmatrix} V_{an} - V_{d'n'} \\ V_{bn} - V_{b'n'} \\ V_{cn} - V_{c'n'} \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} (182.0 - 154.0) + j(70.0 - 28.0) \\ (72.24 - 44.24) - j(32.62 - 74.62) \\ -(170.24 - 198.24) + j(88.62 - 46.62) \end{bmatrix} \\ &= \mathbf{A}^{-1} \begin{bmatrix} 28.0 + j42.0 \\ 28.0 + j42.0 \\ 28.0 + j42.0 \end{bmatrix} = \begin{bmatrix} 28.0 + j42.0 \\ 0 \\ 0 \end{bmatrix} \text{ kV} \end{aligned}$$

Substituting in Eq. (11.59), we obtain

$$\begin{aligned} V_{aa}^{(0)} &= 28,000 + j42,000 = j160 I_a^{(0)} \\ V_{aa}^{(1)} &= 0 = j40 I_a^{(1)} \\ V_{aa}^{(2)} &= 0 = j40 I_a^{(2)} \end{aligned}$$

from which we determine the symmetrical components of the currents in phase a,

$$I_a^{(0)} = 262.5 - j175 \text{ A} \quad I_a^{(1)} = I_a^{(2)} = 0$$

The line currents are therefore

$$I_a = I_b = I_c = 262.5 - j175 \text{ A}$$

The self- and mutual impedances of Eq. (11.52) have values

$$\begin{aligned} Z_1 &= Z_{aa} + Z_{nn} - 2Z_{an} = j60 + j80 - j60 = j80 \Omega \\ Z_2 &= Z_{aa} + Z_{nn} - 2Z_{an} = j20 + j80 - j60 = j40 \Omega \end{aligned}$$

and so line currents can be calculated from Eq. (11.53) without symmetrical components as follows:

$$\begin{aligned} \begin{bmatrix} V_{aa} \\ V_{bb} \\ V_{cc} \end{bmatrix} &= \begin{bmatrix} 28 + j42 \\ 28 + j42 \\ 28 + j42 \end{bmatrix} \times 10^3 = \begin{bmatrix} j80 & j40 & j40 \\ j40 & j80 & j40 \\ j40 & j40 & j80 \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \\ \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} &= \begin{bmatrix} j80 & j40 & j40 \\ j40 & j80 & j40 \\ j40 & j40 & j80 \end{bmatrix}^{-1} \begin{bmatrix} 28 + j42 \\ 28 + j42 \\ 28 + j42 \end{bmatrix} \times 10^3 = \begin{bmatrix} 262.5 - j175 \\ 262.5 - j175 \\ 262.5 - j175 \end{bmatrix} \text{ A} \end{aligned}$$

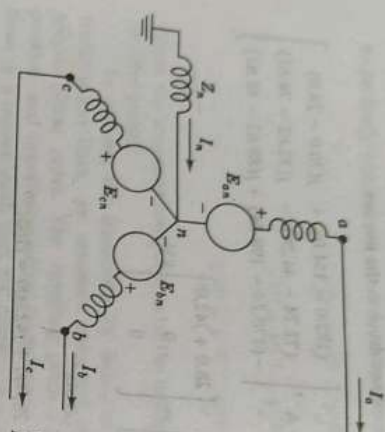


FIGURE 11.13
Circuit diagram of a generator grounded through a reactance. The phase *a* is designated as the reference phase. E_{an} , E_{bn} , and E_{cn} are positive sequence

11.7 SEQUENCE CIRCUITS OF THE SYNCHRONOUS MACHINE

A synchronous generator, grounded through a reactor, is shown in Fig. 11.13. When a fault (not indicated in the figure) occurs at the terminals of the generator, currents I_a , I_b , and I_c flow in the lines. If the fault involves ground, the current flowing into the neutral of the generator is designated I_n and the line currents can be resolved into their symmetrical components regardless of how unbalanced they may be.

The equations developed in Sec. 3.2 for the idealized synchronous machine are all based on the assumption of balanced instantaneous armature currents. In Eq. (3.7) we assumed that $i_a + i_b + i_c = 0$, and then set $i_a = -(i_b + i_c)$ in Eq. (3.5) in order to arrive at Eq. (3.11) for the terminal voltage of phase *a* in the form

$$v_{an} = -Ri_a - (L_s + M_s) \frac{di_a}{dt} + e_{an} \quad (11.61)$$

The steady-state counterpart of this equation is given in Eq. (3.24) as

$$V_{an} = -RI_a - j\omega(L_s + M_s)I_a + E_{an} \quad (11.62)$$

where E_{an} is the synchronous internal voltage of the machine. The subscripts of the voltages in Eqs. (11.61) and (11.62) differ slightly from those of Chap. 3 so as to emphasize that the voltages are with respect to neutral. If the substitution $i_a = -(i_b + i_c)$ had not been made as indicated, then we would have found

$$v_{an} = -Ri_a - L_s \frac{di_a}{dt} + M_s \frac{d}{dt}(i_b + i_c) + e_{an} \quad (11.63)$$

Assuming for now that steady-state sinusoidal currents and voltages of nominal

frequency ω continue to exist in the armature, we can write Eq. (11.63) in the phasor form

$$V_{an} = -RI_a - j\omega L_s I_a + j\omega M_s (I_b + I_c) + E_{an} \quad (11.64)$$

where E_{an} again designates the phasor equivalent of e_{an} . The armature phases *b* and *c* of the idealized machine have similar equations

$$\begin{aligned} V_{bn} &= -RI_b - j\omega L_s I_b + j\omega M_s (I_a + I_c) + E_{bn} \\ V_{cn} &= -RI_c - j\omega L_s I_c + j\omega M_s (I_a + I_b) + E_{cn} \end{aligned} \quad (11.65)$$

We can arrange Eqs. (11.64) and (11.65) in vector-matrix form as follows:

$$\begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \end{bmatrix} = -[R + j\omega(L_s + M_s)] \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} + j\omega M_s \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} + \begin{bmatrix} E_{an} \\ E_{bn} \\ E_{cn} \end{bmatrix} \quad (11.66)$$

Following the procedure demonstrated in the two preceding sections, we now express the *a-b-c* quantities of the machine in terms of symmetrical components of phase *a* of the armature

$$\begin{aligned} \begin{bmatrix} V_{an}^{(0)} \\ V_{an}^{(1)} \\ V_{an}^{(2)} \end{bmatrix} &= -[R + j\omega(L_s + M_s)] \begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} \\ &\quad + j\omega M_s \mathbf{A}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} + \mathbf{A}^{-1} \begin{bmatrix} E_{an} \\ a^2 E_{an} \\ a E_{an} \end{bmatrix} \end{aligned} \quad (11.67)$$

Since the synchronous generator is designed to supply balanced three-phase voltages, we have shown the generated voltages E_{an} , E_{bn} , and E_{cn} as a positive-sequence set of phasors in Eq. (11.67), where the operator $a = 1/120^\circ$ and $a^2 = 240^\circ$. The matrix multiplications of Eq. (11.67) are similar to those of Eqs. (11.56), and so we obtain

$$\begin{bmatrix} V_{an}^{(0)} \\ V_{an}^{(1)} \\ V_{an}^{(2)} \end{bmatrix} = -[R + j\omega(L_s + M_s)] \begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} + j\omega M_s \begin{bmatrix} 3 & \cdot & \cdot \\ \cdot & 1 & \cdot \\ \cdot & \cdot & 1 \end{bmatrix} \begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} + \begin{bmatrix} 0 \\ E_{an} \\ 0 \end{bmatrix} \quad (11.68)$$

ground) are written from Fig. 11.14 as

$$\begin{aligned} V_a^{(0)} &= -I_a^{(0)}Z_0 \\ V_a^{(1)} &= E_{an} - I_a^{(1)}Z_1 \\ V_a^{(2)} &= -I_a^{(2)}Z_2 \end{aligned} \quad (11.72)$$

where E_{an} is the positive-sequence voltage to neutral, Z_1 and Z_2 are the positive- and negative-sequence impedances of the generator, respectively, and Z_0 is defined by Eq. (11.71).

The equations developed to this point are based on a simple machine model which assumes the existence of only fundamental components of currents; on this basis the positive- and negative-sequence impedances of a machine are equal to one another but quite different from the zero-sequence impedances. In fact, however, the impedances of rotating machines to currents of the three negative-sequence armature current rotates in the direction opposite to that of the rotor which has the dc field winding. Unlike the flux produced by the sequence current, which is stationary with respect to the rotor, the flux produced by the negative-sequence current is sweeping rapidly over the face of the rotor. The currents induced in the field and damper windings counteract the occurrence of a short circuit at the terminals of a machine. The flux paths are the same as those encountered in evaluating *subtransient reactance*. So, in a cylindrical-rotor machine subtransient and negative-sequence reactances are equal. Values given in Table A.2 in the Appendix confirm this statement. The reactances in both the positive- and negative-sequence circuits are often taken to be equal to the subtransient or transient reactance, depending on whether subtransient or transient conditions are being studied.

When only zero-sequence current flows in the armature winding of a three-phase machine, the current and mmf of one phase are a maximum at the same time as the current and mmf of each of the other phases. The windings are so distributed around the circumference of the armature that the point of maximum mmf produced by one phase is displaced 120 electrical degrees in space from the point of maximum mmf of each of the other phases. If the mmf produced by the current of each phase had a perfectly sinusoidal distribution in space, a plot of mmf around the armature would result in three sinusoidal curves whose sum would be zero at every point. No flux would be produced across the air gap, and the only reactance of any phase winding would be that due to leakage and end turns. In an actual machine the winding is not distributed to produce perfectly sinusoidal mmf. The flux resulting from the sum of the mmfs is very small, which makes the zero-sequence reactance the smallest

of the machine's reactances—just somewhat higher than zero of the ideal case where there is no air-gap flux due to zero-sequence current.

Equations (11.72), which apply to any generator carrying unbalanced currents, are the starting points for the derivation of equations for the component equivalent circuits at any bus of the system as well as to the case of a loaded generator under steady-state conditions. When computing transient or subtransient conditions, the equations apply to a loaded generator if E' or E'' is substituted for E_{an} .

Example 11.6. A salient-pole generator without dampers is rated 20 MVA, 13.8 kV and has a direct-axis subtransient reactance of 0.25 per unit. The negative- and zero-sequence reactances are, respectively, 0.35 and 0.10 per unit. The neutral of the generator is solidly grounded. With the generator operating unloaded at the machine terminals, which then have per-unit voltages to ground,

$$\begin{aligned} V_a &= 0 & V_b &= 1.013 \angle -102.25^\circ & V_c &= 1.013 \angle 102.25^\circ \end{aligned}$$

Determine the subtransient current in the generator and the line-to-line voltages for subtransient conditions due to the fault.

Solution. Figure 11.15 shows the line-to-ground fault on phase a of the machine. In rectangular coordinates V_b and V_c are

$$\begin{aligned} V_b &= -0.215 - j0.990 \text{ per unit} \\ V_c &= -0.215 + j0.990 \text{ per unit} \end{aligned}$$

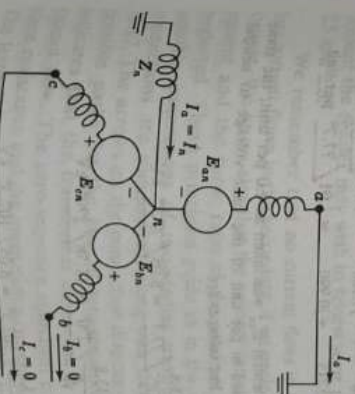


FIGURE 11.15
Circuit diagram for a single line-to-ground fault on phase a at the terminals of an unloaded generator whose neutral is grounded through a reactance.

The symmetrical components of the voltages at the fault point are

$$\begin{bmatrix} V_a^{(0)} \\ V_a^{(1)} \\ V_a^{(2)} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 0 \\ -0.215 - j0.990 \\ -0.215 + j0.990 \end{bmatrix} = \begin{bmatrix} -0.143 + j0 \\ 0.643 + j0 \\ -0.500 + j0 \end{bmatrix} \text{ Per unit}$$

From Eqs. (11.72) and Fig. 11.14 with $Z_n = 0$ we calculate

$$I_a^{(0)} = -\frac{V_a^{(0)}}{Z_0} = -\frac{(-0.143 + j0)}{j0.10} = -j1.43 \text{ Per unit}$$

$$I_a^{(1)} = \frac{E_{an} - V_a^{(1)}}{Z_1} = \frac{(1.0 + j0) - (0.643 + j0)}{j0.25} = -j1.43 \text{ Per unit}$$

$$I_a^{(2)} = -\frac{V_a^{(2)}}{Z_2} = -\frac{(-0.500 + j0)}{j0.35} = -j1.43 \text{ Per unit}$$

Therefore, the fault current into the ground is

$$I_a = I_a^{(0)} + I_a^{(1)} + I_a^{(2)} = 3I_a^{(0)} = -j4.29 \text{ per unit}$$

The base current is $20,000/(\sqrt{3} \times 13.8) = 837 \text{ A}$, and so the subtransient current in line a is

$$I_a = -j4.29 \times 837 = -j3,590 \text{ A}$$

Line-to-line voltages during the fault are

$$V_{ab} = V_a - V_b =$$

$$0.215 \quad +j0.990 = 1.01 \angle 77.7^\circ \text{ per unit}$$

$$V_{bc} = V_b - V_c =$$

$$0 \quad -j1.980 = 1.980 \angle 270^\circ \text{ per unit}$$

$$V_{ca} = V_c - V_a =$$

$$-0.215 \quad +j0.990 = 1.01 \angle 77.7^\circ \text{ per unit}$$

Since the generated voltage-to-neutral E_{an} was taken as 1.0 per unit, the above line-to-line voltages are expressed in per unit of the base voltage to which E_{an} is referred. Expressed in volts, the postfault line voltages are

$$V_{ab} = 1.01 \times \frac{13.8}{\sqrt{3}} \angle 77.7^\circ = 8.05 \angle 77.7^\circ \text{ kV}$$

$$V_{bc} = 1.980 \times \frac{13.8}{\sqrt{3}} \angle 270^\circ = 15.78 \angle 270^\circ \text{ kV}$$

$$V_{ca} = 1.01 \times \frac{13.8}{\sqrt{3}} \angle 102.3^\circ = 8.05 \angle 102.3^\circ \text{ kV}$$

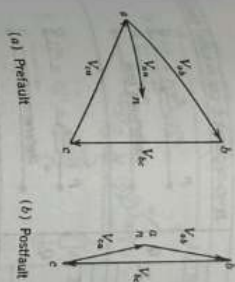


FIGURE 11.16
Phasor diagrams of the line voltages of Example 11.6
before and after the fault.

Before the fault, the line voltages were balanced and equal to 13.8 kV. For comparison with the line voltages after the fault occurs, the prefault voltages, with $V_{an} = E_{an}$ as reference, are given as

$$V_{ab} = 13.8 \angle 30^\circ \text{ kV} \quad V_{bc} = 13.8 \angle 270^\circ \text{ kV} \quad V_{ca} = 13.8 \angle 150^\circ \text{ kV}$$

Figure 11.16 shows phasor diagrams of prefault and postfault voltages.

The preceding example shows that $I_a^{(0)} = I_a^{(1)} = I_a^{(2)}$ in the case of a single line-to-ground fault. This is a general result, which is established in Sec. 12.2.

11.8 SEQUENCE CIRCUITS OF Y-Δ TRANSFORMERS

The sequence equivalent circuits of three-phase transformers depend on the connections of the primary and secondary windings. The different combinations of Δ and Y windings determine the configurations of the zero-sequence circuits and the phase shift in the positive- and negative-sequence circuits. On this account the reader may wish to review some portions of Chap. 2, notably Secs. 2.5 and 2.6.

We remember that no current flows in the primary of a transformer unless current flows in the secondary, if we neglect the relatively small magnetizing current. We know also that the primary current is determined by the secondary current and the turns ratio of the windings, again with magnetizing current neglected. These principles guide us in the analysis of individual cases. Five possible connections of two-winding transformers will be discussed. These connections are summarized, along with their zero-sequence circuits, in Fig. 11.17. The arrows on the connection diagrams of the figures to follow show the possible paths for the flow of zero-sequence current. Absence of an arrow indicates that the transformer connection is such that zero-sequence current cannot flow. The zero-sequence equivalent circuits are approximate as shown since resistance and the magnetizing-current path are omitted from each circuit. The letters P and Q identify corresponding points on the connection diagram

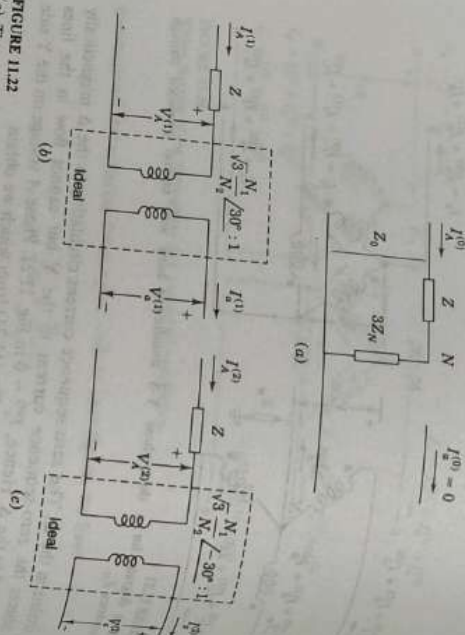


FIGURE 11.22

(a) The zero-sequence circuit of Y-Δ transformer bank with grounding impedance Z_0 in series with the line on the Y side to ground.
 (b) positive-sequence and (c) negative-sequence circuits.

the zero-sequence equivalent circuit must have impedance $3Z_N$ in series with the equivalent resistance and leakage reactance of the transformer to connect the line on the Y side to ground.

CASE 5. Y-Δ Bank, Ungrounded Y

An ungrounded Y is a special case where the impedance Z_N between neutral and ground is infinite. The impedance $3Z_N$ in the zero-sequence equivalent circuit of Case 4 becomes infinite and zero-sequence current cannot flow in the transformer windings.

The positive- and negative-sequence equivalent circuits of the Y-Δ transformer shown in Figs. 11.22(b) and 11.22(c) are based on Eqs. (11.87). We recall from Sec. 2.6 that the multiplier $\sqrt{3} N_1/N_2$ in Eqs. (11.87) is the ratio of the two per-unit calculations, therefore, Eqs. (11.87) become exactly the same as Eqs. (2.35), and again we have the rules

$$\begin{aligned} V_A^{(1)} &= V_a^{(1)} \times 1 / \angle 30^\circ & I_A^{(1)} &= I_a^{(1)} \times 1 / \angle 30^\circ \\ V_A^{(2)} &= V_a^{(2)} \times 1 / \angle -30^\circ & I_A^{(2)} &= I_a^{(2)} \times 1 / \angle -30^\circ \end{aligned} \quad (11.88)$$

that is,

When stepping up from the low-voltage side to the high-voltage side of a Δ-Y or Y-Δ transformer, *advance* the positive-sequence voltages (and currents) by 30° and *retard* the negative-sequence voltages (and currents) by 30° .

The next example shows the numerical application of Eqs. (11.88).

Example 11.7. The resistive Y-connected load bank of Example 11.2 is supplied from the low-voltage Y-side of a Y-Δ transformer. The voltages at the load are the high-voltage side of the transformer.

Solution. In Example 11.2 we found that the positive- and negative-sequence currents flowing toward the resistive load are

$$\begin{aligned} I_A^{(1)} &= 0.9857 / \angle 43.6^\circ \text{ per unit} \\ I_A^{(2)} &= 0.2346 / \angle 250.3^\circ \text{ per unit} \end{aligned}$$

while the corresponding voltages on the low-voltage Y side of the transformer are

$$\begin{aligned} V_{an}^{(1)} &= 0.9857 / \angle 43.6^\circ \text{ per unit (line-to-neutral voltage base)} \\ V_{an}^{(2)} &= 0.2346 / \angle 250.3^\circ \text{ per unit (line-to-neutral voltage base)} \end{aligned}$$

Advancing the phase angle of the low-voltage positive-sequence voltage by 30° and retarding the negative-sequence voltage by 30° give on the high-voltage side

$$\begin{aligned} V_A^{(1)} &= 0.9857 / \angle 43.6 + 30^\circ = 0.9857 / \angle 73.6^\circ = 0.2783 + j0.9456 \\ V_A^{(2)} &= 0.2346 / \angle 250.3 - 30^\circ = 0.2346 / \angle 220.3^\circ = -0.1789 - j0.1517 \\ V_A &= V_A^{(1)} + V_A^{(2)} = 0.0994 + j0.7939 = 0.8 / \angle 82.8^\circ \text{ per unit} \\ V_B^{(1)} &= a^2 V_A^{(1)} = 0.9857 / \angle -46.4^\circ = 0.6798 - j0.7138 \\ V_B^{(2)} &= a V_A^{(2)} = 0.2346 / \angle -19.7^\circ = 0.2209 - j0.0791 \\ V_B &= V_B^{(1)} + V_B^{(2)} = 0.9007 - j0.7929 = 1.20 / \angle -41.4^\circ \text{ per unit} \\ V_C^{(1)} &= a V_A^{(1)} = 0.9857 / \angle 193.6^\circ = -0.9581 - j0.2318 \\ V_C^{(2)} &= a^2 V_A^{(2)} = 0.2346 / \angle 100.3^\circ = -0.0419 + j0.2318 \\ V_C &= V_C^{(1)} + V_C^{(2)} = -1.0 + j0 = 1.0 / \angle 180^\circ \text{ per unit} \end{aligned}$$

Note that the line-to-neutral voltages on the high-voltage Δ side of the transformer are equal in per unit to the line-to-line voltages found in Example 11.2 on the low-voltage Y side. The line-to-line voltages are

$$V_{ab} = V_a - V_b = 0.0994 + j0.7939 - 0.9007 + j0.7929 = -0.8013 + j1.5868 = 1.78 \angle 116.8^\circ \text{ per unit (line-neutral voltage base)}$$

$$= \frac{1.78}{\sqrt{3}} \angle 116.8^\circ = 1.028 \angle 116.8^\circ \text{ per unit (line-line voltage base)}$$

$$V_{bc} = V_b - V_c = 0.9007 - j0.7939 + 1.0 = 1.9007 - j0.7939$$

$$= 2.06 \angle -22.7^\circ \text{ per unit (line-neutral voltage base)}$$

$$= \frac{2.06}{\sqrt{3}} \angle -22.7^\circ = 1.19 \angle -22.7^\circ \text{ per unit (line-line voltage base)}$$

$$V_{ca} = V_c - V_a = -1.0 - 0.0994 - j0.7939 = 1.0994 - j0.7939$$

$$= 1.356 \angle 215.8^\circ \text{ per unit (line-neutral voltage base)}$$

$$= \frac{1.356}{\sqrt{3}} \angle 215.8^\circ = 0.783 \angle 215.8^\circ \text{ per unit (line-line voltage base)}$$

Since the load impedance in each phase is a resistance of $1.0/\angle 0^\circ$ per unit, $I_a^{(0)}$ and $I_b^{(0)}$ are found to have identical per-unit values in this problem. Likewise, $I_c^{(0)}$ and $I_d^{(0)}$ are identical in per unit. Therefore, I_a must be identical to I_b , expressed in per unit. Thus,

$$I_a = 0.80 \angle 82.8^\circ \text{ per unit}$$

$$I_b = 1.20 \angle -41.4^\circ \text{ per unit}$$

$$I_c = 1.0 \angle 180^\circ \text{ per unit}$$

This example emphasizes the fact that in going from one side of the Δ -Y or Y- Δ transformer to the other, the positive-sequence components of currents and voltages on one side must be phase shifted separately from the negative-sequence components on the same side before combining them together to form the actual voltage on the other side.

Remarks on Phase Shift. The American National Standards Institute (ANSI) requires connection of Y- Δ and Δ -Y transformers so that the positive-sequence voltage to neutral, V_{1N} , on the high-voltage side leads the positive-sequence voltage to neutral, V_{1n} , on the low-voltage side by 30° . The wiring

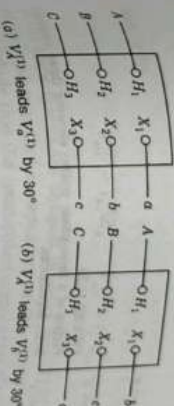


FIGURE 11.21 Labeling of lines connected to a three-phase Y- Δ transformer.

diagram of Fig. 11.21 and the connection diagram of Fig. 11.23(a) both satisfy the ANSI requirement, and because the connections of the phases to the transformer terminals $H_1, H_2, H_3 - X_1, X_2, X_3$ are respectively marked $a, b, c - a, b, c$ as shown in those figures, we find that the positive-sequence voltage to neutral $V_{1N}^{(0)}$ leads the positive-sequence voltage to neutral $V_{1n}^{(0)}$ by 30° .

It is not absolutely necessary, however, to label lines attached to the transformer terminals X_1, X_2 , and X_3 by the letters a, b , and c , respectively, as we have done, since no standards have been adopted for such labeling. In fact, in calculations the designation of lines could be chosen as shown in Fig. 11.23(b), which shows the letters b, c , and a associated, respectively, with X_1, X_2 , and X_3 . If the scheme of Fig. 11.23(b) is preferred, it is necessary only to exchange b for a , c for b , and a for c in the wiring and phasor diagrams of Fig. 11.21, which would then show that $V_{1n}^{(0)}$ leads $V_{1N}^{(0)}$ by 90° and that $V_{2n}^{(0)}$ lags $V_{2N}^{(0)}$ by 90° . It is easy to show that similar statements also apply to the corresponding currents.

We shall continue to follow the labeling scheme of Fig. 11.23(a), and Eqs. (11.88) then become identical to the ANSI requirement. When problems involving unsymmetrical faults are solved, positive- and negative-sequence components are found separately and phase shift is taken into account, if necessary, by applying Eqs. (11.88). Computer programs can be written to incorporate the effects of phase shift.

A transformer in a three-phase circuit may consist of three individual single-phase units, or it may be a three-phase transformer. Although the zero-sequence series impedances of three-phase units may differ slightly from the positive- and negative-sequence values, it is customary to assume that series impedances of all sequences are equal regardless of the type of transformer. Table A.1 in the Appendix lists transformer reactances. Resistance and impedance are almost equal for transformers of 1000 kVA or larger. For simplicity in our calculations we omit the shunt admittance which accounts for exciting current.

11.9 UNSYMMETRICAL SERIES IMPEDANCES

In the previous sections we have been concerned particularly with systems that are normally balanced. Let us look, however, at the equations of a three-phase

- In any part of the network impedance to zero-sequence current, Z_0 , generally different from Z_1 and Z_2 .
- Only positive-sequence circuits of rotating machines contain sources which are of positive-sequence voltages.
- Neutral is the reference for voltages in positive- and negative-sequence circuits, and such voltages to neutral are the same as voltages to ground if a physical connection of zero or other finite impedance exists between neutral and ground in the actual circuit.
- No positive- or negative-sequence currents flow between neutral points and ground.
- Impedances Z_n in the physical connections between neutral and ground are not included in positive- and negative-sequence circuits but are represented by impedances $3Z_n$ between the points for neutral and ground in the zero-sequence circuits only.

These characteristics of individual sequence circuits guide the construction of corresponding *sequence networks*. The object of obtaining the values of the sequence impedances of the various parts of a power system is to enable us to construct the sequence networks for the complete system. The network of a particular sequence—constructed by joining together all the corresponding sequence circuits of the separate parts—shows all the paths for the corresponding current of that sequence in one phase of the actual system.

In a balanced three-phase system the currents flowing in the three phases under normal operating conditions constitute a symmetrical positive-sequence set. These positive-sequence currents cause voltage drops of the same sequence only. Because currents of only *one* sequence occurred in the preceding chapter, we considered them to flow in an independent *per-phase* network which combined the positive-sequence emfs of rotating machines and the impedances of other static circuits to positive-sequence currents only. That same per-phase equivalent network is now called the *positive-sequence network*. That same per-phase distinguish it from the networks of the other two sequences.

We have discussed the construction of impedance and admittance representations of some rather complex positive-sequence networks in earlier chapters. Generally, we have not included the phase shift associated with Δ -Y and Y- Δ transformers in positive-sequence networks since practical systems are designed with such phase shifts summing to zero around all loops. In detailed calculations, however, we must remember to *advance* all positive-sequence voltages and currents by 30° when *stepping up* from the low-voltage side to the high-voltage side of a Δ -Y or Y- Δ transformer.

The transition from a positive-sequence network to a *negative-sequence* network is simple. Three-phase synchronous generators and motors have internal voltages of positive sequence only because they are designed to generate the same in a static symmetrical system, conversion of a positive-sequence

network to a negative-sequence network is accomplished by changing, if necessary, electromagnetic forces are omitted on the assumption of balanced generated voltages and the absence of negative-sequence voltages induced from outside sources. Of course, in using the negative-sequence network for detailed calculations, we must also remember to *retard* the negative-sequence voltages and currents by 30° when *stepping up* from the low-voltage side to the high-voltage side of a Δ -Y or Y- Δ transformer.

Since all the neutral points of a symmetrical three-phase system are at the same potential when balanced three-phase currents are flowing, all the neutral points must be at the same potential for either positive- or negative-sequence currents. Therefore, the neutral of a symmetrical three-phase system is the logical reference potential for specifying positive- and negative-sequence voltage drops. Impedance connected between the neutral of a machine and ground is not a part of either the positive- or negative-sequence network because neither positive- nor negative-sequence current can flow in an impedance so connected. Negative-sequence networks, like the positive-sequence networks of previous chapters, may contain the exact equivalent circuits of parts of the system or be simplified by omitting series resistance and shunt admittance.

Example 11.8. Draw the negative-sequence network for the system described in Example 6.1. Assume that the negative-sequence reactance of each machine is equal to its subtransient reactance. Omit resistance and phase shifts associated with the transformer connections.

Solution. Since all the negative-sequence reactances of the system are equal to the positive-sequence reactances, the negative-sequence network is identical to the positive-sequence network of Fig. 6.6 except for the omission of emfs from the negative-sequence network. The required network is drawn without transformer phase shifts in Fig. 11.25.

Zero-sequence equivalent circuits determined for the various separate parts of the system are readily combined to form the complete *zero-sequence network*. A three-phase system operates single phase insofar as the zero-

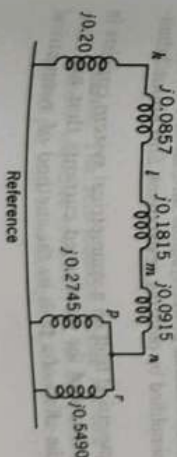


FIGURE 11.25
Negative-sequence network for
Example 11.8.

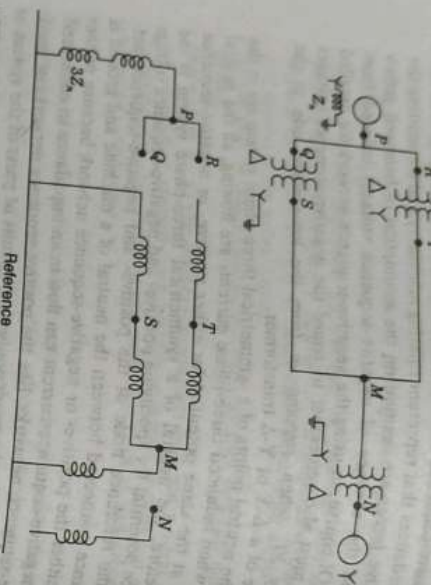


FIGURE 11.26 One-line diagram of a small power system and the corresponding zero-sequence network.

sequence currents are concerned, for the zero-sequence currents are the same in magnitude and phase at any point in all the phases of the system. Therefore, zero-sequence currents will flow only if a return path exists through which a completed circuit is provided. The reference for zero-sequence voltages is the potential of the ground at the point in the system at which any particular voltage is specified. Since zero-sequence currents may be flowing in the ground, the ground is not necessarily at the same potential at all points and the reference node of the zero-sequence network does not represent a ground of uniform potential. We have already discussed the fact that the impedance of the conductor of zero impedance, which is the reference node of the zero-sequence network is a impedance that voltages measured to the reference node of the system. It is network give the correct voltage to equivalent ideal ground. Figures 11.26 and 11.27 show one-line diagrams of two small power systems and their corresponding zero-sequence networks simplified by omitting resistances and shunt admittances.

The analysis of an unsymmetrical fault on a symmetrical system consists in finding the symmetrical components of the unbalanced currents that are flowing. Therefore, to calculate the effect of a fault by the method of symmetrical

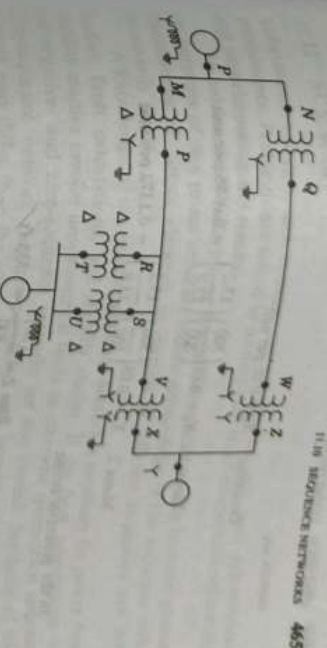


FIGURE 11.27 One-line diagram of a small power system and the corresponding zero-sequence network.

components, it is essential to determine the sequence impedances and to combine them to form the sequence networks. The sequence networks carrying the symmetrical-component currents $I_0^{(0)}$, $I_1^{(1)}$, and $I_2^{(2)}$ are then interconnected to represent various unbalanced fault conditions, as described in Chap. 12.

Example 11.9. Draw the zero-sequence network for the system described in Example 6.1. Assume zero-sequence reactances for the generator and motors of 0.05 per unit. A current-limiting reactor of 0.4 Ω is in each of the neutrals of the generator and the larger motor. The zero-sequence reactance of the transmission line is 1.5 Ω/km .

Solution. The zero-sequence leakage reactance of transformers is equal to the positive-sequence reactance. So, for the transformers $X_0 = 0.0857$ per unit and 0.0915 per unit, as in Example 6.1. Zero-sequence reactances of the generator and

MOTORS ARE

Generator: $X_0 = 0.05$ per unit

$$\text{Motor 1: } X_0 = 0.05 \left(\frac{300}{200} \right) \left(\frac{13.2}{13.8} \right)^2 = 0.0686 \text{ per unit}$$

$$\text{Motor 2: } X_0 = 0.05 \left(\frac{300}{100} \right) \left(\frac{13.2}{13.8} \right)^2 = 0.1372 \text{ per unit}$$

In the generator circuit

$$\text{Base } Z = \frac{(20)^2}{300} = 1.333 \Omega$$

and in the motor circuit

$$\text{Base } Z = \frac{(13.8)^2}{300} = 0.635 \Omega$$

In the impedance network for the generator

$$3Z_n = 3 \left(\frac{0.4}{1.333} \right) = 0.900 \text{ per unit}$$

and for the motor

$$3Z_n = 3 \left(\frac{0.4}{0.635} \right) = 1.890 \text{ per unit}$$

For the transmission line

$$Z_0 = \frac{1.5 \times 64}{176.3} = 0.5445 \text{ per unit}$$

The zero-sequence network is shown in Fig. 11.28.

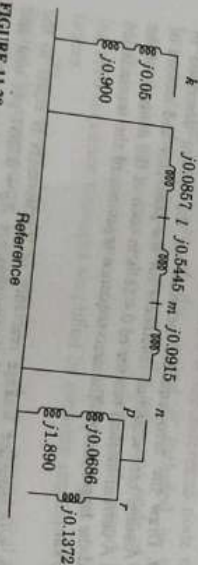


FIGURE 11.28
Zero-sequence network for Example 11.9.

11.11 SUMMARY

PROBLEMS 467

Unbalanced voltages and currents can be resolved into their symmetrical components. Problems are solved by treating each set of components separately and superimposing the results.

In balanced networks having strictly symmetrical coupling between phases the currents of one phase sequence induce voltage drops of like sequence only, the impedances of circuit elements to currents of different sequences are not necessarily equal.

A knowledge of the positive-sequence network is necessary for power-flow studies, fault calculations, and stability studies. If the fault calculations or the negative- and zero-sequence networks are also needed, synthesis of the zero-sequence network may differ from the others considerably.

PROBLEMS

- 11.1. If $V_{an}^{(1)} = 50 \angle 0^\circ$, $V_{an}^{(2)} = 20 \angle 90^\circ$, and $V_{an}^{(0)} = 10 \angle 180^\circ$ V, determine analytically the voltages to neutral V_{an} , V_{bn} , and V_{cn} , and also show graphically the sum of the given symmetrical components which determine the line-to-neutral voltages.
- 11.2. When a generator has terminal *a* open and the other two terminals are connected to each other with a short circuit from this connection to ground, typical values $I_a^{(0)} = 250 \angle 90^\circ$, and $I_b^{(0)} = 350 \angle 90^\circ$ A. Find the current into the ground and the current in each phase of the generator.
- 11.3. Determine the symmetrical components of the three currents $I_a = 10 \angle 0^\circ$, $I_b = 10 \angle 230^\circ$, and $I_c = 10 \angle 130^\circ$ A.
- 11.4. The currents flowing in the lines toward a balanced load connected in Δ are $I_a = 100 \angle 0^\circ$, $I_b = 141.4 \angle 225^\circ$, and $I_c = 100 \angle 90^\circ$. Find the symmetrical components of the given line currents and draw phasor diagrams of the positive- and negative-sequence line and phase currents. What is I_{a0} in amperes?
- 11.5. The voltages at the terminals of a balanced load consisting of three $10\text{-}\Omega$ resistors connected in Y are $V_{ab} = 100 \angle 0^\circ$, $V_{bc} = 80.8 \angle -121.4^\circ$, and $V_{ca} = 90 \angle 130^\circ$ V. Assuming that there is no connection to the neutral of the load, find the line currents from the symmetrical components of the given line voltages.
- 11.6. Find the power expended in the three $10\text{-}\Omega$ resistors of Prob. 11.5 from the symmetrical components of currents and voltages. Check the answer.
- 11.7. If there is impedance in the neutral connection to ground of a Y -connected load, then show that the voltages V_a , V_b , and V_c of Eq. (11.26) must be interpreted as voltages with respect to ground.
- 11.8. A balanced three-phase load consists of Δ -connected impedances Z_a in parallel with solidly grounded Y -connected impedances Z_y .
(a) Express the currents I_a , I_b , and I_c flowing in the lines from the supply source toward the load in terms of the source voltages V_a , V_b , and V_c .

of the Z_{bus} building algorithm. The reader is referred to the reference cited in footnote 1 for further guidance in handling open-circuit and short-circuit (i.e.) branches.

The faults to be discussed in succeeding sections may involve impedances Z_f between lines and from one or two lines to ground. When $Z_f = 0$, we have a direct short circuit, which is called a *bolted fault*. Although such direct circuits result in the highest value of fault current and are therefore the most conservative values to use when determining the effects of anticipated faults, the fault impedance is seldom zero. Most faults are the result of insulator flashover, where the impedance between the line and ground depends on the resistance of

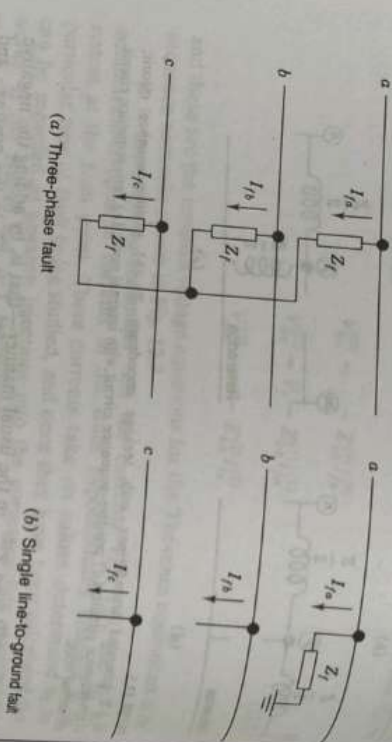


FIGURE 12.4

Connection diagrams of the hypothetical stubs for various faults through impedance.

the air, of the tower itself, and of the tower footing if ground wires are not used. Tower-footing resistances form the major part of the resistance between line and ground and depend on the soil conditions. The resistance between is 10 to 100 times the resistance of swampy ground. Connections of the hypothetical stubs for faults through impedance Z_f are shown in Fig. 12.4.

A balanced system remains symmetrical after the occurrence of a *three-phase fault* having the same impedance between each line and a common point. Only positive-sequence currents flow. With the fault impedance Z_f equal in all phases, as shown in Fig. 12.4(a), we simply add impedance Z_f to the usual (positive-sequence) Thévenin equivalent circuit of the system at the fault bus $\mathcal{K}$ and calculate the fault current from the equation

$$I_a^{(1)} = \frac{V_f}{Z_{kk}^{(1)} + Z_f} \quad (12.8)$$

For each of the other faults shown in Fig. 12.4, formal derivations of the equations for the symmetrical-component currents $I_a^{(0)}$, $I_a^{(1)}$, and $I_a^{(2)}$ are provided in the sections which follow. In each case the fault point P is designated as bus $\mathcal{K}$.

Example 12.1. Two synchronous machines are connected through three-phase transformers to the transmission line shown in Fig. 12.5. The ratings and reactances of the machines and transformers are

Machines 1 and 2: 100 MVA, 20 kV; $X_d'' = X_1' = X_2' = 20\%$,

$X_0 = 4\%$, $X_n = 5\%$

Transformers T_1 and T_2 : 100 MVA, 20Δ/345Y kV, $X = 8\%$

On a chosen base of 100 MVA, 345 kV in the transmission-line circuit the line reactances are $X_1 = X_2 = 15\%$ and $X_0 = 50\%$. Draw each of the three sequence networks and find the zero-sequence bus impedance matrix by means of the Z_{bus} building algorithm.

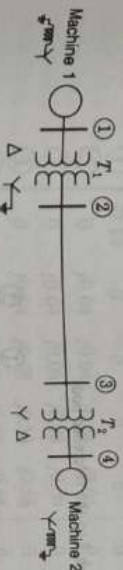


FIGURE 12.5

Single-line diagram of the system of Example 12.1.

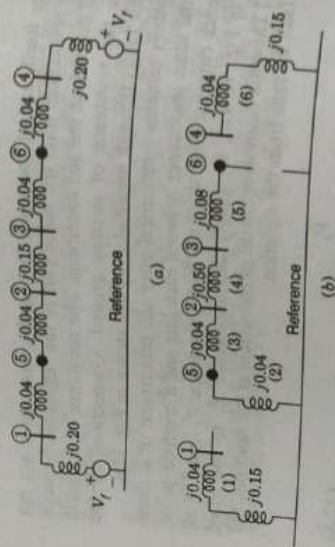


FIGURE 12.6

(a) Positive-sequence and (b) zero-sequence networks of the system of Fig. 12.5. Buses 1 and 2 are internal nodes of the transformers.

Solution. The given per-unit impedance values correspond to the chosen base, so they can be used directly to form the sequence networks. Figure 12.6(a) shows the positive-sequence network, which is identical to the negative-sequence network when the emfs are short-circuited; Fig. 12.6(b) shows the zero-sequence network. Note that each transformer is assigned an internal node—bus 1 for transformer T_1 and bus 2 for transformer T_2 . These internal nodes do not have an active role in the analysis of the system. In order to apply the Z_{bus} building algorithm, which is particularly simple in this example, let us label the zero-sequence branches from 1 to 7 as shown.

Step 1

Add branch 1 to the reference node

$$\begin{array}{c|c} \textcircled{1} & \textcircled{1} \\ \hline \textcircled{1} & [j0.19] \end{array}$$

Step 2

Add branch 2 to the reference node

$$\begin{array}{c|c} \textcircled{1} & \textcircled{5} \\ \hline \textcircled{5} & \begin{bmatrix} j0.19 & 0 \\ 0 & j0.04 \end{bmatrix} \end{array}$$

Step 3

Add branch 3 between buses 5 and 2

$$\begin{array}{c|c} \textcircled{1} & \textcircled{5} & \textcircled{2} \\ \hline \textcircled{5} & \begin{bmatrix} j0.19 & 0 \\ 0 & j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} \\ \textcircled{2} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 \end{array}$$

Step 4

Add branch 4 between buses 2 and 3

$$\begin{array}{c|c} \textcircled{1} & \textcircled{5} & \textcircled{2} & \textcircled{3} \\ \hline \textcircled{5} & \begin{bmatrix} j0.19 & 0 \\ 0 & j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} \\ \textcircled{2} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.08 \\ \textcircled{3} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.58 \end{array}$$

Step 5

Add branch 5 between buses 3 and 6

$$\begin{array}{c|c} \textcircled{1} & \textcircled{5} & \textcircled{2} & \textcircled{3} & \textcircled{6} \\ \hline \textcircled{5} & \begin{bmatrix} j0.19 & 0 \\ 0 & j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} \\ \textcircled{2} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.08 & j0.08 \\ \textcircled{3} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.58 & j0.58 \\ \textcircled{6} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.58 & j0.66 \end{array}$$

Step 6

Add branch 6 from bus 4 to the reference

$$\begin{array}{c|c} \textcircled{1} & \textcircled{5} & \textcircled{2} & \textcircled{3} & \textcircled{6} & \textcircled{4} \\ \hline \textcircled{5} & \begin{bmatrix} j0.19 & 0 \\ 0 & j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} \\ \textcircled{2} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.08 & j0.08 & 0 \\ \textcircled{3} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.58 & j0.58 & 0 \\ \textcircled{6} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & j0.08 & j0.58 & j0.66 & 0 \\ \textcircled{4} & \begin{bmatrix} 0 \\ j0.04 \end{bmatrix} & 0 & 0 & 0 & j0.19 \end{array}$$

Equating both terms and setting $I_f^{(2)} = -I_f^{(1)}$ as in Fig. 12.12, we obtain

$$(a^2 - a)(V_{ka}^{(1)} - V_{ka}^{(2)}) = (a^2 - a)I_f^{(1)}Z_f \quad (12.16)$$

or

$$V_{ka}^{(1)} - V_{ka}^{(2)} = I_f^{(1)}Z_f \quad (12.17)$$

which is precisely the voltage-drop equation for impedance Z_f in Fig. 12.12. Thus, all the fault conditions of Eqs. (12.13) are satisfied by connecting positive- and negative-sequence networks in parallel through impedance Z_f as shown in Fig. 12.12. The zero-sequence network is inactive and does not tie into the line-to-line fault calculations. The equation for the positive-sequence current in the fault can be determined directly from Fig. 12.12 so that

$$I_f^{(1)} = -I_f^{(2)} = \frac{V_f}{Z_{kk}^{(1)} + Z_{kk}^{(2)} + Z_f} \quad (12.18)$$

For a bolted line-to-line fault we set $Z_f = 0$.

Equations (12.17) are the fault current equations for a line-to-line fault through impedance Z_f . Once $I_f^{(1)}$ and $I_f^{(2)}$ are known, they can be treated as injections $-I_f^{(1)}$ and $-I_f^{(2)}$ into the positive- and negative-sequence networks respectively, and the changes in the sequence voltages at the buses of the system due to the fault can be obtained from the bus impedance matrices, as previously demonstrated. When Δ -Y transformers are present, phase shift of the positive- and negative-sequence currents and voltages must be taken into account in the calculations. The following example shows how this is accomplished.

Example 12.3. The same system as in Example 12.1 is operating at nominal system voltage without prefault currents when a bolted line-to-line fault occurs at bus (3). Using the bus impedance matrices of the sequence networks for subtransient conditions, determine the currents in the fault, the line-to-line voltages at the fault bus, and the line-to-line voltages at the terminals of machine 2.

Solution. $Z_{bus}^{(1)}$ and $Z_{bus}^{(2)}$ are already set forth in Example 12.1. Although $Z_{bus}^{(0)}$ is also given, we are not concerned with the zero-sequence network in this solution since the fault is line to line.

To simulate the fault, the Thévenin equivalent circuits at bus (3) of the positive- and negative-sequence networks of Example 12.1 are connected in parallel, as shown in Fig. 12.13. From this figure the sequence currents at calculated as follows:

$$I_f^{(1)} = -I_f^{(2)} = \frac{V_f}{Z_{33}^{(1)} + Z_{33}^{(2)}} = \frac{1 + j0}{j0.1696 + j0.1696} = -j2.9481 \text{ per unit}$$

Uppercase A is used here because the fault is in the high-voltage transmission line.

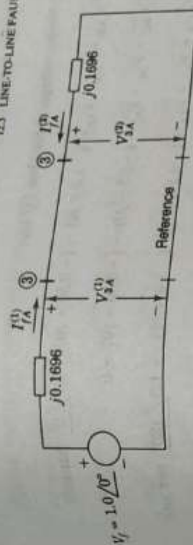


FIGURE 12.13 Connection of the Thévenin equivalent circuits for the line-to-line fault of Example 12.3.

circuit. Since $I_f^{(0)} = 0$, the components of currents in the fault are calculated from

$$I_{fA} = I_f^{(1)} + I_f^{(2)} = -j2.9481 + j2.9481 = 0$$

$$I_{fB} = a^2 I_f^{(1)} + a I_f^{(2)} = -j2.9481(-0.5 - j0.866) + j2.9481(-0.5 + j0.866) \\ = -5.1061 + j0 \text{ per unit}$$

$$I_{fC} = -I_{fB} = 5.1061 + j0 \text{ per unit}$$

As in Example 12.2, base current in the transmission line is 167.35 A, and so

$$I_{fA} = 0$$

$$I_{fB} = -5.1061 \times 167.35 = 855 \angle 180^\circ \text{ A}$$

$$I_{fC} = 5.1061 \times 167.35 = 855 \angle 0^\circ \text{ A}$$

Symmetrical components of phase-A voltage to ground at bus (3) are

$$V_{3A}^{(0)} = 0$$

$$V_{3A}^{(1)} = V_{3A}^{(2)} = 1 - Z_{3A}^{(1)} I_f^{(1)} = 1 - (j0.1696)(-j2.9481) = 0.5 + j0 \text{ per unit}$$

Line-to-ground voltages at fault bus (3) are

$$V_{3A} = V_{3A}^{(0)} + V_{3A}^{(1)} + V_{3A}^{(2)} = 0 + 0.5 + 0.5 = 1.0 \angle 0^\circ \text{ per unit}$$

$$V_{3B} = V_{3A}^{(0)} + a^2 V_{3A}^{(1)} + a V_{3A}^{(2)} = 0 + a^2 0.5 + a 0.5 = 0.5 \angle 180^\circ \text{ per unit}$$

$$V_{3C} = V_{3A}^{(0)} + a V_{3A}^{(1)} + a^2 V_{3A}^{(2)} = 0 + a 0.5 + a^2 0.5 = 0.5 \angle 180^\circ \text{ per unit}$$

The negative- and zero-sequence currents *out* of the system and into the fault can be determined from Fig. 12.15 by simple current division so that

$$I_a^{(2)} = -I_a^{(1)} \left[\frac{Z_{kk}^{(0)} + 3Z_f}{Z_{kk}^{(2)} + Z_{kk}^{(0)} + 3Z_f} \right] \quad (12.26)$$

$$I_a^{(0)} = -I_a^{(1)} \left[\frac{Z_{kk}^{(2)}}{Z_{kk}^{(2)} + Z_{kk}^{(0)} + 3Z_f} \right] \quad (12.27)$$

For a bolted fault Z_f is set equal to 0 in the above equations. When $Z_f = 0$, the zero-sequence circuit becomes an open circuit; no zero-sequence current can then flow and the equations revert back to those for the line-to-line fault discussed in the preceding section.

Again, we observe that the sequence currents $I_a^{(1)}$, $I_a^{(2)}$, and $I_a^{(0)}$ calculated, can be treated as negative injections into the sequence networks at the fault bus (4) and the sequence voltage *changes* at all buses of the system can then be calculated from the bus impedance matrices, as we have done in preceding sections.

Example 12.4. Find the subtransient currents and the line-to-line voltages at the fault under subtransient conditions when a double line-to-ground fault with $Z_f = 0$ occurs at the terminals of machine 2 in the system of Fig. 12.5. Assume that the system is unloaded and operating at rated voltage when the fault occurs. Use the bus impedance matrices and neglect resistance.

Solution. The bus impedance matrices $Z_{bus}^{(1)}$, $Z_{bus}^{(2)}$, and $Z_{bus}^{(0)}$ are the same as in Example 12.1, and so the Thévenin impedances at fault bus (4) are equal in per unit to the diagonal elements $Z_{44}^{(0)} = j0.19$ and $Z_{44}^{(1)} = Z_{44}^{(2)} = j0.1437$. To simulate the double line-to-ground fault at bus (4), we connect the Thévenin equivalents of all three sequence networks in parallel, as shown in Fig. 12.16, from which we

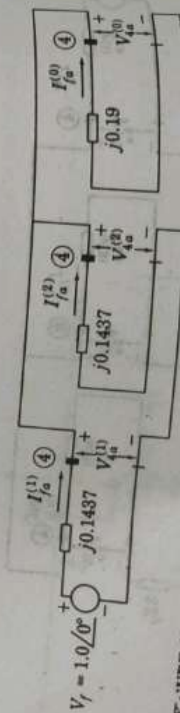


FIGURE 12.16

Connection of the Thévenin equivalents of the sequence networks for the double line-to-line fault of Example 12.4.

obtain

$$I_a^{(1)} = \frac{V_f}{Z_{44}^{(1)} + \left[\frac{Z_{44}^{(2)} Z_{44}^{(0)}}{Z_{44}^{(2)} + Z_{44}^{(0)}} \right]} = \frac{1 + j0}{j0.1437 + \left[\frac{(j0.1437)(j0.19)}{j0.1437 + j0.19} \right]} = -j4.4342 \text{ per unit}$$

Therefore, the sequence voltages at the fault are

$$V_{4a}^{(1)} = V_{4a}^{(2)} = V_{4a}^{(0)} = V_f - I_a^{(1)} Z_{44}^{(1)} = 1 - (-j4.4342)(j0.1437) = 0.3628 \text{ per unit}$$

Current injections into the negative- and zero-sequence networks at the fault bus are calculated by current division as follows:

$$I_a^{(2)} = -I_a^{(1)} \left[\frac{Z_{44}^{(0)}}{Z_{44}^{(2)} + Z_{44}^{(0)}} \right] = j4.4342 \left[\frac{j0.19}{j0.1437 + j0.19} \right] = j2.5247 \text{ per unit}$$

$$I_a^{(0)} = -I_a^{(1)} \left[\frac{Z_{44}^{(2)}}{Z_{44}^{(2)} + Z_{44}^{(0)}} \right] = j4.4342 \left[\frac{j0.1437}{j0.1437 + j0.19} \right] = j1.9095 \text{ per unit}$$

The currents out of the system at the fault point are

$$I_{fa} = I_a^{(0)} + I_a^{(1)} + I_a^{(2)} = j1.9095 - j4.4342 + j2.5247 = 0$$

$$I_{fb} = I_a^{(0)} + a^2 I_a^{(1)} + a I_a^{(2)}$$

$$= j1.9095 + (1 \angle 240^\circ)(4.4342 \angle -90^\circ) + (1 \angle 120^\circ)(2.5247 \angle 90^\circ)$$

$$= -6.0266 + j2.8642 = 6.6726 \angle 154.6^\circ \text{ per unit}$$

$$I_{fc} = I_a^{(0)} + a I_a^{(1)} + a^2 I_a^{(2)}$$

$$= j1.9095 + (1 \angle 120^\circ)(4.4342 \angle -90^\circ) + (1 \angle 240^\circ)(2.5247 \angle 90^\circ)$$

$$= 6.0266 + j2.8642 = 6.6726 \angle 25.4^\circ \text{ per unit}$$

and the current I_f into the ground is

$$I_f = I_{fb} + I_{fc} = 3I_a^{(0)} = j5.7285 \text{ per unit}$$

and from Fig. 12.6(a) the other sequence currents are calculated as

$$I_a^{(1)} = \frac{V_f - V_{fa}^{(1)}}{jX''} = \frac{1.0 - 0.7754}{j0.20} = -j1.123 \text{ per unit}$$

$$I_a^{(2)} = -\frac{V_{fa}^{(2)}}{jX_2} = \frac{0.2246}{j0.20} = -j1.123 \text{ per unit}$$

Note that the machine currents are shown without subscript f , which is reserved exclusively for the (stub) currents and voltages of the fault point. The phase currents in machine 2 are now easily calculated,

$$I_c = I_a^{(0)} + aI_a^{(1)} + a^2I_a^{(2)} = -j6.525 + a(-j1.123) + a^2(-j1.123) = -j5.402 \text{ per unit}$$

The base current in the machine circuits is $100,000/(\sqrt{3} \times 20) = 2886.751 \text{ A}$, and so $|I_c| = 15,594 \text{ A}$. Other voltages and currents in the system can be calculated similarly.

12.3 LINE-TO-LINE FAULTS

To represent a line-to-line fault through impedance Z_f , the hypothetical stubs on the three lines at the fault are connected, as shown in Fig. 12.11. Bus (k) is again the fault point P , and without any loss of generality, the line-to-line fault is regarded as being on phases b and c . The following relations must be satisfied at the fault point

$$I_{fa} = 0 \quad I_{fb} = -I_{fc} \quad V_{kb} - V_{kc} = I_{fb}Z_f \quad (12.13)$$

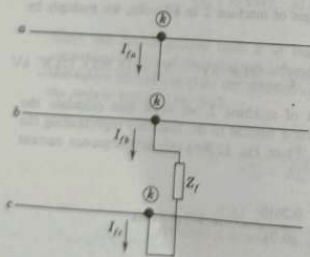


FIGURE 12.11 Connection of the hypothetical stubs for a line-to-line fault. The fault point is called bus (k).

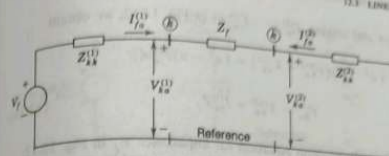


FIGURE 12.12 Connection of the Thévenin equivalents of the positive- and negative-sequence networks for a line-to-line fault between phases b and c at bus (k) of the system.

Since $I_{fb} = -I_{fc}$ and $I_{fa} = 0$, the symmetrical components of current are

$$\begin{bmatrix} I_a^{(0)} \\ I_a^{(1)} \\ I_a^{(2)} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 0 \\ I_{fb} \\ -I_{fb} \end{bmatrix}$$

and multiplying through in this equation shows that

$$I_a^{(0)} = 0 \quad (12.14)$$

$$I_a^{(1)} = -I_a^{(2)} \quad (12.15)$$

The voltages throughout the zero-sequence network must be zero since there are no zero-sequence sources, and because $I_a^{(0)} = 0$, current is not being injected into that network due to the fault. Hence, line-to-line fault calculations do not involve the zero-sequence network, which remains the same as before the fault—a dead network.

To satisfy the requirement that $I_a^{(1)} = -I_a^{(2)}$, let us connect the Thévenin equivalents of the positive- and negative-sequence networks in parallel, as shown in Fig. 12.12. To show that this connection of the networks also satisfies the voltage equation $V_{kb} - V_{kc} = I_{fb}Z_f$, we now expand each side of that equation separately as follows:

$$V_{kb} - V_{kc} = (V_{kb}^{(1)} + V_{kb}^{(2)}) - (V_{kc}^{(1)} + V_{kc}^{(2)}) = (V_{kb}^{(1)} - V_{kc}^{(1)}) + (V_{kb}^{(2)} - V_{kc}^{(2)})$$

$$= (a^2 - a)V_{ka}^{(1)} + (a - a^2)V_{ka}^{(2)} = (a^2 - a)(V_{ka}^{(1)} - V_{ka}^{(2)})$$

$$I_{fb}Z_f = (I_{fb}^{(1)} + I_{fb}^{(2)})Z_f = (a^2I_{fa}^{(1)} + aI_{fa}^{(2)})Z_f$$

Calculating a - b - c voltages at the fault bus, we find that

$$V_{a0} = V_{a0}^{(0)} + V_{a0}^{(1)} + V_{a0}^{(2)} = 3V_{a0}^{(1)} = 3(0.3628) = 1.0884 \text{ per unit}$$

$$V_{ab} = V_{ac} = 0$$

$$V_{a,b} = V_{a0} - V_{ab} = 1.0884 \text{ per unit}$$

$$V_{a,bc} = V_{a0} - V_{ac} = 0$$

$$V_{a,ca} = V_{a0} - V_{aa} = -1.0884 \text{ per unit}$$

Base current equals $100 \times 10^3 / (\sqrt{3} \times 20) = 2887 \text{ A}$ in the circuit of machine 1, and so we find that

$$I_{f0} = 0$$

$$I_{fb} = 2887 \times 6.6726 / 154.6^\circ = 19,262 / 154.6^\circ \text{ A}$$

$$I_{fc} = 2887 \times 6.6726 / 25.4^\circ = 19,262 / 25.4^\circ \text{ A}$$

$$I_f = 2887 \times 5.7285 / 90^\circ = 16,538 / 90^\circ \text{ A}$$

The base line-to-neutral voltage in machine 2 is $20 / \sqrt{3} \text{ kV}$, and so

$$V_{a,b} = 1.0884 \times \frac{20}{\sqrt{3}} = 12.568 / 0^\circ \text{ kV}$$

$$V_{a,bc} = 0$$

$$V_{a,ca} = -1.0884 \times \frac{20}{\sqrt{3}} = 12.568 / 180^\circ \text{ kV}$$

Examples 12.3 and 12.4 show that phase shifts due to Δ -Y transformers do not enter into the calculations of sequence currents and voltages in that part of the system where the fault occurs, provided V_f at the fault point is chosen as the reference voltage for the calculations. However, for those parts of the system which are separated by Δ -Y transformers from the fault point, the sequence currents, and voltages calculated by bus impedance matrix must be shifted in phase before being combined to form the actual voltages. This is because the bus impedance matrices of the sequence networks are formed without consideration of phase shifts, and so they consist of per-unit impedances referred to the part of the network which includes the fault point.

Example 12.5. Solve for the subtransient voltages to ground at bus (2), the end of the transmission line remote from the double line-to-ground fault, in the system of Example 12.4.

Solution. Numerical values of the fault-current components are given in the solution of Example 12.4 and the elements of $Z_{11}^{(1)}$, $Z_{22}^{(2)}$, and $Z_{21}^{(2)}$ are provided in the solution of Example 12.1. Neglecting phase shift of the Δ -Y transformers for the moment and substituting the appropriate values in Eq. (12.6), we obtain for the voltages at bus (2) due to the fault at bus (4),

$$V_{20}^{(0)} = -I_{f0}^{(0)} Z_{20}^{(0)} = -(1.9095)(0) = 0$$

$$V_{21}^{(1)} = V_f - I_{f1}^{(1)} Z_{21}^{(1)} = 1 - (-j4.4342)(j0.0789) = 0.6501 \text{ per unit}$$

$$V_{22}^{(2)} = -I_{f2}^{(2)} Z_{22}^{(2)} = -(j2.5247)(j0.0789) = 0.1992 \text{ per unit}$$

Accounting for phase shift in stepping up to the transmission-line circuit from the fault at bus (4), we have

$$V_{20}^{(0)} = 0$$

$$V_{21}^{(1)} = V_{21}^{(1)} / 30^\circ = 0.6501 / 30^\circ = 0.5630 + j0.3251 \text{ per unit}$$

$$V_{22}^{(2)} = V_{22}^{(2)} / -30^\circ = 0.1992 / -30^\circ = 0.1725 - j0.0996 \text{ per unit}$$

The required voltages can now be calculated:

$$V_{2A} = V_{20}^{(0)} + V_{21}^{(1)} + V_{22}^{(2)} = (0.5630 + j0.3251) + (0.1725 - j0.0996)$$

$$= 0.7355 + j0.2255 = 0.7693 / 17.0^\circ \text{ per unit}$$

$$V_{2B} = V_{20}^{(0)} + a^2 V_{21}^{(1)} + a V_{22}^{(2)} = (1 / 240^\circ)(0.6531 / 30^\circ)$$

$$+ (1 / 120^\circ)(0.1992 / 30^\circ)$$

$$= -0.1725 - j0.5535 = 0.5798 / 107.3^\circ \text{ per unit}$$

$$V_{2C} = V_{20}^{(0)} + a V_{21}^{(1)} + a^2 V_{22}^{(2)} = (1 / 120^\circ)(0.6531 / 30^\circ)$$

$$+ (1 / 240^\circ)(0.1992 / 30^\circ)$$

$$= -0.5656 + j0.1274 = 0.5798 / 167.3^\circ \text{ per unit}$$

These per-unit values can be converted to volts by multiplying by the line-to-neutral base voltage $345 / \sqrt{3} \text{ kV}$ of the transmission line.

12.5 DEMONSTRATION PROBLEMS

Large-scale computer programs based on the bus impedance matrices of the sequence networks are generally used to analyze faults on electric utility transmission systems. Three-phase and single line-to-ground faults are the only types of fault studied. Since circuit-breaker applications are usually according to the symmetrical short-circuit current that must be interrupted, the current is calculated for the two types of fault. The printout includes the fault current and the contributions from each line. The results also list those quantities when each line connected to the faulted bus is opened in turn while all others are in operation.

The program uses the impedances for the lines as provided in the line data for the power-flow program and includes the appropriate reactance for each machine in forming the positive- and zero-sequence bus impedance matrices. As far as impedances are concerned, the negative-sequence network is taken to be the same as the positive-sequence network. So, for a single line-to-ground fault at bus (k), $I_k^{(1)}$ is calculated in per unit as 1.0 divided by the sum $(2Z_{kk}^{(1)} + Z_{kk}^{(0)} + 3Z_f)$. The bus voltages are included in the computer printout, if called for, as well as the current in lines other than those connected to the faulted bus since this information can easily be found from the bus impedance matrices.

The following numerical examples show the analysis of a single line-to-ground fault on (1) an industrial power system and (2) a small electric utility system. Both of these systems are quite small in extent compared to the large-scale systems normally encountered. The calculations are presented without matrices in order to emphasize the circuit concepts which underlie fault analysis. The presentation should allow the reader to become more familiar with the sequence networks and how they are used to analyze faults. The principles demonstrated here are essentially the same as those employed within the large-scale computer programs used by industry. The same examples are to be solved by the bus impedance matrices in the problems at the end of this chapter.

Example 12.6. A group of identical synchronous motors is connected through a transformer to a 4.16-kV bus at a location remote from the generating plants of a power system. The motors are rated 600 V and operate at 89.5% efficiency when carrying a full load at unity power factor and rated voltage. The sum of their output ratings is 4476 kW (6000 hp). The reactances in per unit of each motor based on its own input kilovoltampere rating are $X_d'' = X_1 = 0.20$, $X_2 = 0.25$, $X_0 = 0.04$, and each is grounded through a reactance of 0.02 per unit. The motors are connected to the 4.16-kV bus through a transformer bank composed of three single-phase units, each of which is rated 2400/600 V, 2500 kVA. The 600-V windings are connected in Δ to the motors and the 2400-V windings are connected in Y. The leakage reactance of each transformer is 10%.

The power system which supplies the 4.16-kV bus is represented by a Thévenin equivalent generator rated 7500 kVA, 4.16 kV with reactances of

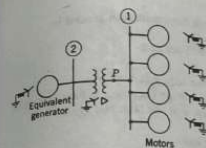


FIGURE 12.17
Single-line diagram of the system of Example 12.6.

$X_d'' = X_2 = 0.10$ per unit, $X_0 = 0.05$ per unit, and X_n from neutral to ground equal to 0.05 per unit.

Each of the identical motors is supplying an equal share of a total load of 3730 kW (5000 hp) and is operating at rated voltage, 85% power-factor lag, and 88% efficiency when a single line-to-ground fault occurs on the low-voltage side of the transformer bank. Treat the group of motors as a single equivalent motor. Draw the sequence networks showing values of the impedances. Determine the subtransient line currents in all parts of the system with prefault current neglected.

Solution. The single-line diagram of the system is shown in Fig. 12.17. The 600-V bus and the 4.16-kV bus are numbered (1) and (2), respectively. Choose the rating of the equivalent generator as base: 7500 kVA, 4.16 kV at the system bus. Since

$$\sqrt{3} \times 2400 = 4160 \text{ V} \quad 3 \times 2500 = 7500 \text{ kVA}$$

the three-phase rating of the transformer is 7500 kVA, 4160 V/600 V. So, the base for the motor circuit is 7500 kVA, 600 V.

The input rating of the single equivalent motor is

$$\frac{6000 \times 0.746}{0.895} = 5000 \text{ kVA}$$

and the reactances of the equivalent motor in percent are the same on the base of the combined rating as the reactances of the individual motors on the base of the rating of an individual motor. The reactances of the equivalent motor in per unit on the selected base are

$$X_d'' = X_1 = X_2 = 0.2 \frac{7500}{5000} = 0.3 \quad X_0 = 0.04 \frac{7500}{5000} = 0.06$$

In the zero-sequence network the reactance between neutral and ground of the equivalent motor is

$$3X_n = 3 \times 0.02 \frac{7500}{5000} = 0.09 \text{ per unit}$$

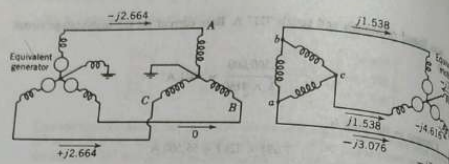


FIGURE 12.19 Per-unit values of subtransient line currents in all parts of the system of Example 12.6, prefault current neglected.

phase *a* is

$$-0.565 + j0.350 - j1.026 = -0.565 - j0.676$$

These values are shown in Fig. 12.18. The remainder of the calculation, using these new values, proceeds as in the example.

Figure 12.19 gives the per-unit values of subtransient line currents in all parts of the system when the fault occurs at no load. Figure 12.20 shows the values for the fault occurring on the system when the load specified in the example is considered. In a larger system where the fault current is much higher in comparison with the load current the effect of neglecting the load current is less than is indicated by comparing Figs. 12.19 and 12.20. In the large system, however, the prefault currents determined by a power-flow study could simply be added to the fault current found with the load neglected.

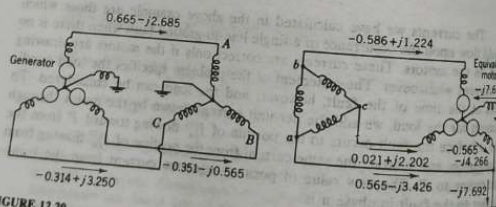


FIGURE 12.20 Per-unit values of subtransient line currents in all parts of the system of Example 12.6, prefault current considered.

Example 12.7. The single-line diagram of a small power system is shown in Fig. 12.21. A bolted single line-to-ground fault at point *P* is to be analyzed. The ratings and reactances of the generator and the transformers are

Generator: 100 MVA, 20 kV: $X'' = X_2 = 20\%$, $X_0 = 4\%$,
 $X_n = 5\%$

Transformers T_1 and T_2 : 100 MVA, 20Δ/345Y kV: $X = 10\%$

On a chosen base of 100 MVA, 345 kV in the transmission-line circuit the line reactances are

From T_1 to P : $X_1 = X_2 = 20\%$, $X_0 = 50\%$

From T_2 to P : $X_1 = X_2 = 10\%$, $X_0 = 30\%$

To simulate the fault, the sequence networks of the system with reactances marked in per unit are connected in series, as shown in Fig. 12.22. Verify the values of the currents shown in the figure and draw a complete three-phase circuit diagram with all current flows marked in per unit. Assume that the transformers are lettered so that Eqs. (11.88) apply.

Solution. With switch *S* open, the prefault currents are zero and the open-circuit voltage of phase *A* at point *P* can be taken as the reference voltage $1.0 + j0.0$ per unit. The impedances seen looking into the sequence networks at the fault point are

$$Z_{pp}^{(0)} = \frac{(j0.6)(j0.4)}{j0.6 + j0.4} = j0.24 \text{ per unit}$$

$$Z_{pp}^{(1)} = Z_{pp}^{(2)} = j0.5 \text{ per unit}$$

The sequence currents in the hypothetical stub of phase *A* at *P* are

$$I_{fA}^{(0)} = I_{fA}^{(1)} = I_{fA}^{(2)} = \frac{1.0 + j0.0}{j0.5 + j0.5 + j0.24} = -j0.8065 \text{ per unit}$$

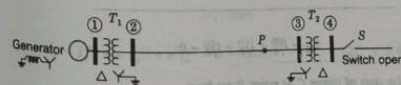


FIGURE 12.21 Single-line diagram of the system of Example 12.7. Single line-to-ground fault is at point *P*.

Resolving the line currents into their symmetrical components gives

$$I_a^{(0)} = I_a^{(1)} = I_a^{(2)} = \frac{I_a}{3} \quad (12.40)$$

Equations (12.39) and (12.41) are both satisfied by connecting the Thevenin equivalent of the negative- and zero-sequence networks in series between points p and p' , as shown in Fig. 12.30. The sequence currents are now expressed by

$$I_a^{(0)} = I_a^{(2)} = I_a^{(1)} = I_{mn} \frac{Z_{pp'}^{(1)}}{Z_{pp'}^{(0)} + Z_{pp'}^{(1)} + Z_{pp'}^{(2)}} \quad (12.42)$$

where I_{mn} is again the prefault current in phase a of the line (7)–(8) before the open circuits occur in phases b and c . The sequence voltage drops are now given by

$$\begin{aligned} V_a^{(1)} &= I_a^{(1)} (Z_{pp'}^{(2)} + Z_{pp'}^{(0)}) = I_{mn} \frac{Z_{pp'}^{(1)} (Z_{pp'}^{(2)} + Z_{pp'}^{(0)})}{Z_{pp'}^{(1)} + Z_{pp'}^{(2)} + Z_{pp'}^{(0)}} \\ V_a^{(2)} &= -I_a^{(2)} Z_{pp'}^{(2)} = -I_{mn} \frac{-Z_{pp'}^{(1)} Z_{pp'}^{(2)}}{Z_{pp'}^{(1)} + Z_{pp'}^{(2)} + Z_{pp'}^{(0)}} \end{aligned} \quad (12.43)$$

$$V_a^{(0)} = -I_a^{(0)} Z_{pp'}^{(0)} = -I_{mn} \frac{-Z_{pp'}^{(1)} Z_{pp'}^{(0)}}{Z_{pp'}^{(1)} + Z_{pp'}^{(2)} + Z_{pp'}^{(0)}}$$

In each of these equations the right-hand side quantities are all known before the fault occurs. Therefore, Eq. (12.38) can be used to evaluate the symmetrical components of the voltage drops between the fault points p and p' when an open-conductor fault occurs; and Eq. (12.43) can be similarly used when a fault due to two open conductors occurs.

The net effect of the open conductors on the positive-sequence network is to increase the transfer impedance across the line in which the open-conductor fault occurs. For one open conductor this increase in impedance equals the parallel combination of the negative- and zero-sequence networks between points p and p' ; for two open conductors the increase in impedance equals the series combination of the negative- and zero-sequence networks between points p and p' .

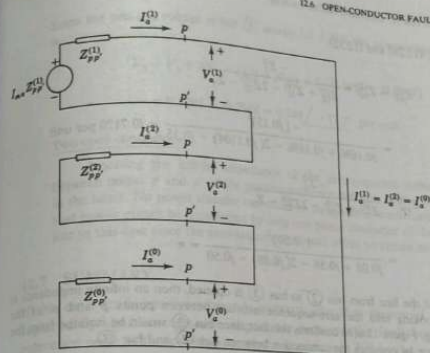


FIGURE 12.30 Connection of the sequence networks of the system to simulate opening phases b and c between points p and p' .

Example 12.8. In the system of Fig. 12.5 consider that machine 2 is a motor drawing a load equivalent to 50 MVA at 0.8 power-factor lagging and nominal system voltage of 345 kV at bus (3). Determine the change in voltage at bus (3) when the transmission line undergoes (a) a one-open-conductor fault and (b) a two-open-conductor fault along its span between buses (2) and (3). Choose a base of 100 MVA, 345 kV in the transmission line.

Solution. All the per-unit parameters given in Example 12.1 apply directly here. By choosing the voltage at bus (3) as $1.0 + j0.0$ per unit, we can calculate the prefault current in line (2)–(3) as follows:

$$I_{23} = \frac{P - jQ}{V_3^*} = \frac{0.5(0.8 - j0.6)}{1.0 + j0.0} = 0.4 - j0.3 \text{ per unit}$$

The sequence networks of Fig. 12.6 show that line (2)–(3) has parameters

$$Z_1 = Z_2 = j0.15 \text{ per unit} \quad Z_0 = j0.50 \text{ per unit}$$

The bus impedance matrices $Z_{bus}^{(0)}$ and $Z_{bus}^{(1)} = Z_{bus}^{(2)}$ are also given in Example 12.1. Designating the open-circuit points of the line as p and p' , we can calculate from

Eqs. (12.28) and (12.32)

$$Z_{pp'}^{(1)} = Z_{pp'}^{(2)} = \frac{-Z_1^2}{Z_{22}^{(1)} + Z_{33}^{(1)} - 2Z_{23}^{(1)} - Z_1}$$

$$= \frac{-(j0.15)^2}{j0.1696 + j0.1696 - 2(j0.1104) - j0.15} = j0.7120 \text{ per unit}$$

$$Z_{pp'}^{(0)} = \frac{-Z_0^2}{Z_{22}^{(0)} + Z_{33}^{(0)} - 2Z_{23}^{(0)} - Z_0}$$

$$= \frac{-(j0.50)^2}{j0.08 + j0.58 - 2(j0.08) - j0.50} = \infty$$

Thus, if the line from bus (2) to bus (3) is opened, then an infinite impedance is seen looking into the zero-sequence network between points p and p' of the opening. Figure 12.6(b) confirms this fact since bus (3) would be isolated from the reference by opening the connection between bus (2) and bus (3).

One open conductor

In this example Eq. (12.38) becomes

$$V_a^{(0)} = V_a^{(2)} = V_a^{(1)} = I_{22} \frac{Z_{pp'}^{(1)} Z_{pp'}^{(2)}}{Z_{pp'}^{(1)} + Z_{pp'}^{(2)}}$$

$$= (0.4 - j0.3) \frac{(j0.7120)(j0.7120)}{j0.7120 + j0.7120}$$

$$= 0.1068 + j0.1424 \text{ per unit}$$

and from Eqs. (12.27) we now calculate the symmetrical components of the voltage at bus (3):

$$\Delta V_3^{(1)} = \Delta V_3^{(2)} = \frac{Z_{32}^{(1)} - Z_{33}^{(1)}}{Z_1} V_a^{(1)} = \left(\frac{j0.1104 - j0.1696}{j0.15} \right) (0.1068 + j0.1424)$$

$$= -0.0422 - j0.0562 \text{ per unit}$$

$$\Delta V_3^{(0)} = \frac{Z_{32}^{(0)} - Z_{33}^{(0)}}{Z_0} V_a^{(0)} = \left(\frac{j0.08 - j0.58}{j0.50} \right) (0.1068 + j0.1424)$$

$$= -0.1068 - j0.1424 \text{ per unit}$$

$$\Delta V_3 = \Delta V_3^{(0)} + \Delta V_3^{(1)} + \Delta V_3^{(2)} = -0.1068 - j0.1424 - 2(0.0422 + j0.0562)$$

$$= -0.1912 - j0.2548 \text{ per unit}$$

Since the prefault voltage at bus (3) equals $1.0 + j0.0$, the new voltage at bus (3) is

$$V_3' = V_3 + \Delta V_3 = (1.0 + j0.0) + (-0.1912 - j0.2548)$$

$$= 0.8088 - j0.2548 = 0.848 \angle -17.5^\circ \text{ per unit}$$

Two open conductors

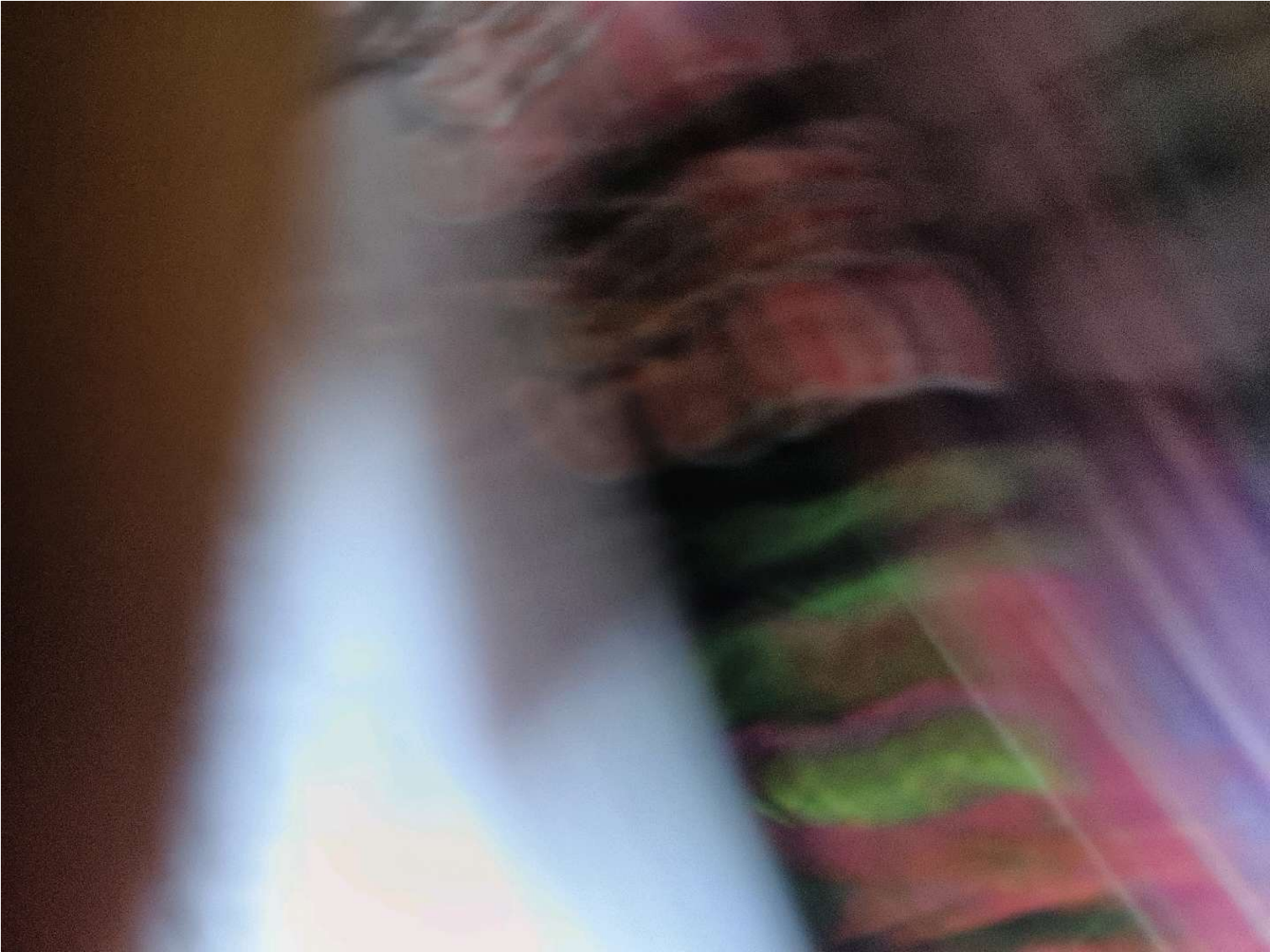
Inserting the infinite impedance of the zero-sequence network in series between points p and p' of the positive-sequence network causes an open circuit in the latter. No power transfer can occur in the system—confirmation of the fact that power cannot be transferred by only one phase conductor of the transmission line in this case since the zero-sequence network offers no return path for current.

12.7 SUMMARY

If, the emfs in a positive-sequence network like that shown in Fig. 12.2 are replaced by short circuits, the impedance between the fault bus (k) and the reference node is the positive-sequence impedance $Z_{kk}^{(1)}$ in the equation developed for faults on a power system and is the series impedance of the Thévenin equivalent of the circuit between bus (k) and the reference node. Thus, we can regard $Z_{kk}^{(1)}$ as a single impedance or the entire positive-sequence network between bus (k) and the reference with no emfs present. If the voltage V_f is connected in series with this modified positive-sequence network, the resulting circuit, shown in Fig. 12.2(e), is the Thévenin equivalent of the original positive-sequence network. The circuits shown in Fig. 12.2 are equivalent only in their effect on any external connections made between bus (k) and the reference node of the original networks. We can easily see that no current flows in the branches of the equivalent circuit in the absence of an external connection, but current will flow in the branches of the original positive-sequence network if any difference exists in the phase or magnitude of the two emfs in the network. In Fig. 12.2(b) the current flowing in the branches in the absence of an external connection is the prefault load current.

When the other sequence networks are interconnected with the positive-sequence network of Fig. 12.2(b) or its equivalent shown in Fig. 12.2(e), the current flowing out of the network or its equivalent is $I_a^{(1)}$ and the voltage between bus (k) and the reference is $V_a^{(1)}$. With such an external connection, the current in any branch of the original positive-sequence network of Fig. 12.2(b) is the positive-sequence current in phase a of that branch during the fault. The prefault component of this current is included. The current in any branch of the Thévenin equivalent of Fig. 12.2(e), however, is only that portion of the actual positive-sequence current found by apportioning $I_a^{(1)}$ of the fault among the branches represented by $Z_{kk}^{(1)}$ according to their impedances and does not include the prefault component.

In the preceding sections we have seen that the Thévenin equivalents of the sequence networks of a power system can be interconnected so that solving



equation, which can be written as the two first-order differential equations

$$\frac{2H}{\omega_s} \frac{d\omega}{dt} = P_m - P_e \text{ per unit} \quad (16.15)$$

$$\frac{d\delta}{dt} = \omega - \omega_s \quad (16.16)$$

in which ω , ω_s , and δ involve electrical radians or electrical degrees.

We use the various equivalent forms of the swing equation throughout this chapter to determine the stability of a machine within a power system. When the swing equation is solved, we obtain the expression for δ as a function of time. A graph of the solution is called the *swing curve* of the machine and inspection of the swing curves of all the machines of the system will show whether the machines remain in synchronism after a disturbance.

16.3 FURTHER CONSIDERATIONS OF THE SWING EQUATION

The megavoltampere (MVA) base used in Eq. (16.11) is the machine rating S_{mach} which is introduced by the definition of H . In a stability study of a power system with many synchronous machines only one MVA base common to all parts of the system can be chosen. Since the right-hand side of the swing equation for each machine must be expressed in per unit on this common system base, it is clear that H on the left-hand side of each swing equation must also be consistent with the system base. This is accomplished by converting H for each machine based on its own individual rating to a value determined by the system base, S_{system} . Equation (16.11), multiplied on each side by the ratio (S_{mach}/S_{system}) , leads to the conversion formula

$$H_{system} = H_{mach} \frac{S_{mach}}{S_{system}} \quad (16.17)$$

in which the subscript for each term indicates the corresponding base being used. In industry studies the system base that is usually chosen is 100 MVA.

The inertia constant M is rarely used in practice and the forms of the swing equation involving H are more often encountered. This is because the value of M varies widely with the size and type of the machine, whereas H assumes a much narrower range of values, as shown in Table 16.1. Machine manufacturers also use the symbol WR^2 to specify for the rotating parts of a generating unit (including the prime mover) the weight in pounds multiplied by the square of the radius of gyration in feet. Hence, $WR^2/32.2$ is the moment of inertia of the machine in slug-feet squared.

TABLE 16.1
Typical inertia constants of synchronous machines†

Type of machine	Inertia constant, H , MJ/MVA
Turbine generator:	
Condensing, 1800 r/min	9–6
3600 r/min	7–4
Noncondensing, 3600 r/min	4–3
Waterwheel generator:	
Slow-speed, < 200 r/min	2–3
High-speed, > 200 r/min	2–4
Synchronous condenser‡	
Large	1.25
Small	1.00
Synchronous motor with load	2.00
varies from 1.0 to 5.0 and higher for heavy flywheels	

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‡Where range is given, the first figure applies to machines of smaller megavoltampere rating.

§Hydrogen-cooled, 25% less.

Example 16.1. Develop a formula to calculate the H constant from WR^2 and then evaluate H for a nuclear generating unit rated at 1333 MVA, 1800 r/min with $WR^2 = 5,820,000 \text{ lb-ft}^2$.

Solution. The kinetic energy (KE) of rotation in foot-pounds at synchronous speed

$$KE = \frac{1}{2} \frac{WR^2}{32.2} \left[\frac{2\pi(r/\min)}{60} \right]^2 \text{ ft-lb}$$

Since 550 ft-lb/s equals 746 W, it follows that 1 ft-lb equals 746/550 J. Hence, converting foot-pounds to megajoules and dividing by the machine rating in megavoltamperes, we obtain

$$H = \frac{\left(\frac{746}{550} \times 10^{-6} \right) \frac{1}{2} \frac{WR^2}{32.2} \left[\frac{2\pi(r/\min)}{60} \right]^2}{S_{mach}}$$

which yields upon simplification

$$H = \frac{2.31 \times 10^{-10} WR^2 (r/\min)^2}{S_{mach}}$$

Inserting the given machine data in this equation, we obtain

$$H = \frac{2.31 \times 10^{-10} (5.82 \times 10^9)(1800)^2}{1333} = 3.27 \text{ MJ/MVA}$$

Converting H to a 100-MVA system base, we obtain

$$H = 3.27 \times \frac{1333}{100} = 43.56 \text{ MJ/MVA}$$

In a stability study for a large system with many machines geographically dispersed over a wide area it is desirable to minimize the number of swing equations to be solved. This can be done if the transmission-line fault, or other disturbance on the system, affects the machines within a power plant so that their rotors swing together. In such cases the machines within the plant can be combined into a single equivalent machine just as if their rotors were mechanically coupled and only one swing equation must be written for them. Consider a power plant with two generators connected to the same bus which is electrically remote from the network disturbances. The swing equations on the common system base are

$$\frac{2H_1}{\omega_s} \frac{d^2\delta_1}{dt^2} = P_{m1} - P_{e1} \text{ per unit} \quad (16.18)$$

$$\frac{2H_2}{\omega_s} \frac{d^2\delta_2}{dt^2} = P_{m2} - P_{e2} \text{ per unit} \quad (16.19)$$

Adding the equations together and denoting δ_1 and δ_2 by δ since the rotors swing together, we obtain

$$\frac{2H}{\omega_s} \frac{d^2\delta}{dt^2} = P_m - P_e \text{ per unit} \quad (16.20)$$

where $H = (H_1 + H_2)$, $P_m = (P_{m1} + P_{m2})$, and $P_e = (P_{e1} + P_{e2})$. This single equation, which is in the form of Eq. (16.12), can be solved to represent the plant dynamics.

Example 16.2. Two 60-Hz generating units operate in parallel within the same power plant and have the following ratings:

Unit 1: 500 MVA, 0.85 power factor, 20 kV, 3600 r/min

$$H_1 = 4.8 \text{ MJ/MVA}$$

Unit 2: 1333 MVA, 0.9 power factor, 22 kV, 1800 r/min

$$H_2 = 3.27 \text{ MJ/MVA}$$

Calculate the equivalent H constant for the two units on a 100-MVA base.

Solution. The total kinetic energy (KE) of rotation of the two machines is

$$\text{KE} = (4.8 \times 500) + (3.27 \times 1333) = 6759 \text{ MJ}$$

Therefore, the H constant for the equivalent machine on 100-MVA base is

$$H = 67.59 \text{ MJ/MVA}$$

and this value can be used in a single swing equation, provided the machines swing together so that their rotor angles are in step at each instant of time.

Machines which swing together are called *coherent* machines. It is noted that when both ω_s and δ are expressed in electrical degrees or radians, the swing equations for coherent machines can be combined together even though, as in the example, the rated speeds are different. This fact is often used in stability studies involving many machines in order to reduce the number of swing equations which need to be solved.

For any pair of noncoherent machines in a system swing equations similar to Eqs. (16.18) and (16.19) can be written. Dividing each equation by its left-hand-side coefficient and subtracting the resultant equations, we obtain

$$\frac{d^2\delta_1}{dt^2} - \frac{d^2\delta_2}{dt^2} = \frac{\omega_s}{2} \left(\frac{P_{m1} - P_{e1}}{H_1} - \frac{P_{m2} - P_{e2}}{H_2} \right) \quad (16.21)$$

Multiplying each side by $H_1H_2/(H_1 + H_2)$ and rearranging, we find that

$$\frac{2}{\omega_s} \left(\frac{H_1H_2}{H_1 + H_2} \right) \frac{d^2(\delta_1 - \delta_2)}{dt^2} = \frac{P_{m1}H_2 - P_{m2}H_1}{H_1 + H_2} - \frac{P_{e1}H_2 - P_{e2}H_1}{H_1 + H_2} \quad (16.22)$$

which also may be written more simply in the form of the basic swing equation, Eq. (16.12), as follows:

$$\frac{2}{\omega_s} \frac{d^2\delta_{12}}{dt^2} = P_{m12} - P_{e12} \quad (16.23)$$

Here the relative angle δ_{12} equals $\delta_1 - \delta_2$, and an equivalent inertia and

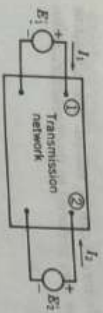


FIGURE 16.3 Schematic diagram for stability studies. Transmission reactances associated with E_1 and E_2 are included in the transmission network.

chronous reactance X_d is in series with the synchronous internal or no-load voltage E_i . Armature resistance is negligible in most cases so that the phasor diagram of Fig. 16.2(b) applies. Since each machine must be considered relative to the system of which it is part, the phasor angles of the machine quantities are measured with respect to the common system reference.

Figure 16.3 schematically represents a generator supplying power through a transmission system to a receiving-end system at bus ②. The rectangle represents the transmission system of linear passive components, such as transformers, transmission lines, and capacitors, and includes the transient reactance of the generator. Therefore, the voltage E_1' represents the transient internal voltage of the generator at bus ①. The voltage E_2' at the receiving end is regarded here as that of an infinite bus or as the transient internal voltage of a synchronous motor whose transient reactance is included in the network. Later we shall consider the case of two generators supplying constant-impedance loads within the network. The bus admittance matrix for the network reduced to two nodes in addition to the reference node is

$$\mathbf{Y}_{\text{bus}} = \begin{matrix} & \text{①} & \text{②} \\ \begin{matrix} \text{①} \\ \text{②} \end{matrix} & \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \end{matrix} \quad (16.28)$$

From Eq. (9.4) we have

$$P_k + jQ_k = V_k \sum_{n=1}^N (Y_{kn} V_n)^* \quad (16.29)$$

Letting k and N equal 1 and 2, respectively, and substituting E' for V , we obtain

$$P_1 + jQ_1 = E_1'(Y_{11}E_1')^* + E_1'(Y_{12}E_2')^* \quad (16.30)$$

If we define

$$\begin{aligned} E_1' &= |E_1'| \angle \delta_1 & E_2' &= |E_2'| \angle \delta_2 \\ Y_{11} &= G_{11} + jB_{11} & Y_{12} &= |Y_{12}| \angle \theta_{12} \end{aligned}$$

Eq. (16.30) yields

$$P_1 = |E_1'|^2 G_{11} + |E_1'| |E_2'| |Y_{12}| \cos(\delta_1 - \delta_2 - \theta_{12}) \quad (16.31)$$

$$Q_1 = -|E_1'|^2 B_{11} + |E_1'| |E_2'| |Y_{12}| \sin(\delta_1 - \delta_2 - \theta_{12}) \quad (16.32)$$

Similar equations apply at bus ② by substituting subscripts 1 for 2, and 2 for 1 in the two preceding equations.

If we let

$$\delta = \delta_1 - \delta_2$$

and define a new angle γ such that

$$\gamma = \theta_{12} - \frac{\pi}{2}$$

we obtain from Eqs. (16.31) and (16.32)

$$P_1 = |E_1'|^2 G_{11} + |E_1'| |E_2'| |Y_{12}| \sin(\delta - \gamma) \quad (16.33)$$

$$Q_1 = -|E_1'|^2 B_{11} - |E_1'| |E_2'| |Y_{12}| \cos(\delta - \gamma) \quad (16.34)$$

Equation (16.33) may be written more simply as

$$P_g = P_e + P_{\text{bus}} \sin(\delta - \gamma) \quad (16.35)$$

where

$$P_g = |E_1'|^2 G_{11} \quad P_{\text{bus}} = |E_1'| |E_2'| |Y_{12}| \quad (16.36)$$

Since P_1 represents the electric power output of the generator (armature loss neglected), we have replaced it by P_g in Eq. (16.35), which is often called the *power-angle equation*; its graph as a function of δ is called the *power-angle curve*. The parameters P_g , P_{bus} , and γ are constants for a given network configuration and constant voltage magnitudes $|E_1'|$ and $|E_2'|$. When the network is considered without resistance, all the elements of $\mathbf{Y}_{\text{bus}}$ are susceptances, and then G_{11} and γ are both zero. The power-angle equation which then applies for the pure reactance network is simply the familiar equation

$$P_g = P_{\text{bus}} \sin \delta \quad (16.37)$$

where $P_{\text{bus}} = |E_1'| |E_2'| / X$ and X is the transfer reactance between E_1' and E_2' .

Example 16.3. The single-line diagram of Fig. 16.4 shows a generator connected through parallel transmission lines to a large metropolitan system considered as an infinite bus. The machine is delivering 1.0 per-unit power and both the terminal voltage and the infinite-bus voltage are 1.0 per unit. Numbers on the diagram

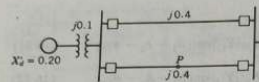


FIGURE 16.4
One-line diagram for Examples 16.3 and 16.4. Point P is at the center of the line.

indicate the values of the reactances on a common system base. The transient reactance of the generator is 0.20 per unit as indicated. Determine the power-angle equation for the given system operating conditions.

Solution. The reactance diagram for the system is shown in Fig. 16.5(a). The series reactance between the terminal voltage and the infinite bus is

$$X = 0.10 + \frac{0.4}{2} = 0.3 \text{ per unit}$$

and therefore the 1.0 per-unit power output of the generator is determined by

$$\frac{|V_t||V|}{X} \sin \alpha = \frac{(1.0)(1.0)}{0.3} \sin \alpha = 1.0$$

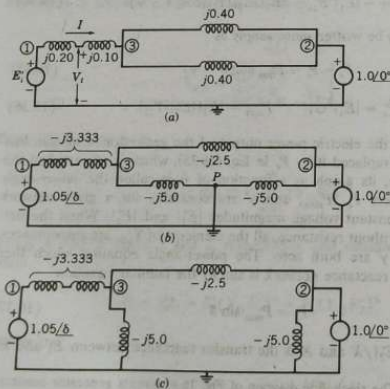


FIGURE 16.5
Reactance diagram for: (a) the prefault network for Example 16.3 with impedances in per unit; (b) and (c), the faulted network for Example 16.4 with the same impedances converted to admittances and marked in per unit.

where V is the voltage of the infinite bus and α is the angle of the terminal voltage relative to the infinite bus. Solving for α , we obtain

$$\alpha = \sin^{-1} 0.3 = 17.458^\circ$$

so that the terminal voltage is

$$V_t = 1.0 / 17.458^\circ = 0.954 + j0.300 \text{ per unit}$$

The output current from the generator is now calculated as

$$I = \frac{1.0 / \delta - 1.0 / 0^\circ}{j0.3}$$

$$= 1.0 + j0.1535 = 1.012 / 8.729^\circ \text{ per unit}$$

and the transient internal voltage is then found to be

$$\begin{aligned} E_t' &= (0.954 + j0.30) + j(0.2)(1.0 + j0.1535) \\ &= 0.923 - j0.5 = 1.050 / 28.44^\circ \text{ per unit} \end{aligned}$$

The power-angle equation relating the transient internal voltage E_t' and the infinite-bus voltage V is determined by the total series reactance

$$X = 0.2 + 0.1 + \frac{0.4}{2} = 0.5 \text{ per unit}$$

Hence, the desired equation is

$$P_e = \frac{(1.050)(1.0)}{0.5} \sin \delta = 2.10 \sin \delta \text{ per unit}$$

where δ is the machine rotor angle with respect to the infinite bus.

The power-angle equation of the preceding example is plotted in Fig. 16.6. Note that the mechanical input power P_m is constant and intersects the

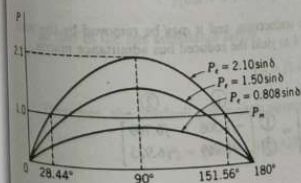


FIGURE 16.6
Plot of power-angle curves found in Examples 16.3 to 16.5.

sinusoidal power-angle curve at the operating angle $\delta_0 = 24.44^\circ$. This is the initial angular position of the generator rotor corresponding to the given operating conditions. The swing equation for the machine may be written

$$\frac{H}{180f} \frac{d^2\delta}{dt^2} = 1.0 - 2.10 \sin \delta \text{ per unit} \quad (16.38)$$

where H is in megajoules per megavoltampere, f is the electrical frequency of the system, and δ is in electrical degrees. We can easily check the results of Example 16.3 since under the given operating conditions, $P_e = 2.10 \sin 28.44^\circ = 1.0$ per unit, which corresponds exactly to the mechanical power input P_m and the acceleration is zero.

In the next example we determine the power-angle equation for the same system with a three-phase fault at P , the midpoint of one of the transmission lines. Positive acceleration is shown to exist due to the fault.

Example 16.4. The system of Example 16.3 is operating under the indicated conditions when a three-phase fault occurs at point P of Fig. 16.4. Determine the power-angle equation for the system with the fault on and the corresponding swing equation. Take $H = 5 \text{ MJ/MVA}$.

Solution. The reactance diagram is shown in Fig. 16.5(b) with the fault on the system at point P . Values shown are admittances in per unit. The effect of the short circuit caused by the fault is clearly shown by redrawing the reactance diagram, as in Fig. 16.5(c). As calculated in Example 16.3, transient internal voltage of the generator remains at $E'_t = 1.05 \angle 28.44^\circ$ based on the assumption of constant flux linkage in the machine. The net transfer admittance connecting the voltage sources remains to be determined. The buses are numbered as shown and the Y_{bus} is formed by inspection of Fig. 16.5(c) as follows:

$$Y_{bus} = \begin{array}{c} \textcircled{1} \quad \textcircled{2} \quad \textcircled{3} \\ \begin{bmatrix} -j3.333 & 0.00 & j3.333 \\ 0.000 & -j7.50 & j2.500 \\ j3.333 & j2.50 & -j10.833 \end{bmatrix} \end{array}$$

Bus $\textcircled{3}$ has no external source connection, and it may be removed by the node elimination procedure of Sec. 7.4 to yield the reduced bus admittance matrix

$$\begin{array}{c} \textcircled{1} \quad \textcircled{2} \\ \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{array}{c} \textcircled{1} \quad \textcircled{2} \\ \begin{bmatrix} -j2.308 & j0.769 \\ j0.769 & -j6.923 \end{bmatrix} \end{array} \end{array}$$

The magnitude of the transfer admittance is 0.769, and thus

$$P_{\max} = |E'_t| |E'_r| |Y_{12}| = (1.05)(1.0)(0.769) \approx 0.808 \text{ per unit}$$

The power-angle equation with the fault on the system is therefore

$$P_e = 0.808 \sin \delta \text{ per unit}$$

and the corresponding swing equation is

$$\frac{5}{180f} \frac{d^2\delta}{dt^2} = 1.0 - 0.808 \sin \delta \text{ per unit} \quad (16.39)$$

Due to its inertia, the rotor cannot change position instantly upon occurrence of the fault. Therefore, the rotor angle δ is initially 28.44° , the same as in Example 16.3, and the electrical power output is $P_e = 0.808 \sin 28.44^\circ = 0.385$. The initial accelerating power is

$$P_a = 1.0 - 0.385 = 0.615 \text{ per unit}$$

and the initial acceleration is positive with the value given by

$$\frac{d^2\delta}{dt^2} = \frac{180f}{5} (0.615) = 22.14 f \text{ elec deg/s}^2$$

where f is the system frequency.

Relaying schemes sensing the fault on the line will act to clear the fault by simultaneous opening of the line-end breakers. When this occurs, another power-angle equation applies because of the network change.

Example 16.5. The fault on the system of Example 16.4 is cleared by simultaneous opening of the circuit breakers at each end of the affected line. Determine the power-angle equation and the swing equation for the postfault period.

Solution. Inspection of Fig. 16.5(a) shows that upon removal of the faulted line, the net transfer admittance across the system is

$$\frac{1}{j(0.2 + 0.1 + 0.4)} = -j1.429 \text{ per unit}$$

so that in the bus admittance matrix

$$Y_{12} = j1.429$$

Therefore, the postfault power-angle equation is

$$P_e = (1.05)(1.0)(1.429) \sin \delta = 1.500 \sin \delta$$

and the corresponding swing equation is

$$\frac{5}{180} \frac{d^2 \delta}{dt^2} = 1.0 - 1.500 \sin \delta$$

The acceleration at the instant of clearing the fault depends on the angular position of the rotor at that time. The power-angle curves for Examples 16.3 through 16.5 are compared in Fig. 16.6.

16.5 SYNCHRONIZING POWER COEFFICIENTS

In Example 16.3 the operating point on the sinusoidal P_e curve of Fig. 16.6 is found to be at $\delta_0 = 28.44^\circ$, where the mechanical power input P_m equals the electrical power output P_e . In the same figure it is also seen that P_e equals P_m at $\delta = 151.56^\circ$, and this might appear to be an equally acceptable operating point. However, this is not the case as now shown.

A commonsense requirement for an acceptable operating point is that the generator should not lose synchronism when small temporary changes occur in the electrical power output from the machine. To examine this requirement for fixed mechanical input power P_m , consider small incremental changes in the operating point parameters; that is, consider

$$\delta = \delta_0 + \delta_\Delta \quad P_e = P_{e0} + P_{e\Delta} \quad (16.40)$$

where the subscript zero denotes the steady-state operating point values and the subscript delta (Δ) identifies the incremental variations from those values. Substituting Eqs. (16.40) in Eq. (16.37), we obtain the power-angle equation for the general two-machine system in the form

$$\begin{aligned} P_{e0} + P_{e\Delta} &= P_{\max} \sin(\delta_0 + \delta_\Delta) \\ &= P_{\max} (\sin \delta_0 \cos \delta_\Delta + \cos \delta_0 \sin \delta_\Delta) \end{aligned}$$

Since δ_Δ is a small incremental displacement from δ_0 ,

$$\sin \delta_\Delta \approx \delta_\Delta \quad \text{and} \quad \cos \delta_\Delta \approx 1 \quad (16.41)$$

and when strict equality is assumed, the previous equation becomes

$$P_{e0} + P_{e\Delta} = P_{\max} \sin \delta_0 + (P_{\max} \cos \delta_0) \delta_\Delta \quad (16.42)$$

At the initial operating point δ_0

$$P_m = P_{e0} = P_{\max} \sin \delta_0 \quad (16.43)$$

and it then follows from Eq. (16.42) that

$$P_m - (P_{e0} + P_{e\Delta}) = -(P_{\max} \cos \delta_0) \delta_\Delta \quad (16.44)$$

Substituting the incremental variables of Eq. (16.40) in the basic swing equation, Eq. (16.12), we obtain

$$\frac{2H}{\omega_s} \frac{d^2(\delta_0 + \delta_\Delta)}{dt^2} = P_m - (P_{e0} + P_{e\Delta}) \quad (16.45)$$

Replacing the right-hand side of this equation by Eq. (16.44) and transposing terms, we obtain

$$\frac{2H}{\omega_s} \frac{d^2 \delta_\Delta}{dt^2} + (P_{\max} \cos \delta_0) \delta_\Delta = 0 \quad (16.46)$$

since δ_0 is a constant value. Noting that $P_{\max} \cos \delta_0$ is the slope of the power-angle curve at the angle δ_0 , we denote this slope as S_p and define it as

$$S_p = \left. \frac{dP_e}{d\delta} \right|_{\delta=\delta_0} = P_{\max} \cos \delta_0 \quad (16.47)$$

where S_p is called the *synchronizing power coefficient*. When S_p is used in Eq. (16.46), the swing equation governing the incremental rotor-angle variations may be rewritten in the form

$$\frac{d^2 \delta_\Delta}{dt^2} + \frac{\omega_s S_p}{2H} \delta_\Delta = 0 \quad (16.48)$$

This is a linear, second-order differential equation, the solution to which depends on the algebraic sign of S_p . When S_p is positive, the solution $\delta_\Delta(t)$ corresponds to that of simple harmonic motion; such motion is represented by the oscillations of an undamped swinging pendulum.³ When S_p is negative, the solution $\delta_\Delta(t)$ increases exponentially without limit. Therefore, in Fig. 16.6 the operating point $\delta_0 = 28.44^\circ$ is a point of stable equilibrium, in the sense that the rotor-angle swing is bounded following a small perturbation. In the physical situation damping will restore the rotor angle to δ_0 following the temporary electrical perturbation. On the other hand, the point $\delta = 151.56^\circ$ is a point

³The equation of simple harmonic motion is $d^2x/dt^2 + \omega_n^2 x = 0$, which has the general solution $A \cos \omega_n t + B \sin \omega_n t$ with constants A and B determined by the initial conditions. The solution when plotted is an undamped sinusoid of angular frequency ω_n .

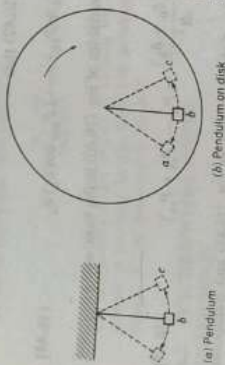


FIGURE 16.7
Pendulum and rotating disk to illustrate a rotor swinging with respect to an infinite bus.

unstable equilibrium since S_p is negative there. So, this point is not a valid operating point.

The changing position of the generator rotor swinging with respect to the infinite bus may be visualized by an analogy. Consider a pendulum swinging from a pivot on a stationary frame, as shown in Fig. 16.7(a). Points a and c are the maximum points of the oscillation of the pendulum about the equilibrium point b . Damping will eventually bring the pendulum about the equilibrium point b rotating in a clockwise direction about the pivot of the pendulum, as shown in Fig. 16.7(b), and superimpose the motion of the pendulum on the motion of the disk. When the pendulum is moving from a to c , the combined angular velocity is slower than that of the disk. When the pendulum is moving from c to a , the combined angular velocity is faster than that of the disk. At points a and c the angular velocity of the pendulum alone is zero and the combined angular velocity equals that of the disk. If the angular velocity of the disk corresponds to the synchronous speed of the rotor, and if the motion of the pendulum alone represents the swinging of the rotor, and if the motion of the infinite bus, the superimposed motion of the rotor with respect to an infinite bus, represents the actual angular motion of the rotor.

From the above discussion we conclude that the solution of Eq. (16.48) represents sinusoidal oscillations, provided the synchronizing power coefficient S_p is positive. The angular frequency of the undamped oscillations is given by

$$\omega_n = \sqrt{\frac{\omega_s S_p}{2H}} \text{ elec rad/s} \quad (16.49)$$

which corresponds to a frequency of oscillation given by

$$f_n = \frac{1}{2\pi} \sqrt{\frac{\omega_s S_p}{2H}} \text{ Hz} \quad (16.50)$$

Example 16.6. The machine of Example 16.3 is operating at $\delta = 28.44^\circ$ when it is subjected to a slight temporary electrical system disturbance. Determine the

frequency and period of oscillation of the machine rotor if the disturbance is removed before the prime mover responds. Take $H = 5 \text{ MJ/MVA}$.

Solution. The applicable swing equation is Eq. (16.48) and the synchronizing power coefficient at the operating point is

$$S_p = 2.10 \cos 28.44^\circ = 1.8466$$

The angular frequency of oscillation is therefore

$$\omega_n = \sqrt{\frac{\omega_s S_p}{2H}} = \sqrt{\frac{377 \times 1.8466}{2 \times 5}} = 8.343 \text{ elec rad/s}$$

The corresponding frequency of oscillation is

$$f_n = \frac{8.343}{2\pi} = 1.33 \text{ Hz}$$

and the period of oscillation is

$$T = \frac{1}{f_n} = 0.753 \text{ s}$$

The above example is an important one from the practical viewpoint since it indicates the order of magnitude of the frequencies which can be superimposed upon the nominal 60-Hz frequency in a large system having many interconnected machines. As load on the system changes randomly throughout the day, intermachine oscillations involving frequencies of the order of 1 Hz tend to arise, but these are quickly damped out by the various damping influences caused by the prime mover, the system loads, and the machine itself. It is worthwhile to note that even if the transmission system in our example contains resistance, nonetheless, the swinging of the rotor is harmonic and undamped. Problem 16.8 examines the effect of resistance on the synchronizing power coefficient and the frequency of oscillations. In a later section we again discuss the concept of synchronizing coefficients. In the next section we examine a method of determining stability under transient conditions caused by large disturbances.

16.6 EQUAL-AREA CRITERION OF STABILITY

In Sec. 16.4 we developed swing equations which are nonlinear in nature. Formal solutions of such equations cannot be explicitly found. Even in the case of a single machine swinging with respect to an infinite bus, it is very difficult to obtain literal-form solutions, and computer methods are therefore normally

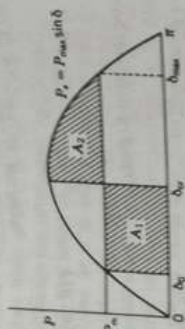


FIGURE 16.10
Power-angle curve showing the critical clearing angle δ_{cr} . Areas A_1 and A_2 are equal.

accelerating, whereas areas A_2 and A_3 are proportional to the decrease in kinetic energy of the rotor while it is decelerating. This can be seen by inspection of both sides of Eq. (16.61). Therefore, the equal-area criterion states that the kinetic energy added to the rotor following a fault must be removed after the fault in order to restore the rotor to synchronous speed.

The shaded area A_1 is dependent on the time taken to clear the fault. If there is delay in clearing, the angle δ_c is increased; likewise, the area A_1 increases and the equal-area criterion requires that area A_2 also increase to restore the rotor to synchronous speed at a larger angle of maximum swing δ_{max} . If the delay in clearing is prolonged so that the rotor angle δ swings beyond the angle δ_{max} in Fig. 16.9, then the rotor speed at that point on the power-angle curve is above synchronous speed when positive accelerating power is again encountered. Under the influence of this positive accelerating power, the angle δ will increase without limit and instability results. Therefore, there is a critical angle for clearing the fault in order to satisfy the requirements of the equal-area criterion for stability. This angle, called the *critical clearing angle* δ_{cr} , is shown in Fig. 16.10. The corresponding critical time for removing the fault is called the *critical clearing time* t_{cr} . Thus, the critical clearing time is the maximum elapsed time from the initiation of the fault until its isolation such that the power system is transiently stable.

In the particular case of Fig. 16.10 both the critical clearing angle and the critical clearing time can be calculated as follows. The rectangular area A_1 is

$$A_1 = \int_{\delta_0}^{\delta_{cr}} P_m d\delta = P_m(\delta_{cr} - \delta_0) \quad (16.65)$$

while the area A_2 is

$$\begin{aligned} A_2 &= \int_{\delta_{cr}}^{\delta_{max}} (P_{max} \sin \delta - P_m) d\delta \\ &= P_{max}(\cos \delta_{cr} - \cos \delta_{max}) - P_m(\delta_{max} - \delta_{cr}) \end{aligned} \quad (16.66)$$

equating the expressions for A_1 and A_2 and transposing terms, yield

$$\cos \delta_{cr} = (P_m/P_{max})(\delta_{max} - \delta_0) + \cos \delta_{max} \quad (16.67)$$

we see from the sinusoidal power-angle curve that

$$\delta_{max} = \pi - \delta_0 \text{ elec rad} \quad (16.68)$$

and

$$P_m = P_{max} \sin \delta_0 \quad (16.69)$$

Substituting for δ_{max} and P_m in Eq. (16.67), simplifying the result, and solving for the critical clearing angle δ_{cr} , we obtain

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0) \sin \delta_0 - \cos \delta_0] \quad (16.70)$$

Substituting this value of δ_{cr} in the left-hand side of Eq. (16.55) yields

$$\delta_{cr} = \frac{\omega P_m t_{cr}^2}{4H} + \delta_0 \quad (16.71)$$

from which we find the critical clearing time

$$t_{cr} = \sqrt{\frac{4H(\delta_{cr} - \delta_0)}{\omega P_m}} \quad (16.72)$$

Example 16.7. Calculate the critical clearing angle and the critical clearing time for the system of Fig. 16.8 when the system is subjected to a three-phase fault at point P on the short transmission line. The initial conditions are the same as those in Example 16.3, and $H = 5 \text{ MJ/MVA}$.

Solution. In Example 16.3 the power-angle equation and initial rotor angle are

$$P_e = P_{max} \sin \delta = 2.10 \sin \delta$$

$$\delta_0 = 28.44^\circ = 0.496 \text{ elec rad}$$

Mechanical input power P_m is 1.0 per unit, and Eq. (16.70) then gives

$$\begin{aligned} \delta_{cr} &= \cos^{-1}[(\pi - 2 \times 0.496) \sin 28.44^\circ - \cos 28.44^\circ] \\ &= 81.69^\circ = 1.426 \text{ elec rad} \end{aligned}$$

Using this value with the other known quantities in Eq. (16.72) yields

$$t_{cr} = \sqrt{\frac{4 \times 5(1.426 - 0.496)}{377 \times 1}} = 0.222 \text{ s}$$

which is equivalent to a critical clearing time of 13.3 cycles on a frequency base of 60 Hz.

This example serves to establish the concept of critical clearing time which is essential to the design of proper relaying schemes for fault clearing. In more general cases the critical clearing time cannot be explicitly found without solving the swing equations by computer simulation.

16.7 FURTHER APPLICATIONS OF THE EQUAL-AREA CRITERION

The equal-area criterion is a very useful means for analyzing stability of a system of two machines, or of a single machine supplied from an infinite bus. However, the computer is the only practical way to determine the stability of a large system. Because the equal-area criterion is so helpful in understanding transient stability, we continue to examine it briefly before discussing the determination of swing curves by the computer approach.

When a generator is supplying power to an infinite bus over parallel transmission lines, opening one of the lines may cause the generator to lose synchronism even though the load could be supplied over the remaining line under steady-state conditions. If a three-phase short circuit occurs on the bus to which two parallel lines are connected, no power can be transmitted over either line. This is essentially the case in Example 16.7. However, if the fault is at the end of one of the lines, opening breakers at both ends of the line will isolate the fault from the system and allow power to flow through the other parallel line. When a three-phase fault occurs at some point on a double-circuit line other than on the paralleling buses or at the extreme ends of the line, there is some impedance between the paralleling buses and the fault. Therefore, some power is transmitted while the fault is still on the system. The power-angle equation in Example 16.4 demonstrates this fact.

When power is transmitted during a fault, the equal-area criterion is applied, as shown in Fig. 16.11, which is similar to the power-angle diagram of Fig. 16.6. Before the fault $P_{\max} \sin \delta$ is the power which can be transmitted; during the fault $r_1 P_{\max} \sin \delta$ is the power which can be transmitted; and $r_2 P_{\max} \sin \delta$ is the power which can be transmitted after the fault is cleared by switching at the instant when $\delta = \delta_{cr}$. Examination of Fig. 16.11 shows that δ_{cr} is the critical clearing angle in this case. By evaluating the areas A_1 and A_2

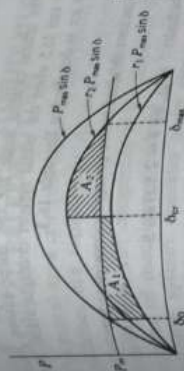


FIGURE 16.11
Equal-area criterion applied to fault clearing when power is transmitted during the fault. Areas A_1 and A_2 are equal.

using the procedural steps of the previous section, we find that

$$\cos \delta_{cr} = \frac{(P_m/P_{\max})(\delta_{\max} - \delta_0) + r_2 \cos \delta_{\max} - r_1 \cos \delta_0}{r_2 - r_1} \quad (16.73)$$

A literal-form solution for the critical clearing time t_{cr} is not possible in this case. For the particular system and fault location shown in Fig. 16.8 the applicable values are $r_1 = 0$, $r_2 = 1$, and Eq. (16.73) then reduces to Eq. (16.67).

Short-circuit faults which do not involve all three phases allow some power to be transmitted over the unaffected phases. Such faults are represented by connecting an impedance (rather than a short circuit) between the fault point and the reference node in the positive-sequence impedance diagram. The larger the impedance shunted across the positive-sequence network to represent the fault, the larger the power transmitted during the fault. The amount of power transmitted during the fault affects the value of A_1 for any given clearing angle. Thus, smaller values of r_1 result in greater disturbances to the system, as smaller amounts of power are transmitted during the fault. Consequently, the area A_1 of acceleration is larger. In order of increasing severity (that is, decreasing $r_1 P_{\max}$) the various faults are:

1. A single line-to-ground fault
2. A line-to-line fault
3. A double line-to-ground fault
4. A three-phase fault

The single line-to-ground fault occurs most frequently, and the three-phase fault is the least frequent. For complete reliability a system should be designed for transient stability for three-phase faults at the worst locations, and this is virtually the universal practice.

Example 16.8. Determine the critical clearing angle for the three-phase fault described in Examples 16.4 and 16.5 when the initial system configuration and prefault operating conditions are as described in Example 16.3.

Solution. The power-angle equations obtained in the previous examples are

$$\text{Before the fault: } P_{\max} \sin \delta = 2.100 \sin \delta$$

$$\text{During the fault: } r_1 P_{\max} \sin \delta = 0.808 \sin \delta$$

$$\text{After the fault: } r_2 P_{\max} \sin \delta = 1.500 \sin \delta$$

Hence,

$$r_1 = \frac{0.808}{2.100} = 0.385 \quad r_2 = \frac{1.500}{2.100} = 0.714$$

From Example 16.3 we have

$$\delta_0 = 28.44^\circ = 0.496 \text{ rad}$$

and from Fig. 16.11 we calculate

$$\delta_{\max} = 180^\circ - \sin^{-1} \left[\frac{1.000}{1.500} \right] = 138.190^\circ = 2.412 \text{ rad}$$

Therefore, inserting numerical values in Eq. (16.73), we obtain

$$\begin{aligned} \cos \delta_{cr} &= \frac{\left(\frac{1.0}{2.10} \right) (2.412 - 0.496) + 0.714 \cos(138.19^\circ) - 0.385 \cos(28.44^\circ)}{0.714 - 0.385} \\ &= 0.127 \end{aligned}$$

Hence,

$$\delta_{cr} = 82.726^\circ$$

To determine the critical clearing time for this example, we must obtain the swing curve of δ versus t . In Sec. 16.9 we discuss one method of computing such swing curves.

The equal-area criterion cannot be used directly in systems where three or more machines are represented. Although the physical phenomena observed in the two-machine problems are basically the same as in the multimachine case, nonetheless, the complexity of the numerical computations increases with the number of machines considered in a transient stability study. When a multimachine system operates under electromechanical transient conditions, intermachine oscillations occur through the medium of the transmission system connecting the machines. If any one machine could be considered to act alone as the single oscillating source, it would tend into the interconnected system an electromechanical oscillation determined by its inertia and synchronizing power. A typical frequency of such an oscillation is of the order of 1–2 Hz, and this is superimposed upon the nominal 60-Hz frequency of the system. When many machine rotors are simultaneously undergoing transient oscillations, the swing curves reflect the combined presence of many such oscillations. Therefore, the transmission system frequency is not unduly perturbed from nominal frequency, and the assumption is made that the 60-Hz network parameters are still applicable. To ease the complexity of system modeling, and thereby the computational burden, the following additional assumptions are commonly made in transient stability studies:

1. The mechanical power input to each machine remains constant during the entire period of the swing curve computation.
2. Damping power is negligible.
3. Each machine may be represented by a constant transient reactance in series with a constant transient internal voltage.
4. The mechanical rotor angle of each machine coincides with δ , the electrical phase angle of the transient internal voltage.
5. All loads may be considered as shunt impedances to ground with values determined by conditions prevailing immediately prior to the transient conditions.

The system stability model based on these assumptions is called the *classical stability model*, and studies which use this model are called *classical stability studies*. These assumptions, which we shall adopt, are in addition to the fundamental assumptions set forth in Sec. 16.1 for all stability studies. Of course, detailed computer programs with more sophisticated machine and load models are available to modify one or more of assumptions 1 to 5. Throughout this chapter, however, the classical model is used to study system disturbances originating from three-phase faults.

The system conditions before the fault occurs, and the network configuration both during and after its occurrence, must be known in any transient

stability study, as we have seen. Consequently, in the multimachine case two preliminary steps are required:

1. The steady-state prefault conditions for the system are calculated using a production-type power-flow program.
2. The prefault network representation is determined and then modified to account for the fault and for the postfault conditions.

From the first preliminary step we know the values of power, reactive power, and voltage at each generator terminal and load bus, with all angles measured with respect to the slack bus. The transient internal voltage of each generator is then calculated using the equation

$$E = V_t + jX_d' I \quad (16.74)$$

where V_t is the corresponding terminal voltage and I is the output current. Each load is converted into a constant admittance to ground at its bus using the equation

$$Y_L = \frac{P_L - jQ_L}{|V_t|^2} \quad (16.75)$$

where $P_L + jQ_L$ is the load and $|V_t|$ is the magnitude of the corresponding bus voltage. The bus admittance matrix which is used for the prefault power-flow calculation is now augmented to include the transient reactance of each generator and the shunt admittance of each load, as suggested in Fig. 16.12. Note that the injected current is zero at all buses except the internal buses of the generators.

In the second preliminary step the bus admittance matrix is modified to correspond to the faulted and postfault conditions. Since only the generator internal buses have injections, all other buses can be eliminated by Kron

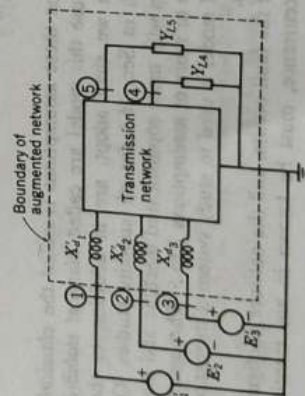


FIGURE 16.12 Augmented network of a power system.

reduction. The dimensions of the modified matrices then correspond to the number of generators. During and after the fault the power flow into the network from each generator is calculated by the power flow into the network. For example, in Fig. 16.12 the power out of generator 1 is given by

$$P_{e1} = |E_1|^2 G_{11} + |E_1||E_2|Y_{12}\cos(\delta_{12} - \theta_{12}) + |E_1||E_3|Y_{13}\cos(\delta_{13} - \theta_{13}) \quad (16.76)$$

where δ_{12} equals $\delta_1 - \delta_2$. Similar equations are written for P_{e2} and P_{e3} using the Y_{ij} elements of the 3×3 bus admittance matrices appropriate to the fault or postfault condition. The P_{ei} expressions form part of the swing equations

$$2H_i \frac{d^2\delta_i}{\omega_s^2} = P_{mi} - P_{ei} \quad i = 1, 2, 3 \quad (16.77)$$

which represent the motion of each rotor during the fault and postfault periods. The solutions depend on the location and duration of the fault, and the Y_{bus} resulting when the faulted line is removed. The basic procedures used in computer programs for classical stability studies are revealed in the following examples.

Example 16.9. A 60-Hz, 230-kV transmission system shown in Fig. 16.13 has two generators of finite inertia and an infinite bus. The transformer and line data are given in Table 16.2. A three-phase fault occurs on line (4)-(5) near bus (4). Using the prefault power-flow solution given in Table 16.3, determine the swing equation for each machine during the fault period. The generators have reactances

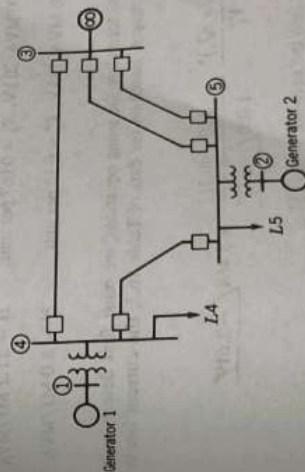


FIGURE 16.13 One-line diagram for Example 16.9.

TABLE 16.2
Line and transformer data for Example 16.9†

Bus to bus	Series Z		Shunt Y
	R	X	B
Transformer ①-④	—	0.022	—
Transformer ②-⑤	—	0.040	—
Line ③-④	0.007	0.040	0.082
Line ③-⑤(1)	0.008	0.047	0.098
Line ③-⑤(2)	0.008	0.047	0.098
Line ④-⑤	0.018	0.110	0.226

†All values in per unit on 230-kV, 100-MVA base.

TABLE 16.3
Bus data and prefault load-flow values†

Bus	Voltage	Generation		Load	
		P	Q	P	Q
①	1.030/8.88°	3.500	0.712	—	—
②	1.020/6.38°	1.850	0.298	—	—
③	1.000/0°	—	—	—	—
④	1.018/4.68°	—	—	1.00	0.44
⑤	1.011/2.27°	—	—	0.50	0.16

†Values are in per unit on 230-kV, 100-MVA base.

and H values expressed on a 100-MVA base as follows:Generator 1: 400 MVA, 20 kV, $X_d' = 0.067$ per unit, $H = 11.2$ MJ/MVAGenerator 2: 250 MVA, 18 kV, $X_d' = 0.10$ per unit, $H = 8.0$ MJ/MVA**Solution.** In order to formulate the swing equations, we must first determine the transient internal voltages. Based on the data of Table 16.3, the current into the network at bus ① is

$$I_1 = \frac{(P_1 + jQ_1)^*}{V_1^*} = \frac{3.50 - j0.712}{1.030 \angle -8.88^\circ} = 3.468 \angle -2.619^\circ$$

Similarly, the current into the network at bus ② is

$$I_2 = \frac{(P_2 + jQ_2)^*}{V_2^*} = \frac{1.850 - j0.298}{1.020 \angle -6.38^\circ} = 1.837 \angle -2.771^\circ$$

16.8 MULTIMACHINE STABILITY STUDIES CLASSICAL REPRESENTATION 731
from Eq. (16.74) we then calculate

$$E_1' = 1.030 \angle 8.88^\circ + j0.067 \times 3.468 \angle -2.619^\circ = 1.100 \angle 20.82^\circ$$

$$E_2' = 1.020 \angle 6.38^\circ + j0.10 \times 1.837 \angle -2.771^\circ = 1.065 \angle 16.19^\circ$$

At the infinite bus we have

$$E_3' = E_3 = 1.000 \angle 0^\circ$$

$$\delta_{13} = \delta_1 \quad \delta_{23} = \delta_2$$

and so

The P - Q loads at buses ④ and ⑤ are converted into equivalent shunt admittances using Eq. (16.75), which yields

$$Y_{L4} = \frac{1.00 - j0.44}{(1.018)^2} = 0.9649 - j0.4246 \text{ per unit}$$

$$Y_{L5} = \frac{0.50 - j0.16}{(1.011)^2} = 0.4892 - j0.1565 \text{ per unit}$$

The prefault bus admittance matrix is now modified to include the load admittances and the transient reactances of the machines. Buses ① and ② designate the fictitious internal nodes behind the transient reactances of the machines. So, in the prefault bus admittance matrix, for example,

$$Y_{11}' = \frac{1}{j0.067 + j0.022} = -j11.236 \text{ per unit}$$

$$Y_{54} = -\frac{1}{0.007 + j0.040} = -4.2450 + j24.2571 \text{ per unit}$$

The sum of the admittances connected to buses ③, ④, and ⑤ must include the shunt capacitances of the transmission lines. So, at bus ④ we have

$$Y_{44} = -j11.236 + \frac{j0.082}{2} + \frac{j0.226}{2} + 4.2450 - j24.2571$$

$$+ \frac{1}{0.018 + j0.110} + 0.9649 - j0.4246$$

$$= 6.6587 - j44.6175 \text{ per unit}$$

The new prefault bus admittance matrix is displayed as Table 16.4 on page 733.

Bus ④ must be short-circuited to the reference to represent the fault. Row 4 and column 4 of Table 16.4 thereby disappear because node ④ is now merged with the reference node. Next, the row and column representing bus ⑤ are eliminated by Kron reduction, and we obtain the bus admittance matrix shown in

the upper half of Table 16.5. The faulted-system Y_{bus} shows that bus ① decouples from the other buses during the fault and that bus ② is connected directly to bus ③. This reflects the physical fact that the short circuit at bus ④ reduces to bus ③ the power injected into the system from generator 1 and causes generator 2 to deliver its power radially to bus ③. Under fault conditions, the power-angle equations based on values from Table 16.5 are

$$\begin{aligned} P_{e1} &= 0 \\ P_{e2} &= |E_2|^2 G_{22} + |E_2||E_3||Y_{23}|\cos(\delta_{23} - \theta_{23}) \\ &= (1.065)^2(0.1362) + (1.065)(1.0)(5.1665)\cos(\delta_2 - 90.755^\circ) \\ &= 0.1545 + 5.5023\sin(\delta_2 - 0.755^\circ) \text{ per unit} \end{aligned}$$

Therefore, while the fault is on the system, the desired swing equations (values of P_{m1} and P_{m2} from Table 16.3) are

$$\begin{aligned} \frac{d^2\delta_1}{dt^2} &= \frac{180f}{H_1}(P_{m1} - P_{e1}) = \frac{180f}{H_1}P_{e1} \\ &= \frac{180f}{11.2}(3.5) \text{ elec deg/s}^2 \\ \frac{d^2\delta_2}{dt^2} &= \frac{180f}{H_2}(P_{m2} - P_{e2}) = \frac{180f}{H_2}P_{e2} \\ &= \frac{180f}{8.0} \left\{ 1.85 - \left[\frac{P_e}{0.1545} + 5.5023\sin\left(\delta_2 - \frac{\gamma}{0.755^\circ}\right) \right] \right\} \\ &= \frac{180f}{8.0} \left\{ \frac{1.6955}{P_m - P_e} - \frac{5.5023\sin(\delta_2 - 0.755^\circ)}{P_{\text{max}}} \right\} \text{ elec deg/s}^2 \end{aligned}$$

Example 16.10. The three-phase fault in Example 16.9 is cleared by simultaneously opening the circuit breakers at the ends of the faulted line. Determine the swing equations for the postfault period.

Solution. Since the fault is cleared by removing line ④–⑤, the prefault Y_{bus} of Table 16.4 must be modified again. This is accomplished by substituting zero for Y_{45} and Y_{54} and by subtracting the series admittance of line ④–⑤ and the capacitive susceptance of one-half the line from elements Y_{44} and Y_{55} of Table 16.4. The reduced bus admittance matrix applicable to the postfault network is shown in the lower half of Table 16.5. The zero elements in the first and second rows reflect the fact that generators 1 and 2 are not interconnected when line

TABLE 16.4
Elements of prefault bus admittance matrix for Example 16.9†

bus	①	②	③	④	⑤
①	-j11.2360	0.0	0.0	0.0	0.0
②	0.0	-j7.1429	0.0	0.0	0.0
③	0.0	0.0	-j11.2360	0.0	0.0
④	j11.2360	0.0	0.0	-j11.2360	0.0
⑤	0.0	0.0	0.0	0.0	-j11.2360

†Admittances in per unit.

TABLE 16.5
Elements of faulted and postfault bus admittance matrices for Example 16.9†

Bus	Faulted network				
	①	②	③	④	⑤
①	0.0000	-j11.2360	0.0 + j0.0	0.0 + j0.0	0.0 + j0.0
②	(11.2360 / -90°)	0.0 + j0.0	0.1362 - j6.2737	-0.0881 + j5.1661	(5.1665 / 90.7552°)
③	0.0 + j0.0	0.0 + j0.0	(6.2737 / -88.7553°)	-0.0881 + j5.1661	(5.1665 / 90.7552°)
④	0.0 + j0.0	0.0 + j0.0	(5.1665 / 90.7552°)	0.1362 - j6.2737	(5.1665 / 90.7552°)
⑤	0.0 + j0.0	0.0 + j0.0	(5.1665 / 90.7552°)	(6.2737 / -88.7553°)	0.1362 - j6.2737

†Admittances in per unit.

④–⑤ is removed. Accordingly, each generator is connected radially to the infinite bus, and we can write power-angle equations for the postfault conditions as follows:

$$\begin{aligned} P_{e1} &= |E_1|^2 G_{11} + |E_1||E_3||Y_{13}|\cos(\delta_{13} - \theta_{13}) \\ &= (1.100)^2(0.5005) + (1.100)(1.0)(7.6323)\cos(\delta_1 - 91.664^\circ) \\ &= 0.6056 + 8.3955\sin(\delta_1 - 1.664^\circ) \text{ per unit} \end{aligned}$$

$$\begin{aligned}
 \text{and } P_{22} &= |E_2|^2 G_{22} + |E_2| |E_3| Y_{23} [\cos(\delta_{23} - \theta_{23})] \\
 &= (1.065)^2 (0.1591) + (1.065)(1.0)(6.0982) \cos(\delta_2 - 90.847^\circ) \\
 &= 0.1804 + 6.4934 \sin(\delta_2 - 0.847^\circ) \text{ per unit}
 \end{aligned}$$

For the postfault period the applicable swing equations are given by

$$\begin{aligned}
 \frac{d^2 \delta_1}{dt^2} &= \frac{180f}{11.2} \{ 3.5 - [0.6056 + 8.3955 \sin(\delta_1 - 1.664^\circ)] \} \\
 &= \frac{180f}{11.2} [2.8944 - 8.3955 \sin(\delta_1 - 1.664^\circ)] \text{ elec deg/s}^2 \\
 \text{and } \frac{d^2 \delta_2}{dt^2} &= \frac{180f}{8.0} \{ 1.85 - [0.1804 + 6.4934 \sin(\delta_2 - 0.847^\circ)] \} \\
 &= \frac{180f}{8.0} [1.6696 - 6.4934 \sin(\delta_2 - 0.847^\circ)] \text{ elec deg/s}^2
 \end{aligned}$$

The power-angle equations obtained in Examples 16.9 and 16.10 are of the form of Eq. (16.35), and the corresponding swing equations assume the form

$$\frac{d^2 \delta}{dt^2} = \frac{180f}{H} [P_m - P_c - P_{\max} \sin(\delta - \gamma)] \quad (16.78)$$

The bracketed right-hand term represents the accelerating power of the rotor. Accordingly, we may write Eq. (16.78) as

$$\frac{d^2 \delta}{dt^2} = \frac{180f}{H} P_a \text{ elec deg/s}^2 \quad (16.79)$$

where

$$P_a = P_m - P_c - P_{\max} \sin(\delta - \gamma) \quad (16.80)$$

In the next section we discuss how to solve equations of the form of Eq. (16.79) in order to obtain δ as a function of time for specified clearing times.

16.9 STEP-BY-STEP SOLUTION OF THE SWING CURVE

For large systems we depend on the computer, which determines δ versus t for all machines in which we are interested; and δ may be plotted versus t for a machine to obtain the swing curve of that machine. The angle δ is calculated as a function of time over a period long enough to determine whether δ will increase without limit or reach a maximum and start to decrease. Although the

result usually indicates stability, on an actual system where a number of variables are taken into account it may be necessary to plot δ versus t over a long enough interval to be sure δ will not increase again without returning to a low value.

By determining swing curves for various clearing times, we can find the length of time permitted before clearing a fault. Standard interrupting times for circuit breakers and their associated relays are commonly 8, 5, 3, or 2 cycles after a fault occurs, and thus breaker speeds may be specified. Calculations should be made for a fault in the position which will allow the least transfer of power from the machine and for the most severe type of fault for which protection against loss of stability is justified.

A number of different methods is available for the numerical evaluation of second-order differential equations in step-by-step computations for small increments of the independent variable. The more elaborate methods are practical only when the computations are performed on a computer. The step-by-step method used for hand calculations is necessarily simpler than some of the methods recommended for computers. In the methods for hand calculation the change in the angular position of the rotor during a short interval of time is computed by making the following assumptions:

1. The accelerating power P_a computed at the beginning of an interval is constant from the middle of the preceding interval to the middle of the interval considered.
2. Throughout any interval the angular velocity is constant at the value computed for the middle of the interval.

Of course, neither of the above assumptions is accurate since δ is changing continuously and both P_a and ω are functions of δ . As the time interval is decreased, the computed swing curve becomes more accurate. Figure 16.14(a) will help in visualizing the assumptions. The accelerating power is computed for the points enclosed in circles at the ends of the $n-2$, $n-1$, and n intervals, which are the beginnings of the $n-1$, n , and $n+1$ intervals. The step curve of P_a in Fig. 16.14(a) results from the assumptions that P_a is constant between the midpoints of the intervals. Similarly, ω , the excess of the angular velocity ω over the synchronous angular velocity ω_s , is shown in Fig. 16.14(b) as a step curve that is constant throughout the interval at the value computed for the midpoint. Between the ordinates $n-\frac{1}{2}$ and $n+\frac{1}{2}$ there is a change of speed caused by the constant accelerating power. The change in speed is the product of the acceleration and the time interval, and so

$$\omega_{r,n-1/2} - \omega_{r,n-3/2} = \frac{d^2 \delta}{dt^2} \Delta t = \frac{180f}{H} P_{a,n-1} \Delta t \quad (16.81)$$

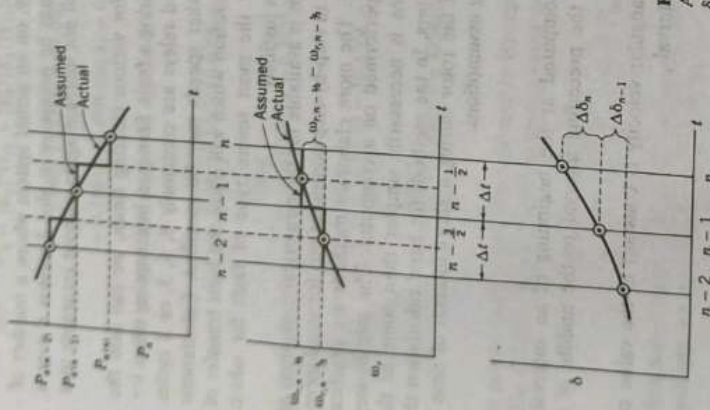


FIGURE 16.14

Actual and assumed values of P_a , ω_r , and δ as functions of time.

The change in δ over an interval is the product of ω_r for the interval and the time of the interval. Thus, the change in δ during the $n-1$ interval is

$$\Delta\delta_{n-1} = \delta_{n-1} - \delta_{n-2} = \Delta t \times \omega_{r, n-3/2} \quad (16.82)$$

and during the n th interval

$$\Delta\delta_n = \delta_n - \delta_{n-1} = \Delta t \times \omega_{r, n-1/2} \quad (16.83)$$

Subtracting Eq. (16.82) from Eq. (16.83) and substituting Eq. (16.81) in the resulting equation eliminate all values of ω_r , and we find that

$$\Delta\delta_n = \delta_{n-1} + kP_{a, n-1} \quad (16.84)$$

$$k = \frac{180f}{H} (\Delta t)^2 \quad (16.85)$$

where

Equation (16.84) is important for the step-by-step solution of the swing equation, for it shows how to calculate the change in δ during an interval based on the accelerating power for that interval and the change in δ for the preceding interval. The accelerating power is calculated at the beginning of each new interval and the solution progresses until enough points are obtained for plotting the swing curve. Greater accuracy is obtained when Δt is small. A value of $\Delta t = 0.05$ s is usually satisfactory.

The occurrence of a fault causes a discontinuity in the accelerating power P_a , which has a zero value before the fault and a nonzero value immediately following the fault. The discontinuity occurs at the beginning of the interval when $t = 0$. Reference to Fig. 16.14 shows that our method of calculation assumes that the accelerating power computed at the beginning of an interval is constant from the middle of the preceding interval to the middle of the interval being considered. When the fault occurs, we have two values of P_a at the beginning of an interval, and we must take the average of these two values as the constant accelerating power. The procedure is illustrated in the following example.

Example 16.11. Prepare a table showing the steps taken to plot the swing curve of machine 2 for the fault on the 60-Hz system of Examples 16.9 and 16.10. The fault is cleared at 0.225 s by simultaneously opening the circuit breakers at the ends of the faulted line.

Solution. Without loss of generality, we consider the detailed computations for machine 2. Computations to plot the swing curve for machine 1 are left to the student. Accordingly, we drop the subscript 2 as the indication of the machine number from all symbols in what follows. All calculations are made in per unit on a 100-MVA base. For the time interval $\Delta t = 0.05$ s the parameter k applicable to machine 2 is

$$k = \frac{180f}{H} (\Delta t)^2 = \frac{180 \times 60}{8.0} \times (0.05)^2 = 3.375 \text{ elec deg}$$

When the fault occurs at $t = 0$, the rotor angle of machine 2 cannot change instantly. Hence, from Example 16.9

$$\delta_0 = 16.19^\circ$$

and during the fault

$$P_e = 0.1545 + 5.5023 \sin(\delta - 0.755^\circ) \text{ per unit}$$

Therefore, as already seen in Example 16.9,

$$P_a = P_m - P_e = 1.6955 - 5.5023 \sin(\delta - 0.755^\circ) \text{ per unit}$$

At the beginning of the first interval there is a discontinuity in the accelerating power of each machine. Immediately before the fault occurs $P_a = 0$. Immediately after the fault occurs

$$P_a = 1.6955 - 5.5023 \sin(16.19^\circ - 0.755^\circ) = 0.231 \text{ per unit}$$

The average value of P_a at $t = 0$ is $\frac{1}{2} \times 0.2310 = 0.1155$ per unit. We then find that

$$kP_a = 3.375 \times 0.1155 = 0.3898^\circ$$

Identifying the time intervals by numerical subscripts, we find that the change in the rotor angle of machine 2 as time advances over the first interval from 0 to Δt is given by

$$\Delta\delta_1 = 0 + 0.3898 = 0.3898^\circ$$

At the end of the first time interval we then have

$$\delta_1 = \delta_0 + \Delta\delta_1 = 16.19^\circ + 0.3898^\circ = 16.5798^\circ$$

and

$$\delta_1 - \gamma = 16.5798^\circ - 0.755^\circ = 15.8248^\circ$$

At $t = \Delta t = 0.05$ s we find that

$$\begin{aligned} kP_{a,1} &= 3.375[(P_m - P_e) - P_{\max} \sin(\delta_1 - \gamma)] \\ &= 3.375[1.6955 - 5.5023 \sin(15.8248^\circ)] = 0.6583^\circ \end{aligned}$$

and it follows that the increase in the rotor angle over the second time interval is

$$\Delta\delta_2 = \Delta\delta_1 + kP_{a,1} = 0.3898^\circ + 0.6583^\circ = 1.0481^\circ$$

At the end of the second time interval

$$\delta_2 = \delta_1 + \Delta\delta_2 = 16.5798^\circ + 1.0481^\circ = 17.6279^\circ$$

The subsequent steps in the computations are shown in Table 16.6. Note that the postfault equation found in Example 16.10 is needed.

In Table 16.6 the terms $P_{\max} \sin(\delta - \gamma)$, P_a , and δ_n have values computed at the time t shown in the first column but $\Delta\delta_n$ is the change in the rotor angle during the interval that begins at the time indicated. For example, in the row for $t = 0.10$ s the angle 17.6279° is the first value calculated and is found by adding the change in angle during the preceding time interval (0.05 to 0.10 s) to the angle at $t = 0.05$ s. Next, $P_{\max} \sin(\delta - \gamma)$ is calculated for $\delta = 17.6279^\circ$. Then, $P_a = (P_m - P_e) - P_{\max} \sin(\delta - \gamma)$ and kP_a are calculated. The value of kP_a is 0.3323° , which is added to the angular change of 1.0481° during the preceding interval to find the

TABLE 16.6
16.9 STEP-BY-STEP SOLUTION OF THE SWING CURVE 739

Computation of swing curve for machine 2 of Example 16.11 for clearing at 0.225 s
 $1 = (180/(\pi \times \Delta t))^2 = 3.375$ elec deg. Before clearing $P_m - P_e = 1.6955$ p.u., $P_{\max} = 5.5023$ p.u., and $\gamma = 0.755^\circ$. After clearing these values become 1.6955, 0.6934, and 0.947° , respectively.

t , s	$(\delta_n - \gamma)$, elec deg	$P_{\max} \sin(\delta_n - \gamma)$, per unit	P_e , per unit	$P_{a,n-1}$, elec deg	$\Delta\delta_n$, elec deg	δ_n , elec deg
0-	—	—	0.00	—	—	—
0+	15.435	1.4644	0.2310	—	—	16.19
0 ^{av}	—	—	0.1155	0.3898	—	16.19
0.05	15.8248	1.5005	0.1950	0.6583	0.3898	16.5798
0.10	16.8729	1.5970	0.0985	0.3323	1.0481	17.6279
0.15	18.2533	1.7234	-0.0279	-0.9942	1.3604	19.0083
0.20	19.5395	1.8403	-0.1448	-0.6886	1.2862	20.2945
0.25	20.2451	2.2470	-0.5774	-1.9487	0.7976	21.0921
0.30	19.0940	2.1241	-0.6345	-1.534	-1.1511	19.9410
0.35	16.4088	1.8343	-0.1647	-0.5559	-2.6852	17.2558
0.40	13.1678	1.4792	0.1904	0.6425	-3.2410	14.0148
0.45	10.5693	1.1911	0.4785	1.6151	-2.9883	11.4163
0.50	9.5860	1.0813	0.5883	1.9854	-0.9833	10.4330
0.55	10.5880	1.1931	0.4765	1.6081	1.0020	11.4350
0.60	13.1981	1.4826	0.1870	0.8312	2.6101	14.0451
0.65	16.4395	1.8376	-0.1680	-0.5072	3.2414	17.2865
0.70	19.1137	2.1262	-0.4566	-1.5411	2.6742	19.9607
0.75	20.2468	2.2471	-0.5775	-1.9492	1.1331	21.0938
0.80	19.4307	2.1601	-0.4905	-1.6556	-0.8161	20.2777
0.85	—	—	—	—	-2.4716	17.8061

change of 1.3804° during the interval beginning at $t = 0.10$ s. This value added to 17.6279° gives the value $\delta = 19.0083^\circ$ at $t = 0.15$ s. Note that the value of $P_m - P_e$ changes at 0.25 s because the fault is cleared at 0.225 s. The angle γ has also changed from 0.755° to 0.847° .

Whenever a fault is cleared, a discontinuity occurs in the accelerating power P_a . When clearing is at 0.225 s, as in Table 16.6, no special approach is